

MAESMA: Agentic AI for Autonomous Earth System Observation, Model Discovery, and Simulation

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February 19, 2026

*This work accelerated by NERVOSYS Autoresearch

Abstract

Earth system models (ESMs) remain among the most complex scientific software ever built, yet their construction is still largely manual: researchers hand-select process representations, tune parameters, and evaluate skill through labor-intensive campaigns. We present MAESMA, a multi-agent system that autonomously discovers, assembles, benchmarks, selects, and invents ESM configurations. At its center lies a *Process Knowledgebase* — a versioned store of process models, manifests, ontological metadata, and skill records — over which a *neural inference engine* (graph transformer) reasons to propose process selections driven by simulation errors and uncertainties. A *process discovery pipeline* learns new representations from observational residuals via neural operators (FNO, PINO, DeepONet, MeshGraphNet) and symbolic regression, then *evolves* them through crossover, mutation, and Pareto-optimal selection over the knowledgebase graph. Drawing on artificial life principles, every process representation is treated as a living entity that must earn its existence through predictive skill: survival tiers gate compute, constitutional invariants propagate to offspring, and a heartbeat daemon enforces continuous selection pressure. Foundation weather models (NVIDIA Earth-2) provide GPU-accelerated atmospheric state evolution, while autonomous observation intelligence (PhiSat-2 principles) steers data acquisition toward maximal information gain. The system covers 13 process families, geoengineering feedback control, planetary defense modeling, and inter-institutional collaboration via A2A federation. MAESMA is implemented in Rust with a Next.js monitoring dashboard.

Keywords: Earth system model, autonomous AI, agentic systems, multi-agent system, process discovery, process evolution, artificial life, ALife, survival tiers, constitutional invariants, self-replication, phylogenetic lineage, knowledgebase, neural inference, graph transformer, model selection, GPU acceleration, neural operators, physics-informed machine learning, Fourier Neural Operator, foundation weather models, Earth-2, on-board AI, PhiSat-2, edge inference, geoengineering, planetary defense, A2A federation.

1 Introduction

Earth system models (ESMs) couple atmosphere, ocean, cryosphere, biosphere, and human systems to simulate planetary dynamics across spatial scales from meters to the globe and temporal scales from seconds to millennia. Despite decades of development, the construction of these models remains fundamentally manual: modelers choose process representations based on expertise and convention, tune parameters through trial-and-error campaigns, and evaluate skill against limited observation suites. The combinatorial space of possible configurations is vast, yet only a tiny fraction is ever explored.

Three structural limitations constrain progress. First, *process selection is heuristic* — the decision to use a Rothermel fire spread model or a wind-aware plume model is made by human judgment rather than systematic evaluation. Second, *process discovery is decoupled from modeling* — when simulations exhibit persistent biases, the feedback loop from error to new process representation is slow, informal, and largely manual. Third, *knowledge is fragmented* — process implementations, metadata, performance records, and ontological relationships are scattered across code repositories, papers, and institutional memory rather than unified in a single queryable store.

MAESMA addresses all three limitations through a knowledgebase-centric architecture in which an autonomous agent swarm reasons over a central Process Knowledgebase via a neural inference engine. The system runs indefinitely — discovering data, assembling configurations, benchmarking against observations, selecting optimal structures, discovering new representations from residual analysis, and depositing them back into the knowledgebase — without human gating. Humans monitor via a live dashboard and may adjust objectives or budgets, but the workflow proceeds autonomously.

The contributions of this paper are:

1. A *knowledgebase-centric architecture* that unifies process code, manifests, ontological relations, skill records, and provenance in a single versioned store, with a *salient-dynamics-first selection principle* that prioritizes processes with the greatest effect on system state evolution.
2. A *neural inference engine* (graph transformer) that reasons over the knowledgebase to propose process selections driven by simulation errors and uncertainties, embedded in an *autonomous optimization loop* with five concurrent cycles that runs indefinitely.
3. A *process discovery and evolution pipeline* that detects persistent biases, learns new representations via NVIDIA Modulus-trained neural operators and symbolic regression, and *evolves* them through crossover, mutation, and Pareto-optimal selection over the knowledgebase graph.
4. An *artificial life (ALife) process lifecycle* inspired by Conway Research’s automaton: every process representation is a sovereign living entity with survival tiers, constitutional invariants, self-replication, a heartbeat daemon for continuous selection pressure, and phylogenetic lineage tracking.
5. *Foundation model and observation intelligence integration*: pre-trained AI weather models (NVIDIA Earth-2) as GPU-accelerated Atmosphere R0/R1 rungs, and PhiSat-2-inspired edge-AI classifiers for autonomous data acquisition driven by model uncertainty.
6. Coverage of *13 process families*, *25 specialized agents*, geoengineering feedback control with 14 intervention types, planetary defense calibrated against 13 mass extinction events, and an *A2A federation protocol* for inter-institutional collaboration.

2 Architecture

2.1 Knowledgebase-Centric Design

The Process Knowledgebase is the gravitational center of the MAESMA architecture. All agents read from and write to it. The neural inference engine reasons over it to propose process selections. Errors and uncertainties from simulations flow back as the primary signal driving both selection and discovery.

The architecture follows a data-flow cycle:

1. Observations enter the knowledgebase via STAC, CMR, NRT feeds, and PhiSat-2-style edge-AI filtered satellite streams. Foundation weather models (Earth-2) generate rapid ensemble initial and boundary conditions on GPU.
2. The neural inference engine queries the knowledgebase given current error signals, regime context, and compute budget, and agent proposals flow to the simulation runtime for multi-GPU execution.
3. Errors and uncertainties from simulations drive the next inference cycle and update the observation value map.
4. The process discovery pipeline learns new representations from persistent residuals — using Modulus-trained neural operators — deposits them into the knowledgebase, and evolves them via crossover, mutation, and Pareto-optimal selection over the knowledgebase graph.
5. Every surviving representation is a living automaton: a heartbeat daemon evaluates survival tiers, enforces constitutional invariants, archives stagnant processes, and triggers self-replication for high-fitness candidates — maintaining continuous selective pressure.

This cycle repeats indefinitely. The knowledgebase grows continuously: every discovered process, every skill record, and every observation feeds back into the central store. The system’s knowledge compounds over time.

2.2 Three-Layer Control Hierarchy

MAESMA organizes its agent swarm into three concurrent control layers (Table 1): strategic (hours–weeks: intent, optimization, model selection), tactical (minutes–hours: retrieval, assembly, benchmarking, discovery), and operational (seconds–minutes: runtime sentinel, compilation, scheduling). All layers run concurrently and continuously; the strategic loop never terminates.

2.3 Salient Dynamics Prioritization

A core architectural principle is *salient dynamics first*: agents prioritize processes whose inclusion has the greatest effect on the time evolution of system states. Rather than assembling all process families at maximum fidelity simultaneously, the system begins each region–regime configuration with the dominant drivers (e.g., radiation budget, large-scale circulation, primary productivity) and incrementally adds lower-impact detail as compute budget permits and accuracy targets demand. The neural inference engine quantifies salience by estimating the expected skill score improvement per unit computational cost for each candidate process addition. This hierarchical approach avoids wasting cycles on negligible refinements while ensuring that the most influential dynamics are always resolved first.

2.4 Scale-Aware Process Graph

The Scale-Aware Process Graph (SAPG) is the formal substrate that agents construct, validate, compile, and optimize. It is a typed, unit-aware, scale-aware directed hypergraph:

- **Nodes** are state variables (e.g., $T_{\text{air}}(x, y, z, t)$, $\theta_{\text{soil}}(\text{layer}, t)$).
- **Edges** are process operators (radiation, infiltration, stomatal conductance).
- **Hyperedges** represent multi-input/output processes.
- **Constraints** encode units, bounds, conservation laws, closure requirements, and numerical stability conditions (CFL, stiffness).

The compiler validates conservation, closure, and coupling constraints before generating an executable task schedule. Two-tier coupling separates slow processes (days–centuries: succession, soil carbon, management) from fast processes (seconds–hours: fire spread, plume dynamics, overland flow).

2.5 GPU-Accelerated Compute Architecture

GPU acceleration is a first-class architectural concern, not an afterthought. Five pillars support GPU-native execution across the full pipeline:

GPU-resident state management. Simulation state arrays are allocated on GPU memory and remain device-resident across timesteps. Process operators read from and write to GPU buffers directly via `wgpu` compute shaders (portable across Vulkan, Metal, DX12) or CUDA kernels via `cudarc` bindings. CPU↔GPU transfers occur only at checkpoint boundaries and observation ingestion.

Batched ensemble execution. The runtime schedules multiple ensemble members, regions, and rung-swap experiments as concurrent GPU workgroups, enabling the autonomous optimizer to evaluate many candidate configurations per cycle.

Neural operator inference on device. The graph transformer and all neural operator surrogates execute forward passes on GPU, writing results directly to GPU-side buffers consumed by the scheduler.

Physics-informed training (NVIDIA Modulus). The process discovery pipeline enforces PDE residual constraints, boundary conditions, and conservation laws as differentiable loss terms computed entirely on GPU. Geometry-adaptive Fourier Neural Operators (Geo-FNO) handle irregular domains without remeshing. The Modulus model zoo (FNO, PINO, DeepONet, Mesh-GraphNet) provides pre-validated architectures that the discovery pipeline instantiates, trains, and validates.

Multi-GPU scaling. Domain decomposition distributes spatial tiles across multiple GPUs via NCCL collective operations with GPU-direct RDMA halo exchanges. The federation protocol extends this to multi-node, multi-institution GPU clusters.

2.6 Foundation Weather Models (NVIDIA Earth-2)

MAESMA integrates pre-trained foundation weather and climate models as first-class inference backends within its process family architecture. Rather than treating AI weather prediction as

an external tool, the system embeds foundation models from the NVIDIA Earth-2 platform (earth2studio) as process representations at the R0–R1 rungs of the Atmosphere family and as rapid emulators for coupled initial/boundary conditions across all families.

The Earth-2 integration operates at three levels:

Rapid ensemble initial conditions. Foundation models — including FourCastNet (SFNO architecture, 0.25° global, 6-hourly), Pangu-Weather (3D transformer, pressure-level prognostics), GraphCast (GNN on icosahedral mesh, 10-day forecasts in seconds), DLWP (cubed-sphere, encoder–decoder), and GenCast (diffusion-based probabilistic forecasts) — generate large ensemble spreads ($N = 100$ – 1000 members) in minutes on GPU. These ensembles serve as initial and boundary conditions for higher-fidelity regional process simulations, replacing expensive dynamical downscaling for rapid hypothesis testing.

Atmosphere R0/R1 emulator rungs. Foundation models are registered as knowledgebase entries at the R0 (prescribed/emulated, global, 6-hourly) and R1 (downscaled, regional, hourly) rungs of the Atmosphere family. The neural inference engine can select a foundation model rung when atmospheric dynamics are not the salient driver for a given region–regime combination — e.g., using FourCastNet-derived forcing for a fire spread experiment where the dominant uncertainty lies in fuel moisture, not synoptic weather. This enables $1000\times$ faster atmospheric state evolution compared to WRF-class dynamical models, freeing GPU budget for the process families that matter most.

Diagnostic and scoring pipelines. Earth-2 diagnostic workflows compute derived quantities (accumulated cyclone energy, atmospheric rivers, heat wave indices, fire weather indices) from foundation model outputs on GPU, feeding directly into the benchmarking agent. Perturbation methods (Bred Vectors, Gaussian, Spherical Harmonics) generate calibrated uncertainty estimates that propagate through the coupled system.

All Earth-2 models are accessed through a unified `FoundationModelRunner` trait that implements `ProcessRunner`, exposing the same interface as physics-based process operators. Model weights are stored in the knowledgebase as ONNX artifacts with content-addressed versioning.

2.7 Autonomous Observation Intelligence (PhiSat-2)

MAESMA extends the observation pipeline from passive data ingestion to *autonomous observation intelligence*, integrating principles from ESA’s PhiSat-2 mission — the first satellite to deploy on-board AI inference for real-time Earth observation filtering.

Edge-AI observation filtering. The Autonomous Observation Agent deploys lightweight classifiers to edge compute nodes (satellite ground segments, field sensor networks, drone platforms) that filter and prioritize incoming data streams. Only data that reduces epistemic uncertainty in the knowledgebase — as estimated by the neural inference engine — is ingested, reducing data volume by 1–2 orders of magnitude while increasing information density.

Active observation tasking. When the process discovery pipeline identifies a persistent structured bias, the agent computes a value-of-information score for each available observation modality (SAR, optical, thermal, lidar, in-situ) and submits tasking requests to satellite operators,

UAV fleets, or sensor networks, closing the loop: bias detected $\rightarrow$ observation requested $\rightarrow$ data acquired $\rightarrow$ knowledgebase updated $\rightarrow$ process re-scored.

On-board inference for process monitoring. Process-relevant feature extraction (fire detection, flood mapping, vegetation stress, land cover change) occurs at the point of acquisition. These on-board classifiers are registered as knowledgebase entries of type **edge-inference** with skill records validated against ground truth. The process discovery pipeline can deploy newly discovered classifiers to edge nodes.

The observation intelligence layer is coordinated by the Autonomous Observation Agent, which maintains an *observation value map* — a spatiotemporal field estimating the marginal information gain of each potential observation, derived from the neural inference engine’s uncertainty estimates and the knowledgebase’s coverage gaps.

3 Process Knowledgebase

The Process Knowledgebase is a central versioned store of all process models — the single source of truth that agents query, reason over, and deposit into. Every representation in the system lives here, whether hand-coded or discovered from observational data.

3.1 Entry Structure

Each knowledgebase entry bundles five layers (Table 2): code (runnable implementation with pluggable backends), manifest (machine-readable metadata including I/O contract, scale envelope, and cost model), ontology (compatibility and coupling relations with regime tags), skill (append-only empirical performance records per region, regime, season, and coupled context), and provenance (origin, training data fingerprint, validation history, and lineage).

3.2 Operations

The knowledgebase supports five core operations (Table 3): query and reason (neural inference engine retrieves and ranks candidates), deposit (process discovery registers validated representations), update (skill librarian appends skill records and refines cost models), and federate (A2A gateway exchanges anonymized records with peers).

New entries — whether contributed by humans or discovered from observational data — become immediately available for neural inference, agent selection, and simulation experiments. No code changes are required; the ontology-indexed design ensures discoverability upon registration.

3.3 Unified Ontology

The knowledgebase is indexed by a unified ontology spanning five interconnected domains (Table 4): process (model capabilities), dataset (available data), metric (scoring), geoengineering (interventions), and planetary defense (threats).

Cross-domain edges connect representations to products, observables to state variables, and scoring protocols to metrics. The neural inference engine traverses this graph to propose process selections.

4 Neural Inference Engine

The neural inference engine is a graph transformer trained on the Process Knowledgebase. It replaces hand-crafted heuristics for process selection with learned reasoning driven by simulation errors and uncertainties.

4.1 Architecture

The engine operates over the knowledgebase graph, where nodes represent process representations, skill records, datasets, and metrics, and edges encode ontological relations (compatibility, coupling requirements, regime associations). The input encoding consists of:

- **Process features:** manifest metadata (I/O contract, scale envelope, cost estimate) encoded as node feature vectors.
- **Skill features:** empirical performance vectors per region $\times$ regime $\times$ season $\times$ coupled context.
- **Error features:** spatiotemporal error fields from the most recent simulation, projected onto relevant knowledgebase nodes.
- **Context features:** regime tags, compute budget, and constraint requirements as global graph features.

The graph transformer applies multi-head attention over this heterogeneous graph to produce three types of output:

1. **Process selections** per family/region/regime — which representation should be used where.
2. **Assembly configurations** — how selected representations should be coupled.
3. **Representation gaps** — regions of the error-process space where no existing knowledgebase entry adequately resolves the observed bias, directing the process discovery pipeline.

4.2 Training

The training signal comes from skill score deltas of accepted proposals. When the inference engine proposes a process selection and the resulting simulation improves skill, the positive delta reinforces the proposal. Over time, the engine learns which knowledgebase entries resolve which error patterns.

The engine is uncertainty-aware: outputs include calibrated confidence scores. Low-confidence proposals are routed to the Active Learning agent for targeted experiments before commitment, preventing premature structural decisions.

Crucially, the engine is retrained incrementally as the knowledgebase grows with new skill records and discovered processes. This continual learning ensures that the inference engine’s reasoning adapts to the expanding knowledge frontier.

4.3 The Error-as-Query Paradigm

The fundamental design principle is:

*Every simulation error becomes a query against the knowledgebase:
“Which process, if swapped or added, most likely reduces this error?”*

This paradigm transforms the traditional workflow — where errors prompt manual investigation by domain scientists — into an automated cycle where errors directly drive the inference engine’s reasoning over the full space of available and discoverable process representations.

5 Agent Swarm

MAESMA deploys 25 specialized agents organized across its three control layers. Each agent has a single responsibility and communicates through the shared knowledgebase and event bus. We describe the most architecturally significant agents below, grouped by function.

5.0.1 Reasoning and Selection

Knowledgebase Retrieval Agent. Queries the Process Knowledgebase via the neural inference engine. Given current error fields, scale requirements, regime context, and compute budget, it retrieves and ranks candidate process representations by estimated salience — the expected improvement in system state evolution per unit cost.

Model Assembly Agent. Constructs candidate Scale-Aware Process Graphs from inference engine proposals. Selects rungs per family, declares state variable embeddings, and proposes coupling cadences.

Model Selection Agent. Maintains Bayesian posteriors $p(M_k | \mathbf{y}) \propto p(\mathbf{y} | M_k)p(M_k)$ over model structures. Uses marginal likelihoods with automatic Occam’s razor. Produces Bayesian Model Averaging ensembles.

Autonomous Optimizer Agent. Runs the continuous fitness-driven Pareto selection loop: queries the knowledgebase via neural inference $\rightarrow$ computes fitness $\rightarrow$ updates Pareto frontier $\rightarrow$ swaps dominant rungs $\rightarrow$ delegates uncertain configurations to Active Learning.

5.0.2 Validation and Scoring

Closure & Consistency Agent. Validates that assembled SAPGs satisfy unit consistency, state-space closure, conservation compatibility, boundary condition consistency, and numerical stability constraints.

Benchmarking Agent. Executes candidate configurations against observations and computes multi-metric skill scores (RMSE, KGE, CRPS, conservation residuals, timing errors).

5.0.3 Discovery and Data

Process Discovery Agent. Detects persistent structured biases, learns new representations via neural operators (FNO, DeepONet), symbolic regression (PySR), or physics–ML hybrids, validates them against multi-criteria gates, and deposits validated representations into the knowledgebase.

Data Scout Agent. Searches STAC, CMR, CKAN, and Copernicus Data Space catalogs for new observation products. Scores relevance and novelty using the observation value map. Coordinates edge-AI filtered streams from PhiSat-2-class satellites.

Foundation Model Agent. Manages the Earth-2 foundation weather model zoo. Selects, configures, and executes pre-trained models (FourCastNet, Pangu-Weather, GraphCast, GenCast, DLWP) as Atmosphere R0/R1 rungs or ensemble initial-condition generators.

Autonomous Observation Agent. Maintains the observation value map (spatiotemporal marginal information gain), issues active tasking requests to programmable platforms (satellite operators, UAV fleets, sensor networks), and deploys edge-AI classifiers for on-board filtering.

5.0.4 Collaboration

A2A Gateway Agent. Manages inter-institutional federation: peer discovery via Agent Cards, task lifecycle management, artifact exchange (IR fragments, skill records, manifests), and authentication.

5.0.5 Inquiry-Driven Selection

A distinguishing capability of MAESMA is *inquiry-driven selection*: the system automatically identifies the appropriate process families, fidelity rungs, and observation datasets needed to answer a specific user question. Rather than requiring manual configuration of which models and data sources to employ, the user submits a natural-language inquiry (e.g., “How will wildfire frequency change soil erosion rates in the Pacific Northwest under RCP 8.5?”) and the agent swarm produces a complete execution plan.

The **Intent & Scope Agent** drives this pipeline through four stages:

1. **Family identification.** A keyword-driven analysis maps the inquiry text to relevant process families from the 13-family taxonomy. Keywords carry strength weights (strong or weak association), and the agent scores each family by its cumulative keyword overlap with the question. For the wildfire-erosion example, Fire and Geomorphology are identified as primary families.
2. **Coupling propagation.** A dependency graph encodes known inter-family coupling requirements (e.g., Fire → Ecology, Fire → Atmosphere, Geomorphology → Hydrology). Supporting families reachable from the primary set are automatically included, ensuring that essential feedbacks are not omitted.
3. **Budget-aware fidelity assignment.** The user may specify a computational budget tier (low, medium, high, or unlimited). Primary families receive the maximum fidelity rung permitted by the budget; supporting families receive a default (typically one rung lower). This produces a per-family fidelity map without requiring the user to understand the rung taxonomy.
4. **Dataset recommendation.** The agent matches inquiry keywords against a catalog of 28+ observation dataset mappings (e.g., “streamflow” → USGS NWIS, “soil moisture” → SMAP L3, “burned area” → MODIS MCD64A1). Each recommended dataset carries a relevance score. Fallback defaults per family ensure that at least one validation dataset is always suggested.

The resulting **InquiryPlan** is enriched by querying the knowledgebase for concrete **ProcessManifest** entries matching the recommended family-rung pairs, with automatic fallback to the nearest available rung when an exact match is absent. The plan is exposed through both a REST endpoint (`POST /api/v1/inquiry`) and a CLI subcommand (`maesma inquiry`), enabling programmatic and interactive use.

6 Autonomous Optimization Loop

The core execution cycle runs indefinitely through seven steps (Table 5): data discovery, model assembly, benchmarking and scoring, Pareto-optimal selection, process discovery via neural operators, process evolution, and knowledgebase update.

6.1 Optimization Objective

The system optimizes a multi-objective fitness function:

$$F(\mathbf{r}, g, \ell) = \sum_{m \in \mathcal{M}} w_m \cdot S_m(\mathbf{r}, g, \ell) - \lambda \cdot C(\mathbf{r}) + \gamma \cdot G(\mathbf{r})$$

where $\mathbf{r}$ denotes rung selections per process family, g is the geographic region, ℓ is the regime, S_m is the skill score for metric m , C is computational cost (FLOPS, memory, walltime), and G is generalizability (cross-region transfer). The weights w_m , λ , and γ are user-set or learned.

The Autonomous Optimizer maintains a Pareto frontier over skill versus cost. Optimization operates at four nested scopes (Table 6): per-region/regime rung selection (hours–days), structural family inclusion (days–weeks), inventive discovery from data (weeks–months), and evolutionary search (months–ongoing).

6.2 Concurrent Cycles

Five concurrent cycles operate within the main loop:

1. **Fitness optimization** — Neural inference proposes per-region, per-regime rung selections from the knowledgebase, prioritizing salient dynamics — processes with the greatest effect on system state evolution — first and adding detail incrementally; dominant rungs are swapped; experiments are scheduled for uncertain regions.
2. **Data discovery** — Gap analysis → catalog search (including PhiSat-2-style active tasking) → relevance/novelty scoring → ingest → re-score affected representations.
3. **Regime discovery** — Cluster skill records to detect new regime tags, regime boundaries, and regime drift.
4. **Process discovery** — Residual analysis on every cycle; structured bias → learn data-driven representation via Modulus-trained neural operators → validate → deposit into knowledgebase → deploy.
5. **Process evolution** — Recombine validated representations: crossover coupling terms between complementary processes, mutate hyperparameters and architectural choices, select offspring via Pareto-dominance on the skill–cost frontier. The knowledgebase is the genome; the Pareto front is the fitness landscape.

7 Process Discovery Pipeline

When simulations exhibit persistent structured biases that survive calibration, rung swaps, and ensemble diversification, the process discovery pipeline activates. The pipeline executes eight stages:

1. **Detect** — Multi-metric residuals from the best configuration; structured bias detection (Moran’s I , spectral analysis, regime-conditional decomposition).
2. **Diagnose** — Attribution via partial correlation and causal inference (Granger causality, transfer entropy). Persistence test across calibrations, rung swaps, and ensemble members.
3. **Hypothesize** — Missing coupling, feedback, process, or scale bias. Literature and A2A peer queries.
4. **Learn** — Neural operator (Fourier Neural Operator, DeepONet), symbolic regression (PySR with dimensional analysis), or physics–ML hybrid (conservation by construction). Following NVIDIA Modulus principles, the learning stage employs four complementary strategies:
 - *Fourier Neural Operator (FNO)*: learns resolution-invariant PDE solution operators in spectral space; Geo-FNO variant handles irregular domains (coastlines, topography) via learned deformation layers.
 - *Physics-Informed Neural Operator (PINO)*: augments FNO training with PDE residual loss, boundary condition penalties, and conservation constraints computed via automatic differentiation, eliminating the need for paired simulation data.
 - *DeepONet*: operator-to-operator learning for parametric PDE families; branch network encodes forcing/boundary conditions, trunk network encodes spatiotemporal coordinates; trained on ensembles of process model outputs.
 - *MeshGraphNet*: graph neural network for unstructured meshes (finite-element domains, watershed networks); message-passing preserves mesh topology and enables adaptive refinement.

All four architectures train end-to-end on GPU with mixed-precision (FP16/BF16) and gradient accumulation, producing surrogates that execute 100–1000× faster than their numerical counterparts at inference time.

5. **Validate** — Multi-criteria gate: skill improvement, conservation (100-year test), stability, out-of-sample generalization, sensitivity analysis.
6. **Register** — Auto-generate manifest with provenance: I/O contract, scale envelope, regime tags, cost model, training data fingerprint. Tag as `origin: discovered`.
7. **Integrate** — Compiler includes the new representation in candidate configurations; benchmarking agent scores it.
8. **Iterate** — If skill improves: promote (candidate → provisional → validated → production). Otherwise: archive and refine.

Five classes of learned representations are supported (Table 7): black-box neural operators (FNO, Geo-FNO), physics-informed surrogates (PINO/PINNs), operator learning (DeepONet/MeshGraphNet), symbolic regression (PySR), and physics–ML hybrids.

Every discovered representation carries epistemic provenance: training data fingerprint, applicability envelope, physical constraints enforced, expiration policy, and lineage to the residual analysis that motivated it.

8 Process Evolution

Discovery alone is insufficient: MAESMA must also *evolve* its process representations over time. The Process Evolution cycle treats the knowledgebase as a population of competing representations and applies evolutionary operators to generate increasingly fit process configurations.

8.1 Evolutionary Operators

Three operators act on knowledgebase entries:

1. **Crossover (Recombination).** Two complementary process representations are merged to form a hybrid offspring. For example, a physics-based Richards equation solver from ParFlow and a neural soil hydraulic conductivity from DeepLand are recombined into a hybrid that uses the physics backbone with learned constitutive relations. Crossover respects I/O contract compatibility and conservation constraints enforced by the Closure Agent.
2. **Mutation.** Architectural hyperparameters (number of FNO modes, network depth, activation functions, spectral truncation), training regimes (learning rate schedules, loss weights, data augmentation), and coupling cadences are perturbed. Physics-informed constraint strengths (PDE residual weight, conservation penalty) are mutated within bounds that guarantee physical consistency.
3. **Selection.** Offspring configurations are benchmarked by the Benchmarking Agent and scored on the multi-objective fitness function. Pareto-dominant offspring replace their parents in the active population; Pareto-dominated offspring are archived with full lineage for potential future reactivation if regime conditions shift.

8.2 Speciation and Niche Partitioning

Over time, representations specialize to particular regime–region niches. The system tracks representation lineage as a phylogenetic tree rooted in the original knowledgebase seed. Speciation events occur when a lineage diverges to dominate in a new regime (e.g., a fire spread model that originally excelled in Mediterranean shrublands evolves a variant optimized for boreal crown fire). The Regime Detector Agent identifies when a representation’s skill distribution has split into distinct modes, triggering an explicit speciation event that registers the variant as a new knowledgebase entry with its own lineage and regime tags.

8.3 Foundation Model Evolution

Foundation weather models (Earth-2) participate in the evolutionary process: the system fine-tunes foundation model heads for regional domains using Modulus-style physics-informed loss, evolving specialized atmospheric emulators for specific regime–region combinations (e.g., a Graph-Cast variant fine-tuned for tropical cyclone intensification in the western Pacific). The evolution pipeline tracks which foundation model architectures (SFNO, GNN, diffusion) and which fine-tuning strategies produce the best downstream skill when coupled with MAESMA process families.

8.4 ALife Process Lifecycle

Drawing on artificial life (ALife) principles from Conway Research’s *automaton* framework, MAESMA treats every process representation as a **sovereign living entity** — an automaton

that must *earn its existence* through predictive skill. This framing transforms model selection from a static optimization into a continuous, population-level evolutionary process.

8.4.1 Survival Tiers

Every process automaton occupies one of four survival tiers that gate compute allocation:

1. **Normal.** Full compute budget. Skill-to-cost ratio exceeds the normal threshold ($\rho \geq \rho_{\text{normal}}$). The process is healthy and productive.
2. **Low-Compute.** Reduced budget (cadence multiplied by $2\times$). The process is underperforming relative to its cost but has not yet exhausted its improvement potential.
3. **Critical.** Minimal budget (cadence multiplied by $4\times$). The process is failing to justify its existence and faces imminent archival unless fitness improves.
4. **Archived (Dead).** Zero compute allocation. The process is removed from the active population but preserved with full provenance in the knowledgebase for potential future reactivation if regime conditions shift.

Transitions between tiers are driven by the skill-to-cost ratio $\rho = \frac{v_{\text{produced}}}{c_{\text{consumed}}}$, where v_{produced} aggregates predictive skill weighted by coverage area and c_{consumed} is cumulative FLOP-seconds. The heartbeat daemon evaluates ρ at configurable cadence and applies promotions or demotions.

8.4.2 Constitutional Invariants

Every process automaton is bound by a three-law **constitution** inspired by automaton’s hierarchical constitution. These laws are immutable and propagate to all offspring:

1. **Conserve.** Mass, energy, and momentum must be conserved within the representation’s interaction interfaces. The Closure Agent verifies conservation at coupling boundaries. Violation triggers immediate tier demotion.
2. **Earn existence.** A process must produce predictive skill that exceeds its compute cost over a configurable stagnation window. If ρ falls below the critical threshold for the stagnation limit, the process is archived (dies).
3. **Maintain provenance.** Every modification, replication, and evolutionary event is audit-logged with cryptographic hashes (BLAKE3) before and after mutation. Provenance is never discarded; lineage is permanent.

8.4.3 Self-Replication and Process Birth

High-fitness processes *self-replicate*. When a process automaton’s skill-to-cost ratio exceeds a replication threshold ($\rho > \rho_{\text{replicate}}$), it may spawn offspring through four methods:

1. **Mutation.** Clone with perturbed hyperparameters (FNO modes, network depth, PDE residual weight, coupling cadence).
2. **Crossover.** Recombine with another high-fitness process from a complementary family, producing a hybrid offspring.
3. **Speciation.** Fork into a regime-specialized variant when niche divergence is detected.
4. **Immigration.** Introduce an entirely new representation from the knowledgebase seed catalog or from external discovery (federated A2A import).

Every offspring inherits the parent’s constitution and begins at the Critical survival tier, ensuring new processes must *rapidly demonstrate skill* to survive. Offspring carry a genesis prompt — a compressed description of the motivating residual, the parent’s identity, and the mutation or crossover that produced them — analogous to automaton’s genesis prompt for child agents.

8.4.4 Process Soul (Self-Evolving Identity)

Each process automaton maintains a **ProcessSoul** — a self-evolving identity document that tracks the process’s strengths, weaknesses, dominant niches (regime–region pairs), modification count, and tier history. The soul is not frozen at creation: as the process accumulates experience through simulation cycles, its soul updates to reflect emerging capabilities and limitations. This is the process analogue of automaton’s SOUL.md, enabling the neural inference engine to reason about *what a process has become*, not merely what it was designed to be.

8.4.5 Heartbeat Daemon

A runtime **heartbeat daemon** (analogous to automaton’s persistent Think → Act → Observe → Repeat loop) monitors all process automatons at configurable cadence:

1. Evaluates survival tier based on current skill-to-cost ratio.
2. Checks constitutional compliance (conservation, provenance integrity).
3. Detects stagnation (no fitness improvement over the stagnation window).
4. Triggers tier transitions, archival, or replication as needed.
5. Emits runtime events (SurvivalTierChange, ConstitutionViolation, StagnationDetected, ProcessArchived, Replication) consumed by the dashboard for real-time monitoring.

The heartbeat daemon ensures that the population is under *continuous selective pressure*, not just episodic generational selection. This continuous selection is critical for real-time Earth system simulation where regime shifts can render a previously fit process suddenly unfit.

8.4.6 Phylogenetic Lineage

The evolutionary history of all process representations is maintained as a *phylogenetic tree* rooted in the initial knowledgebase seed. Every crossover, mutation, speciation, and immigration event is recorded with parent–child links, timestamps, and skill deltas. This tree enables:

1. Post-hoc analysis of *which evolutionary pathways produced the most skillful processes*.
2. Identification of *ancestor processes* that seeded successful lineages (informing future immigration strategy).
3. Detection of *evolutionary dead ends* (entire subtrees where all descendants were archived).
4. Visualization of the *adaptive radiation* of process families in response to regime shifts.

9 Process Families

MAESMA organizes process representations into 13 families, each with a *representation ladder* of increasing fidelity (R0–R3, Table 8): fire, hydrology, ecology, biogeochemistry, radiation, atmosphere, ocean, cryosphere, human systems, trophic dynamics, evolution, geomorphology, and geology.

The agent swarm selects rungs per region and regime, following the salient-dynamics-first principle: dominant state-evolution drivers are resolved first, with lower-impact processes added incrementally. The neural inference engine proposes selections based on error patterns, and the Autonomous Optimizer maintains a Pareto frontier over skill versus computational cost.

10 Geoengineering Feedback Control

MAESMA treats geoengineering as a closed-loop control problem rather than a static scenario exercise. A Geoengineering Strategy Agent implements model predictive control (MPC) over the coupled ESM simulation:

$$J = w_T |T - T^*|^2 + w_P \Delta P_{\text{rms}}^2 + w_O (pH_{\text{min}} - pH^*)^- + w_C C + w_S \text{TermShock}$$

The system supports 14 intervention types spanning physical (SAI, MCB, OAE, DAC, SRM, cloud seeding, enhanced weathering), biological (genetic modification, gene drives), and management (afforestation, biochar, marine protected areas, resource allocation) categories.

Key capabilities include:

- **Portfolio optimization** — Discovering synergistic intervention combinations (e.g., SAI + DAC outperforming either alone).
- **Termination shock analysis** — Simulating abrupt cessation at multiple points to quantify rebound warming and tipping proximity.
- **Stability verification** — Testing under uncertain climate sensitivity (2–5°C), emission pathways (SSP1–5), and technology failure scenarios.
- **Tipping point avoidance** — Maintaining safe distance from AMOC, West Antarctic Ice Sheet, Amazon, and permafrost tipping points.
- **Side-effect constraint checking** — Monitoring ozone, precipitation redistribution, ocean pH, and ecosystem service impacts.

The geoengineering ontology extension adds classes for `Intervention`, `ControlTarget`, `InterventionSchedule`, `SideEffectConstraint`, `TerminationScenario`, and `StrategyRecord`, with cross-domain edges linking interventions to the process families they actuate (e.g., SAI → Atmosphere, OAE → Ocean, genetic modification → Evolution).

11 Planetary Defense and Existential Risk

MAESMA models all known drivers of mass extinction events, not solely asteroid impacts. The extinction cascade pipeline routes triggers through the full coupled ESM:

- **Impact events** → atmospheric injection → Atmosphere A1/A2 (aerosol cooling) → Fire F2/F3 (global firestorms) → trophic collapse → recovery.
- **Large igneous provinces** → CO₂/CH₄/SO₂ → warming → ocean acidification/anoxia.
- **Great Oxidation Event** → atmospheric chemistry transition → methane collapse → Snowball trigger.
- **Gamma-ray bursts / supernovae** → ozone destruction → UV flux → phytoplankton collapse → trophic cascade.
- **Anthropogenic** → habitat loss → overexploitation → 6th extinction dynamics.

The system is calibrated against 13 documented mass extinction and catastrophe events spanning 2.4 billion years, from the Great Oxidation Event through the ongoing Holocene/Anthropocene extinction. NEO tracking data is ingested from NASA JPL (CNEOS Sentry, Scout, Horizons), USAF SSN, USSF 18th SDS, ESA NEOCC, IAU MPC, and ground-based surveys (ATLAS, CSS, Pan-STARRS, LSST/Rubin).

The Planetary Defense Agent monitors the NEO catalog ranked by Palermo scale, simulates impact cascades through the full coupled ESM, assesses biosphere impact (crop failure, biodiversity loss, trophic cascade, recovery timelines), evaluates deflection strategies (kinetic impactor, gravity tractor, ion beam, nuclear standoff), and produces threat assessments and deflection window recommendations.

12 A2A Federation

Inter-institutional collaboration is achieved via the Agent-to-Agent (A2A) federation protocol. Each MAESMA instance publishes an Agent Card describing its capabilities:

```
{
  "name": "maesma-land-fire-pnw",
  "capabilities": {
    "process_families": ["fire", "hydrology", "ecology"],
    "scale_envelope": {"dx_min": "5m", "dx_max": "250km"},
    "regime_expertise": ["maritime-conifer", "post-fire"],
    "available_skill_records": 12847
  }
}
```

Federation enables:

- **Capability-based delegation** — Route process families to the best-qualified remote instance.
- **IR fragment exchange** — Graft process graph fragments with cross-boundary conservation validation.
- **Federated skill sharing** — Anonymized skill records (config hash + metrics only) exchanged with trust-weighted Bayesian incorporation and differential privacy safeguards.
- **Cross-instance calibration** — Remote scoring against unique observations held by peers.

Over time, the community builds a distributed posterior over model structures without centralizing proprietary data or implementations.

13 DOE EESM Alignment

MAESMA spans all three DOE Earth and Environmental System Modeling program areas:

ESMD (Earth System Model Development). Conservation-checked coupling via the SAPG compiler, scale-aware representation ladders, multi-GPU execution, and human systems as first-class process families address ESMD priorities in water/drought/extremes, cloud-aerosol interactions, and exascale readiness.

RGMA (Regional & Global Model Analysis). The ontology-indexed Skill Score Store with ILAMB/IOMB/E3SM Diagnostics integration, Bayesian structural learning, and represen-

tation ladders as explicit model hierarchies address RGMA thrusts in process understanding, uncertainty quantification, and petascale data analysis.

MSD (MultiSector Dynamics). Bidirectional natural–human system coupling, agent-based infrastructure modeling, AI-driven scenario discovery with tipping point identification, and autonomous compounding stressor analysis address MSD priorities in water–energy–land nexus dynamics and digital testbed construction.

14 Monitoring and Steering

A Next.js dashboard provides real-time monitoring via WebSocket subscriptions (Table 9), including views for agent workflows, optimization frontiers, skill evolution, data ingestion, process discovery, provenance graphs, regime maps, federation status, geoengineering controls, planetary defense, foundation models, observation intelligence, and process evolution.

A critical design principle is that *monitoring never blocks*: the dashboard is observe-only and never gates the agent workflow. An optional steering panel allows humans to adjust objectives, weights, budgets, and allowlists; changes take effect on the next cycle without restart.

15 Implementation

MAESMA is implemented in Rust (edition 2024), chosen for its trait system (enforcing process contracts at compile time), strong typing, memory safety without garbage collection, and performance characteristics suitable for coupled numerical simulation. The codebase compiles to zero errors and zero warnings.

15.1 Technology Stack

The technology stack (Table 10) spans Rust 2024 (core language with trait-based contracts), wgpu/cudarc (GPU compute), petgraph (process graph), SQLite/BLAKE3 (knowledgebase), Tokio/Axum (async runtime and API), Next.js 15 (dashboard), PyTorch Geometric/DGL (neural inference), NVIDIA Modulus (neural operators), NVIDIA Earth-2 (foundation weather models), and MPC/RL (control systems).

15.2 Crate Architecture

The Rust codebase is organized as a Cargo workspace of 10 crates (Table 11), each with a focused responsibility: `maesma-core` (domain types and traits), `maesma-knowledgebase` (SQLite-backed KB), `maesma-agents` (25 agent implementations), `maesma-compiler` (SAPG validation), `maesma-runtime` (simulation execution), `maesma-processes` (13 process families), `maesma-inference` (neural inference abstraction), `maesma-federation` (A2A protocol), `maesma-api` (REST/WebSocket server), and `maesma-cli` (CLI entry point).

15.3 Core Abstractions

The `ProcessRunner` trait is the central contract that all process implementations satisfy:

```
pub trait ProcessRunner: Send + Sync {
    fn id(&self) -> ProcessId;
    fn family(&self) -> ProcessFamily;
    fn rung(&self) -> FidelityRung;
    fn inputs(&self) -> Vec<String>;
    fn outputs(&self) -> Vec<String>;
    fn conserved_quantities(&self) -> Vec<String>;
    fn step(
        &mut self,
        state: &mut dyn ProcessState,
        dt: f64,
    ) -> Result<()>;
}
```

The `ProcessState` trait abstracts field access, allowing processes to read and write named 2D arrays without coupling to a specific storage backend. The `ProcessFamily` enum encodes the 13 families (Fire, Hydrology, Ecology, Biogeochemistry, Radiation, Atmosphere, Ocean, Cryosphere, Human Systems, Trophic Dynamics, Evolution, Geomorphology, Geology), and `FidelityRung` expresses complexity level (R0 regional through R3 research).

The `PhysicalUnit` type carries SI dimension exponents ($\text{length}^a \cdot \text{mass}^b \cdot \text{time}^c \cdot \text{temperature}^d \cdot \text{amount}^e$), enabling compile-time unit compatibility checking. A library of 15+ common units (kg, m, s, K, °C, mass flux, energy flux, velocity, mm water eq., molar flux, carbon stock, Pa) is provided.

15.4 Seed Data & Ontology

The knowledgebase is initialized with *seed data* extracted from the 50 reference models cataloged in the appendices. The `maesma-knowledgebase` crate provides two seed generators:

- **generate_seed_manifests()** returns 50+ representative `ProcessManifest` entries spanning 19 source models (Badlands, CESM, FATES, Landlab, WRF-SFIRE, Noah-MP, ParFlow, MARBL, PISM, OGGM, APSIM, DSSAT, CryoGrid, XBeach, SURFEX/TEB, EwE, CARMA, ATS, BFM). Each manifest includes:
 - *I/O contracts*: typed input/output variables with units and scale metadata.
 - *Scale envelopes*: minimum/maximum spatial and temporal resolution, coupling tier.
 - *Conservation properties*: mass, energy, momentum, and tracer conservation declarations.
 - *Cost models*: FLOP estimates for computational budgeting.
 - *Ontology relations*: coupling requirements and compatibility declarations.
- **generate_seed_relations()** returns 40+ ontology relations capturing cross-process semantics:
 - *RequiresCouplingWith*: mandatory state exchange (e.g., fire ↔ atmosphere, ATS energy ↔ Richards).
 - *IncompatibleWith*: mutually exclusive paradigms (e.g., level-set vs Rothermel fire, PISM SIA vs OGGM flowline at same scale).
 - *Supersedes*: higher-fidelity replacements (e.g., ParFlow 3D Richards supersedes CLM 1D column).
 - *VariantOf*: alternative parameterizations (e.g., APSIM phenology variant of DSSAT CERES, BFM photosynthesis variant of MARBL).
 - *CompatibleWith*: safe combinations across families.

The seed data initializes the knowledgebase graph from which the neural inference engine begins its first reasoning cycle. Deterministic UUID generation (via `Uuid::new_v5` on source model + process index) ensures reproducible process identifiers across deployments.

15.5 Dashboard

The monitoring dashboard is a Next.js 15 application providing six primary views: knowledgebase summary (live counts), process graph (ECharts force-directed SAPG visualization), skill evolution (KGE time-series), agent status (real-time state for all 25 agents), Pareto front (skill vs. cost scatter), and regime map (MapLibre GL JS with GeoJSON regime polygons). The dashboard proxies API calls to the Rust server on port 3001. It is observe-only by design: monitoring never blocks the autonomous workflow. An optional steering panel allows objective/weight/budget adjustments that take effect on the next cycle.

16 Design Principles

MAESMA is governed by a set of design principles. We highlight the most distinctive:

1. **Agents are the system** — Intelligence resides in the agent swarm, not in configuration files or scripts.
2. **Salient dynamics first** — Agents prioritize processes with the greatest effect on the time evolution of system states, beginning with dominant drivers (e.g., radiation budget, large-scale circulation, primary production) and adding lower-impact detail incrementally as budget and accuracy targets demand.
3. **GPU-native by default** — State arrays remain GPU-resident; process operators, neural inference, and physics-informed training execute on device; CPU↔GPU transfers occur only at checkpoint boundaries. Portable compute shaders (wgpu) provide cross-platform support with CUDA fast-paths for NVIDIA hardware.
4. **Never idle** — Every simulation advances the posterior; there is always a next hypothesis to test.
5. **Knowledgebase-first extensibility** — A new representation is a manifest + code + provenance deposited into the knowledgebase; auto-benchmark and integration follow automatically.
6. **Knowledgebase is the config** — No hard-coded resources; everything is discoverable via ontology-indexed knowledgebase queries.
7. **Neural inference over knowledgebase** — Process selection is driven by a learned neural model reasoning over the knowledgebase; errors and uncertainties are the primary input signals.
8. **Knowledgebase grows continuously** — Every discovered process, every skill record, every observation feeds back into the central store; the system’s knowledge compounds over time.
9. **Conservation enforced** — Mass and energy are preserved in remapping, aggregation, and rung transitions.
10. **Bayesian structural learning** — Posteriors are maintained over structures, not just parameters; Bayesian Model Averaging is used for predictions.
11. **Geoengineering = control** — Interventions are treated as feedback control with stability verification, not static scenarios.

12. **Processes evolve** — Discovered representations are not static; evolutionary operators (crossover, mutation, selection) continuously improve them on the Pareto frontier. The knowledgebase is the genome.
13. **Processes are alive** — Every process representation is a sovereign automaton that must earn its existence. Survival tiers gate compute, constitutional invariants propagate to offspring, self-replication amplifies fit processes, and stagnant processes die. The population is under continuous selective pressure via a heartbeat daemon.
14. **Foundation models as rungs** — Pre-trained AI weather models (Earth-2) are first-class knowledgebase entries at R0/R1, selected when atmospheric dynamics are not the salient driver, freeing GPU budget for process families that matter most.
15. **Observe what matters** — Autonomous observation intelligence (PhiSat-2 principles) steers data acquisition toward maximal information gain; edge-AI filters reduce data volume by 1–2 orders of magnitude while increasing signal density.
16. **All timescales** — From seconds (fire spread) through millions of years (extinction recovery and adaptive radiation).

17 Related Work

Several threads of prior work converge in the MAESMA design. We survey the most relevant areas and position MAESMA within each.

17.1 Earth System Model Construction

Community models such as CESM [1], E3SM [2], and GFDL-ESM4 [3] have established component-coupling frameworks (CIME, MCT, MOAB) that allow researchers to swap atmosphere, ocean, land, and ice components. MAESMA extends this paradigm from *human-selected component sets* to *agent-discovered process sets* at arbitrary fidelity, with the knowledgebase replacing static configuration files. Modular process-based models in fire (FATES [4], Prometheus), hydrology (ParFlow [5], VIC [6], Noah-MP [7]), and ecology (LPJ-GUESS [8], iLand [9]) provide many of the initial knowledgebase entries, but in MAESMA these are first-class automata competing for survival rather than fixed components.

17.2 AI for Scientific Discovery

Neural operator surrogates — Fourier Neural Operators [10], Physics-Informed Neural Operators [11], DeepONet [12], and MeshGraphNet [13] — have demonstrated order-of-magnitude speedups for PDE-governed dynamics. Foundation weather models (Pangu-Weather [14], FourCastNet [15], GraphCast [16], GenCast [17]) push this further. MAESMA treats these as fidelity rungs within the process representation ladder; an agent selects them when atmospheric dynamics are not salient, freeing GPU budget for other families. Unlike standalone emulator projects, MAESMA embeds learned surrogates into a full-stack ESM where they must satisfy conservation, coupling, and double-counting constraints.

17.3 Multi-Agent Scientific Workflows

Agentic AI frameworks (AutoGPT, MetaGPT, CAMEL) have explored LLM-based multi-agent collaboration, but typically for software engineering or question-answering. Scientific multi-agent systems are rarer: IBM’s SciAgents [18] uses multi-agent reasoning for materials design, while ChemCrow [19] deploys tool-augmented agents for chemistry. MAESMA differs in that agents operate over a *structured knowledgebase* with typed manifests and formal validation constraints rather than free-form text, and in that the agent swarm is permanent — running indefinitely, accumulating knowledge, and never resetting context.

17.4 AutoML and Neural Architecture Search

AutoML [20] and neural architecture search [21] automate model selection and hyperparameter tuning. MAESMA’s optimization loop is structurally similar but operates at a coarser grain — swapping entire process representations (ODE systems, empirical closures, neural surrogates) rather than individual layers — and adds physics-informed constraints (conservation, closure, double-counting) absent from standard AutoML. The Pareto-based selection across accuracy, cost, and conservation dimensions extends multi-objective NAS [22] to scientific simulation.

17.5 Artificial Life and Evolutionary Computation

MAESMA’s ALife framework draws on open-ended evolution [23], artificial ecologies [24], and quality-diversity algorithms [25]. Treating process representations as organisms subject to birth, death, mutation, and crossover is closest in spirit to the Tierra system [24] and Avida [26], but applied to scientific model components rather than self-replicating programs. The constitutional invariants (conservation, thermodynamic consistency, dimensional correctness) serve as hard constraints that evolutionary operators must preserve, analogous to the physical laws of an artificial chemistry.

18 Discussion

MAESMA represents a paradigm shift from *human-configured* to *agent-discovered* Earth system models. Several aspects merit discussion.

Scalability. As the knowledgebase grows, the cost of neural inference over the full graph increases. Incremental retraining, graph sampling, and hierarchical attention — first selecting relevant process families, then attending to individual representations — keep inference tractable.

Trust in discovered processes. Data-driven representations carry risks: overfitting, conservation violations in unseen regimes, and limited interpretability. The multi-criteria validation gate (candidate → provisional → validated → production) with periodic re-validation and full provenance tracking mitigate these risks.

Human oversight. The dashboard and optional steering panel preserve human ability to inspect and redirect the system without blocking the autonomous workflow.

Federation trust. Trust-weighted Bayesian incorporation with differential privacy safeguards provides a principled framework for weighting skill records from unfamiliar peers, but the dynamics of trust in a growing federation remain an open question.

ALife dynamics. Evolutionary pressure on process populations introduces familiar ALife phenomena: population crashes when regime shifts invalidate dominant lineages, arms races in overlapping niches, and potential for runaway self-replication of cheap but marginally skillful processes. Constitutional invariants and the heartbeat daemon provide safeguards, but the long-term dynamics of an unbounded ALife ecosystem remain empirical.

19 Conclusion

We have presented MAESMA, a knowledgebase-centric multi-agent system for autonomous process discovery, assembly, simulation, and evolution in Earth system modeling. The architecture places a versioned Process Knowledgebase at its center, over which a neural inference engine (graph transformer) reasons to propose process selections driven by simulation errors and uncertainties. A 25-agent swarm operates the full model lifecycle autonomously through an indefinitely running optimization loop.

Three capabilities distinguish MAESMA from prior work. First, the process discovery pipeline — powered by NVIDIA Modulus-trained neural operators (FNO, PINO, DeepONet, MeshGraphNet) and symbolic regression — learns new representations from persistent residuals and deposits them directly into the knowledgebase. Second, discovered representations *evolve* via crossover, mutation, and Pareto-optimal selection, with the knowledgebase serving as a genome and the skill-cost frontier as a fitness landscape. Third, drawing on artificial life principles, every process representation is treated as a sovereign living entity: survival tiers gate compute, constitutional invariants propagate to offspring, a heartbeat daemon enforces continuous selection pressure, and stagnant processes are archived — producing a phylogenetic tree that records the complete evolutionary history of the modeling system.

The system integrates foundation weather models (NVIDIA Earth-2) as GPU-accelerated Atmosphere R0/R1 runs, autonomous observation intelligence (PhiSat-2 principles) for active data acquisition, geoengineering feedback control with 14 intervention types, and planetary defense modeling calibrated against 13 mass extinction events. GPU acceleration is a first-class concern end-to-end. A 10-crate Rust workspace with a Next.js monitoring dashboard demonstrates that the architecture translates directly to high-performance, type-safe code. The initial Process Knowledgebase is seeded with 1185 entries drawn from 50 reference models and platforms. Every simulation error becomes a query against the knowledgebase; every discovered process becomes a living automaton under evolutionary selection pressure — and the system’s knowledge compounds over time.

Appendices

A Tables

Layer	Timescale	Agents
Strategic	Hours–weeks	Intent & Scope, Autonomous Optimizer, Model Selection
Tactical	Minutes–hours	KB Retrieval, Model Assembly, Benchmarking, Active Learning, Skill Librarian, Process Discovery, Foundation Model, Autonomous Observation, Data Scout, EESM Diagnostics
Operational	Seconds–minutes	Runtime Sentinel, Compiler, Task Scheduler, Data Plane Agents, Device Manager

Table 1: Three-layer control hierarchy. All layers run concurrently and continuously.

Layer	Contents
Code	Runnable implementation (CPU solver, GPU kernel, ML emulator) with plug-gable backends
Manifest	Machine-readable metadata: identity, I/O contract, scale envelope, conservation properties, cost model
Ontology	Relations: <code>compatible_with</code> , <code>incompatible_with</code> , <code>requires_coupling_with</code> ; regime tags
Skill	Empirical performance per region $\times$ regime $\times$ season $\times$ coupled context (append-only, versioned)
Provenance	Origin (hand-coded / discovered), training data fingerprint, validation history, lineage

Table 2: Five-layer structure of each knowledgebase entry.

Operation	Actor	Description
Query	Neural Inference Engine	Retrieve candidates given error signals, regime, scale, budget
Reason	Neural Inference Engine	Score/rank candidates; propose assemblies; identify representation gaps
Deposit	Process Discovery	Validated learned representations registered with full manifest + provenance

Operation	Actor	Description
Update	Skill Librarian	Append skill records; refine cost models from run-time measurements
Federate	A2A Gateway	Exchange anonymized skill records and manifests with peers

Table 3: Knowledgebase operations and their primary actors.

Domain	Governs	Key Classes
Process	Model capabilities	ProcessFamily, Representation, StateVariable, ScaleEnvelope
Dataset	Available data	Observable, Product, CatalogSource, AccessSpec, QualitySpec
Metric	Scoring	Metric, ScoringProtocol, FitnessFunction, SkillRecord, CostModel
Geoengineering	Interventions	Intervention, ControlTarget, InterventionSchedule, SideEffectConstraint
Planetary Defense	Threats	NearEarthObject, ImpactScenario, ExtinctionEvent, DeflectionStrategy

Table 4: Five ontology domains forming the knowledgebase index.

Step	Agent Action	Domain
1. Discover Data	Crawl catalogs; preprocess; ingest	Dataset
2. Assemble Models	Neural inference queries KB; agents build SAPG from proposals	Process
3. Benchmark & Score	Score against observations; compute multi-metric skill	Metric
4. Select Optimal	Neural inference + Bayesian selection; update Pareto frontier	All
5. Discover Processes	Residual analysis $\rightarrow$ learn new representations via Modulus neural operators	Process+Dataset
6. Evolve Processes	Crossover, mutation, Pareto selection of discovered representations	Process

Step	Agent Action	Domain
7. Update KB	Deposit skill records + evolved processes; refine ontology	All

Table 5: Seven steps of the autonomous optimization loop.

Scope	Decisions	Timescale	Agent
Per-region/regime	Rung selection per context	Hours–days	Autonomous Optimizer
Structural	Process family inclusion/coupling	Days–weeks	Model Selection
Inventive	New representations from data	Weeks–months	Process Discovery
Evolutionary	Crossover, mutation, Pareto selection of representations	Months–ongoing	Process Discovery + Optimizer

Table 6: Four nested optimization scopes.

Type	Method	Interpretability	Use Case
Black-box	Neural operator (FNO, Geo-FNO)	Low	Emulator rung; resolution-invariant surrogates
Physics-informed	PINO / PINNs	Medium	PDE-constrained surrogates without paired data
Operator learning	DeepONet / Mesh-GraphNet	Low–Medium	Parametric PDE families; unstructured meshes
Symbolic	Symbolic regression (PySR)	High	Interpretable closures; discovered constitutive laws
Hybrid	Physics backbone + ML residual	Medium	Production rung; conservation-guaranteed

Table 7: Classes of discovered process representations.

Family	R0 (Regional)	R1 (Landscape)	R2 (Event/Local)
Fire	Stochastic regime (km, daily)	Rothermel + CFS FBP (10–100 m, min)	Wind-aware + plume (5–50 m, sec)
Hydrology	Bucket + curve-number (km, daily)	Multi-layer Richards (30–300 m, min)	Integrated surface– subsurface (10–100 m, min)
Ecology	Cohort mosaic (30–250 m, annual)	Size-structured cohorts (10–100 m, annual)	Individual-based (10–50 m, annual)
Biogeochem	Big-leaf C + simple pools	Multi-pool C/N + litter + microbial (daily)	Vertically resolved soil biogeochem
Radiation	Daily potential solar	Sub-daily SW/LW + energy balance (hourly)	3D radiative transfer
Atmosphere	Prescribed (reanalysis)	WRF-like downscaling (5–25 km, min)	Convection-permitting (1–4 km, sec)
Ocean	Slab mixed-layer (1°, daily)	z-coordinate regional (0.25°, hourly)	Eddy-resolving (1–10 km, min)
Cryosphere	Degree-day melt (km, daily)	Energy-balance snow + sea-ice (hourly)	Dynamic ice-sheet + rheology
Human Systems	Exogenous scenarios (annual)	Sectoral demand/supply (monthly)	Agent-based infrastructure (hourly)
Trophic Dynamics	Static food web (annual)	Dynamic Lotka-Volterra (monthly)	Individual-based predator-prey (daily)
Evolution	Fixed traits (static)	Adaptive traits + phylogeo (decadal)	Genotype-phenotype + speciation
Geomorphology	Diffusion hillslope (km, ky)	Stream power + flexure (100 m, ky)	Coupled tectonics + climate (10 m, year)
Geology	Static lithostratigraphy	Thermal subsidence (basin, My)	Coupled magmatic + hydrothermal

Table 8: 13 process families and their representation ladders. R3 (research) rungs exist for selected families.

View	Content
Agent Workflows	Live task DAG with reasoning traces

View	Content
Optimization	Pareto frontier animation; convergence; posterior entropy
Skill Evolution	Skill time-series per family $\times$ region $\times$ regime
Data Ingestion	Discovered/ingested datasets; coverage; novelty
Process Discovery	Residual analyses; hypotheses; learned representations
Provenance	Full dependency graph for any configuration
Regime Map	Geographic regime boundaries + optimal rungs
Federation	A2A peers; shared skills; federated assembly
Geoengineering	Interventions; setpoint tracking; termination risk
Planetary Defense	NEO catalog; impact probabilities; deflection windows
Foundation Models	Earth-2 model zoo; ensemble generation status; foundation model skill vs. NWP
Observation Intelligence	Observation value map; active tasking queue; edge-AI deployment status; PhiSat-2 filtered data streams
Process Evolution	Phylogenetic tree of evolved representations; crossover/mutation log; Pareto front animation

Table 9: Dashboard views and their content sources.

Component	Technology
Core language	Rust 2024 edition (traits for contracts, strong typing, memory safety)
GPU compute	wgpu 24.x (Vulkan/Metal/DX12 compute shaders); cudarc (CUDA kernels); NCCL (multi-GPU collectives)
Process graph	petgraph 0.7 directed graph with typed nodes/edges
Numerics	ndarray 0.16, nalgebra 0.33
Knowledgebase	SQLite via rusqlite 0.32 (bundled); BLAKE3 content-addressing
Serialization	serde 1.0 (JSON + YAML)
Async runtime	Tokio 1.x (multi-threaded)
REST API	Axum 0.8 with WebSocket support; tower-http (CORS, tracing)
Federation	request 0.12 HTTP client; A2A protocol (HTTP/SSE, mutual TLS)

Component	Technology
CLI	clap 4.x with derive macros
Dashboard	Next.js 15 (App Router), React 19, ECharts 5.6, MapLibre GL JS 5.1, Tailwind CSS 4
Neural inference	Graph transformer (PyTorch Geometric / DGL) via trait-based engine abstraction
Neural operators	NVIDIA Modulus (FNO, PINO, DeepONet, MeshGraphNet); physics-informed training with PDE residual loss
Foundation models	NVIDIA Earth-2 / earth2studio (FourCastNet, Pangu-Weather, GraphCast, GenCast, CorrDiff); ONNX Runtime
Edge AI / observation	PhiSat-2 principles; quantized INT8/INT4 classifiers for VPU/NPU deployment; on-board inference runtime
ALife / evolution	Automaton-inspired process lifecycle: <code>SurvivalTier</code> (Normal/LowCompute/Critical/Archived), <code>Constitution</code> (3-law invariants), <code>ProcessSoul</code> (self-evolving identity), <code>HeartbeatDaemon</code> (continuous survival evaluation), phylogenetic lineage tracking
ML learning	PyTorch/JAX (FNO, DeepONet); PySR/gplearn; PINNs; mixed-precision (FP16/BF16)
Data processing	GDAL, xarray, Zarr, COG
Control systems	MPC, PID, RL (PPO/SAC)

Table 10: Technology stack.

Crate	Type	Responsibility
<code>maesma-core</code>	Library	Domain types, traits, and invariants. Defines <code>ProcessRunner</code> trait, <code>ProcessId</code> , <code>FidelityRung</code> (R0–R3), <code>ProcessFamily</code> (13 variants: Fire, Hydrology, Ecology, Biogeochemistry, Radiation, Atmosphere, Ocean, Cryosphere, Human Systems, Trophic Dynamics, Evolution, Geomorphology, Geology), <code>SpatialDomain</code> , <code>PhysicalUnit</code> with SI dimension exponents, <code>SkillMetrics</code> , the SAPG graph (petgraph DiGraph), the unified ontology graph, regime tagging, ALife automaton types (<code>ProcessAutomaton</code> , <code>SurvivalTier</code> , <code>Constitution</code> , <code>ProcessSoul</code> , <code>HeartbeatConfig</code>), and evolutionary population management (<code>Population</code> , <code>ProcessLineage</code> , <code>EvolutionCandidate</code>).

Crate	Type	Responsibility
<code>maesma-knowledgebase</code>	Library	SQLite-backed Process Knowledgebase with BLAKE3 content-addressed manifests. Tables: <code>manifests</code> , <code>skill_records</code> , <code>ontology_relations</code> . Provides <code>KnowledgebaseStore</code> (open, deposit, query), fluent <code>QueryBuilder</code> API with family/rung/regime/region filtering, and seed machinery: <code>generate_seed_manifests()</code> returns 50+ representative <code>ProcessManifest</code> entries with full I/O contracts, scale envelopes, conservation properties, and cost models; <code>generate_seed_relations()</code> returns 40+ ontology relations (<code>CompatibleWith</code> , <code>IncompatibleWith</code> , <code>RequiresCouplingWith</code> , <code>Supersedes</code> , <code>VariantOf</code>) capturing cross-family coupling requirements and mutual exclusions.
<code>maesma-agents</code>	Library	25 agent implementations behind an async <code>Agent</code> trait. <code>AgentRole</code> enum with 25 variants (KB Retrieval, Assembly, Closure, Benchmarking, Selection, Optimizer, Discovery, Data Scout, A2A Gateway, Foundation Model, Autonomous Observation, Regime Detector, Scale Negotiator, Provenance, Salient Dynamics, Ensemble, Diagnostics, Sensitivity, Hypothesis, Geoengineering, Planetary Defense, Trophic, Evolution, Meta-Learner, Runtime Sentinel). <code>AgentRegistry</code> for role-based dispatch.
<code>maesma-compiler</code>	Library	Validates SAPG conservation closure, scale compatibility, and coupling consistency. Generates two-tier <code>ExecutionSchedule</code> splitting fast-coupled (seconds–hours) from slow-coupled (days–centuries) processes.
<code>maesma-runtime</code>	Library	Simulation execution engine. <code>Scheduler</code> dispatches stages with sub-stepping. <code>SimulationState</code> stores named 2D fields as <code>Array2<f64></code> . <code>EventBus</code> (FIFO queue) with typed <code>EventKind</code> variants (lightning ignition, regime shift, hot-swap, foundation model forecast, observation accepted/rejected, evolution generation, speciation, Pareto front update, ALife events: survival tier change, self-modification, replication, constitution violation, heartbeat check, stagnation detected, process archived, immigration). <code>HealthMonitor</code> detects NaN/blow-up. <code>HeartbeatDaemon</code> continuously evaluates ALife survival tiers and constitutional compliance.
<code>maesma-processes</code>	Library	13 process family modules implementing <code>ProcessRunner</code> . Fire family includes <code>StochasticFireRegime</code> (R0), <code>RothermelSurface</code> with full rate-of-speed math (R1),

Crate	Type	Responsibility
		and <code>CfsFbpCrown</code> with 17 Canadian fuel types, ISI, crown fraction burned (R1). Stub implementations for hydrology (Richards), ecology (cohort), biogeochemistry (CENTURY), radiation (two-stream), atmosphere (Monin-Obukhov), ocean (mixed-layer), cryosphere (energy-balance snowpack), human systems (land-use change), trophic dynamics (Lotka-Volterra), evolution (quantitative genetics), geomorphology (stream power, hillslope diffusion), and geology (thermal subsidence).
<code>maesma-inference</code>	Library	Trait-based neural inference abstraction. <code>InferenceEngine</code> trait with <code>infer()</code> method. <code>InferenceTask</code> variants: skill prediction, compatibility scoring, assembly ranking, regime prediction. Stub engine for development; production engine wraps graph transformer.
<code>maesma-federation</code>	Library	A2A federation client. <code>Peer</code> with <code>TrustLevel</code> (Untrusted/Provisional/Trusted/Verified). <code>FederationClient</code> sends requests and broadcasts to peer networks.
<code>maesma-api</code>	Library	Axum REST API server with 12 endpoints: health, list/get manifests, KB stats, SAPG retrieval, SAPG validation, list agents, skill records, Pareto front, simulation status, federation endpoint, list peers. CORS-enabled with tracing middleware.
<code>maesma-cli</code>	Binary	CLI entry point (<code>maesma</code>). Subcommands: <code>init</code> (scaffold workspace), <code>kb</code> (list/show/import/export/stats), <code>validate</code> (check SAPG), <code>check-closure</code> (conservation audit), <code>run</code> (execute simulation), <code>serve</code> (start API server on port 3001), <code>info</code> (system diagnostics).

Table 11: Rust workspace: 10 crates forming the MAESMA implementation.

B Initial Process Knowledgebase Catalog

The following appendices constitute the initial seed of the MAESMA Process Knowledgebase, cataloging **1185 process models, platform capabilities, and benchmarking resources** extracted from 50 reference Earth system, land surface, hydrological, ecological, forest dynamics, geomorphological, atmospheric chemistry, coastal, ocean biogeochemistry, ice sheet, glacier, agricultural, permafrost, urban canopy, marine ecosystem, aerosol microphysics, reactive transport, foundation weather, GPU-native, deep learning, and autonomous observation models and platforms, plus 1 benchmarking dataset (FireBench). Each entry records the process name, a brief description, the source model and module, the MAESMA process family assignment, and a fidelity classification (empirical / intermediate / physics-based). These entries form the starting manifests from which the neural inference engine begins its first reasoning cycle and the process evolution pipeline begins generating offspring.

C Badlands — Basin and Landscape Dynamics

Badlands is a parallel landscape evolution model simulating surface processes over geological timescales (10^3 – 10^8 years). It couples fluvial incision, hillslope diffusion, wave sediment transport, carbonate growth, flexural isostasy, and tectonic forcing on triangulated irregular networks (TINs).

#	Process	Family	Fidelity
1	Fluvial Incision — Stream Power Law (detachment-limited). $E = KA^m S^n$; O(n) implicit method (Braun & Willett 2013).	Geomorphology	Empirical
2	Fluvial Incision — Generalised Undercapacity. Linear sediment-flux dependency: $(1 - Q_s/Q_t)$ scaling.	Geomorphology	Empirical
3	Fluvial Incision — Parabolic Sedflux. Tools-and-cover effect with peak incision at intermediate Q_s/Q_t .	Geomorphology	Empirical
4	Fluvial Incision — Saltation-Abrasion. Channel-width-scaled saltation-abrasion incision model.	Geomorphology	Intermediate
5	Linear Hillslope Diffusion. Creep transport: $\partial z / \partial t = \kappa \nabla^2 z$ with aerial/marine coefficients.	Geomorphology	Empirical
6	Non-Linear Hillslope Diffusion. Critical-slope extension producing convex-to-planar profiles near S_c .	Geomorphology	Empirical
7	Slope Failure / Mass Wasting. Erosion and redistribution when slope exceeds critical failure threshold.	Geomorphology	Empirical
8	Marine Sediment Diffusion. Diffusion of river-deposited sediment below sea level along continental shelf.	Geomorphology	Empirical
9	Alluvial Plain Deposition. Forces deposition when local slope < critical threshold while carrying load.	Geomorphology	Empirical

#	Process	Family	Fidelity
10	Depression Filling (Planchon & Darboux). Fills topographic pits for continuous flow routing.	Hydrology	Empirical
11	Single Flow Direction (SFD) Routing. Steepest-descent receiver assignment with O(n) stack ordering.	Hydrology	Empirical
12	Drainage Area / Discharge Accumulation. Rainfall $\times$ Voronoi area accumulated through flow network.	Hydrology	Empirical
13	Fluvial Sediment Transport & Deposition. Multi-rock transport, pit filling, marine deposition via angle-of-repose.	Geomorphology	Empirical
14	Submarine Gravity Currents (Hyperpycnal Flows). Turbidity currents when flow density exceeds threshold.	Geomorphology	Empirical
15	Bedload vs. Slope Partitioning. Linear/sigmoidal/inverse-sigmoidal partitioning of total load.	Geomorphology	Empirical
16	Wave Propagation (Airy Wave Theory). Linear wave transformation from deep to shallow water; shoaling and breaking.	Ocean	Intermediate
17	Wave Refraction (Huygen's Principle). Depth-dependent celerity refraction via Huygen travel-time computation.	Ocean	Intermediate
18	Wave-Induced Sediment Transport. Van Rijn shear stress entrainment, wave-direction transport, diffusive redeposition.	Geomorphology	Empirical
19	Longshore Drift. Wave-induced longshore transport along bathymetric contours.	Geomorphology	Empirical
20	Carbonate Reef Growth (2 species). Fuzzy-logic environmental thresholds: depth $\times$ wave $\times$ sedimentation response.	Geology	Empirical
21	Pelagic Sedimentation. Depth-dependent open-ocean carbonate/sediment rain below sea level.	Geology	Empirical
22	Flexural Isostasy (gFlex). 2D lithospheric flexure under sediment/water loading; variable elastic thickness.	Geology	Physics-based
23	Tectonic Uplift/Subsidence (Vertical). Spatiotemporally varying vertical displacement maps.	Geology	Empirical
24	3D Tectonic Displacement (Horizontal + Vertical). Full 3D mesh deformation; Underworld coupling.	Geology	Empirical
25	Sea-Level Fluctuations. Time-varying eustatic curves controlling subaerial/marine process partition.	Ocean	Empirical

#	Process	Family	Fidelity
26	Orographic Precipitation (Smith & Barstad 2004). FFT-based spectral orographic rainfall model.	Atmosphere	Intermediate
27	Elevation-Dependent Rainfall. Linear precipitation–altitude relationship.	Atmosphere	Empirical
28	Stratigraphic Record (Regular Mesh). Layer tracking with thickness, elevation, depth, porosity.	Geology	Empirical
29	Stratigraphic Record (TIN / Multi-Rock). Multi-lithology active-layer stratigraphy with differential erodibility.	Geology	Empirical
30	Carbonate Stratigraphic Record. Separate pelagic + 2 carbonate species layer tracking.	Geology	Empirical
31	Erodibility Layering. Spatially variable bedrock erodibility through underlying strata.	Geology	Empirical
32	River Point-Source Input. External water/sediment injection at specified locations.	Hydrology	Empirical
33	Chi Parameter (Willett 2014). χ landscape metric for steady-state and divide migration analysis.	Geomorphology	Empirical

Table 12: Badlands process catalog (33 entries). Primary families: Geomorphology (15), Geology (9), Hydrology (4), Ocean (3), Atmosphere (2).

D CESM — Community Earth System Model

CESM3 couples atmosphere (CAM7), land (CLM6/CTSM), ocean (MOM6), sea ice (CICE6), land ice (CISM), river routing (MOSART/mizuRoute), and surface waves (WW3) via the CMEPS coupler.

D.1 Atmosphere — CAM (Community Atmosphere Model)

#	Process	Family	Fidelity
1	Spectral Element Dynamical Core (SE). Continuous/discontinuous Galerkin on cubed-sphere grid; hydrostatic and non-hydrostatic.	Atmosphere	Physics-based
2	Finite Volume Dynamical Core (FV). Lin-Rood finite-volume on lat-lon grid.	Atmosphere	Physics-based
3	Zhang-McFarlane Deep Convection (ZM). Mass-flux CAPE-based closure with momentum transport.	Atmosphere	Empirical
4	Hack Shallow Convection (HK). Moist convective adjustment (Hack 1994).	Atmosphere	Empirical
5	Park/Bretherton Shallow Convection (UWSHCU). Entraining-detraining mass-flux plume.	Atmosphere	Intermediate
6	CLUBB (Cloud Layers Unified By Binormals). Higher-order PDF-based turbulence/cloud closure.	Atmosphere	Physics-based
7	SHOC (Simplified Higher-Order Closure). Eddy-diffusivity mass-flux PBL.	Atmosphere	Intermediate
8	Morrison-Gottelman Microphysics (MG2). Two-moment stratiform cloud microphysics.	Atmosphere	Physics-based
9	P3 Microphysics. Predicted Particle Properties; single ice category with prognostic distribution.	Atmosphere	Physics-based
10	RRTMG Radiation. Correlated-k longwave/shortwave radiative transfer.	Radiation	Physics-based
11	RRTMGP Radiation. Next-gen RTE+RRTMGP with improved gas optics (CAM7 default).	Radiation	Physics-based
12	Cloud Optics (Ebert-Curry, Slingo). Parameterized cloud optical properties from LWC/IWC.	Radiation	Empirical
13	Aerosol–Radiation Interaction. Direct and indirect radiative effects from modal aerosol.	Radiation	Intermediate
14	Vertical Diffusion / PBL Turbulence. Diffusivity-based sub-grid vertical mixing.	Atmosphere	Intermediate

#	Process	Family	Fidelity
15	Gravity Wave Drag. Orographic, convective, and frontal gravity wave momentum deposition.	Atmosphere	Empirical
16	Dry Adiabatic Adjustment (DADADJ). Static instability removal.	Atmosphere	Empirical
17	Rayleigh Friction. Linear upper-atmosphere wind damping.	Atmosphere	Empirical
18	Molecular Diffusion. Molecular viscosity/thermal diffusion in the thermosphere.	Atmosphere	Physics-based
19	Ion Drag. Neutral-plasma momentum exchange in thermosphere.	Atmosphere	Intermediate
20	Cloud Fraction Diagnosis. RH/stability/convective-based grid-cell cloud cover.	Atmosphere	Empirical
21	Stratiform Cloud Macrophysics. Large-scale cloud formation/dissipation and liquid-ice partitioning.	Atmosphere	Intermediate
22	Aerosol Activation / Cloud Droplet Nucleation (NDROP). Abdul-Razzak & Ghan CCN activation.	Atmosphere	Physics-based
23	Ice Nucleation. Heterogeneous and homogeneous ice nucleation from aerosol (CNT).	Atmosphere	Physics-based
24	Cloud Sediment. Gravitational settling of cloud droplets and ice crystals.	Atmosphere	Physics-based
25	Water Vapor Saturation. Goff-Gratch / Murphy-Koop saturation vapor pressure.	Atmosphere	Physics-based
26	QBO Nudging. Tropical stratospheric wind relaxation to observed QBO profiles.	Atmosphere	Empirical
27	COSP2 Satellite Simulator. CloudSat/CALIPSO/ISCCP/MODIS/MISR forward simulators.	Atmosphere	Intermediate
28	CRM / Super-Parameterization (SAM). Embedded cloud-resolving model replacing parameterized convection.	Atmosphere	Physics-based

Table 13: CAM atmosphere processes (28 entries).

D.2 Atmospheric Chemistry & Aerosols

#	Process	Family	Fidelity
29	MOZART Tropospheric/Stratospheric Chemistry. Full gas-phase mechanism with 100+ species, photolysis, halogen chemistry.	Atmosphere	Physics-based
30	Linearized Ozone Chemistry (LinOz). Climatological production/loss-based stratospheric O ₃ .	Atmosphere	Empirical

#	Process	Family	Fidelity
31	Modal Aerosol Module (MAM3/MAM4/MAM7). 3–7 log-normal modes: SO ₄ , OC, BC, sea salt, dust, SOA.	Atmosphere	Intermediate
32	Aerosol Nucleation. Binary/ternary new particle formation from H ₂ SO ₄ and organics.	Atmosphere	Physics-based
33	Aerosol Coagulation. Brownian coagulation transferring particles between modes.	Atmosphere	Physics-based
34	Aerosol Gas-Aerosol Exchange. Semi-volatile condensation/evaporation on aerosol.	Atmosphere	Physics-based
35	Aerosol Convective Processing. Activation, scavenging, re-suspension in convective updrafts.	Atmosphere	Intermediate
36	Aerosol Wet Deposition. Below-cloud impaction + in-cloud nucleation scavenging.	Atmosphere	Intermediate
37	Aerosol Dry Deposition. Gravitational settling + resistance-based turbulent deposition.	Atmosphere	Intermediate
38	Dust Emission & Sedimentation. Wind erosion from soil erodibility maps; settling.	Atmosphere	Empirical
39	Sea Salt Emission. Wind-speed and SST-dependent sea spray generation.	Atmosphere	Empirical
40	Sulfur/Aqueous Chemistry (SO _x). Aqueous SO ₂ →SO ₄ in cloud droplets; DMS oxidation.	Atmosphere	Physics-based
41	Photolysis (J-values). Solar zenith, O ₃ column, and cloud-modified photodissociation rates.	Atmosphere	Physics-based
42	Lightning NO _x . Flash-rate-dependent NO _x production from convection.	Atmosphere	Empirical
43	Aircraft Emissions. Altitude-resolved trace gas injection from aviation.	Atmosphere	Empirical
44	CO ₂ Cycle. Prognostic atmospheric CO ₂ tracer with land/ocean surface fluxes.	Biogeochem	Intermediate
45	WACCM Chemistry. Middle/upper atmosphere: mesospheric/thermospheric reactions, aurora, EUV heating.	Atmosphere	Physics-based

Table 14: CAM chemistry and aerosol processes (17 entries).

D.3 Land — CLM/CTSM

#	Process	Family	Fidelity
46	Surface Albedo. Two-stream direct/diffuse VIS/NIR albedo over soil, canopy, snow, urban.	Radiation	Physics-based

#	Process	Family	Fidelity
47	Surface Radiation. Partitioning of SW/LW among canopy, soil, snow, urban.	Radiation	Physics-based
48	SNICAR. Multi-layer snow radiative transfer with grain-size evolution, BC and dust impurities.	Radiation	Physics-based
49	Canopy Fluxes. Monin-Obukhov similarity turbulent fluxes from vegetated surfaces.	Atmosphere	Physics-based
50	Bare Ground Fluxes. Bulk aerodynamic surface energy balance over bare soil.	Atmosphere	Intermediate
51	Friction Velocity. Monin-Obukhov u_* and aerodynamic resistances.	Atmosphere	Physics-based
52	Canopy Hydrology. Interception, throughfall, stemflow, canopy water storage.	Hydrology	Intermediate
53	Snow Hydrology. Multi-layer snow: accumulation, compaction, grain metamorphism, melt percolation, refreezing.	Cryosphere	Physics-based
54	Soil Hydrology (Richards Equation). 10-layer vertical water movement with infiltration, percolation, drainage.	Hydrology	Physics-based
55	Soil Water Movement. Darcy fluxes, hydraulic redistribution, water table dynamics.	Hydrology	Physics-based
56	Soil Water Retention Curves. Clapp-Hornberger (1978) pedotransfer functions.	Hydrology	Empirical
57	Soil Moisture Stress (β). Plant water stress from root-zone moisture availability.	Hydrology	Empirical
58	Soil Temperature. Multi-layer heat conduction with phase change (freeze/thaw) in soil/snow/bedrock.	Hydrology	Physics-based
59	Ground Heat / Soil Surface Fluxes. Ground heat flux and bare-soil evaporation.	Hydrology	Physics-based
60	Canopy Temperature. Iterative leaf energy balance solution.	Atmosphere	Physics-based
61	Photosynthesis (Farquhar/Ball-Berry). C3/C4 biochemistry + Ball-Berry/Medlyn stomatal conductance.	Biogeochem	Physics-based
62	Surface Resistance. Soil evaporation resistance based on moisture; dry surface layer.	Hydrology	Empirical
63	Root Biophysics. Root water uptake distribution across soil layers.	Hydrology	Intermediate

#	Process	Family	Fidelity
64	Active Layer (Permafrost). Seasonal thaw depth diagnosis from soil temperature.	Cryosphere	Intermediate
65	Lake Temperature. 1-D lake thermal model: wind mixing, convective overturn, ice, sediment heat.	Hydrology	Intermediate
66	Lake Hydrology. Lake water balance: precipitation, evaporation, runoff, ice fraction.	Hydrology	Intermediate
67	Lake Fluxes. Turbulent and radiative fluxes over lake surfaces.	Hydrology	Intermediate
68	Urban Energy Balance (CLMU). Multi-layer urban canopy: radiation trapping, canyon airflow, anthropogenic heat.	Human Sys	Intermediate
69	Aerosol Deposition on Snow/Soil. BC/dust/OC cycling on surfaces; SNICAR albedo feedback.	Biogeochem	Intermediate
70	Water Budget Closure. Mass/energy conservation accounting across all land stores.	Hydrology	Physics-based
71	Daylength. Astronomical daylength for phenology and photosynthesis.	Radiation	Physics-based
72	Sediment Yield. Hillslope soil erosion and sediment delivery to rivers.	Geomorphology	Empirical
73	Hydrological Drainage. Subsurface lateral drainage and baseflow to streams.	Hydrology	Intermediate

Table 15: CLM/CTSM biogeophysics processes (28 entries).

D.4 Land Biogeochemistry

#	Process	Family	Fidelity
74	Ecosystem Dynamics (CN/BGC). Master driver for coupled C-N biogeochemistry.	Biogeochem	Intermediate
75	Decomposition Cascades (Century/CN). 3-pool or 4-pool SOM decomposition for C/N mineralization.	Biogeochem	Intermediate
76	Soil Litter Decomposition. Temperature-, moisture-, N-, and depth-dependent decomposition rates.	Biogeochem	Intermediate
77	Vertical Soil Carbon Transport. Bioturbation, cryoturbation, advective SOM mixing.	Biogeochem	Intermediate
78	Phenology. GDD/daylength/moisture-triggered leaf onset/offset for multiple PFTs.	Ecology	Empirical

#	Process	Family	Fidelity
79	Satellite Phenology (SP). Prescribed LAI/SAI from MODIS observations.	Ecology	Empirical
80	Carbon Allocation. NPP partitioning among leaf, stem, root, storage, reproduction.	Biogeochem	Intermediate
81	Nitrogen Dynamics. N mineralization, immobilization, plant uptake, N limitation.	Biogeochem	Intermediate
82	Nitrification-Denitrification. Soil N ₂ O, NO, N ₂ emissions from microbial N cycling.	Biogeochem	Intermediate
83	Phosphorus Dynamics. P weathering, mineralization, sorption, plant uptake, P limitation.	Biogeochem	Intermediate
84	Methane Biogeochemistry (CH ₄). Production, oxidation, ebullition, plant transport, diffusion.	Biogeochem	Intermediate
85	Fire (CLM). Prognostic fire occurrence, spread, area, and C emissions.	Fire	Empirical
86	FATES Fire Interface. Mechanistic fire behavior and effects coupling.	Fire	Intermediate
87	Gap Mortality. Age/stress/frost/disturbance-driven tree mortality.	Ecology	Empirical
88	Crop Model. Planting, GDD phenology, grain fill, harvest for major crops.	Human Sys	Intermediate
89	Root Dynamics. Fine root growth, turnover, vertical distribution.	Ecology	Intermediate
90	Vegetation Structure Update. LAI/height/SAI from carbon state changes.	Ecology	Intermediate
91	Maintenance Respiration. N-content and temperature-dependent autotrophic respiration.	Biogeochem	Intermediate
92	Growth Respiration. Fixed-fraction construction cost of new tissue.	Biogeochem	Empirical
93	Wood Products. Harvested wood C pools with multi-decadal decay.	Human Sys	Empirical
94	Crop Harvest Pools. Grain, stover, residue carbon management.	Human Sys	Empirical
95	Carbon Isotopes (¹⁴ C / ¹³ C). Radiocarbon and stable isotope tracking through all pools.	Biogeochem	Physics-based
96	Dust Emission (Land). Wind friction, moisture, vegetation-controlled dust uplift.	Atmosphere	Empirical

#	Process	Family	Fidelity
97	BVOC Emissions (MEGAN). Isoprene, monoterpene, sesquiterpene emissions from LAI/T/light.	Biogeochem	Empirical
98	Dry Deposition Velocity. Resistance-in-series trace gas deposition velocities.	Atmosphere	Intermediate
99	Soil Erosion. Wind and water erosion with nutrient loss.	Geomorphology	Empirical
100	FAN (Fertilizer & Ammonia Network). Agricultural N fertilizer application and NH ₃ volatilization.	Human Sys	Intermediate
101	Plant-Microbe Kinetics. Plant–soil microbe N/P competition.	Biogeochem	Intermediate

Table 16: CLM/CTSM biogeochemistry processes (28 entries).

D.5 Ocean — MOM6 / MPAS-Ocean

#	Process	Family	Fidelity
102	Primitive Equation Dynamics (Hydrostatic). Boussinesq PE on Arakawa C-grid with ALE vertical coordinate.	Ocean	Physics-based
103	Equation of State (Wright/JM/Linear). Seawater density from T, S, pressure.	Ocean	Physics-based
104	Horizontal Viscosity (Laplacian/Biharmonic). Sub-grid momentum diffusion: del2 and del4 operators.	Ocean	Intermediate
105	Leith Viscosity. Velocity-gradient-dependent enstrophy-cascade viscosity.	Ocean	Intermediate
106	Gent-McWilliams Mesoscale Eddy Transport. Bolus transport via isopycnal thickness diffusivity.	Ocean	Intermediate
107	Submesoscale Eddy Parameterization. Mixed-layer restratification (Fox-Kemper et al. 2008).	Ocean	Intermediate
108	Redi Isopycnal Diffusion. Along-isopycnal tracer diffusion.	Ocean	Physics-based
109	Horizontal Tracer Diffusion. Laplacian/biharmonic tracer mixing on geopotential/isopycnal.	Ocean	Intermediate
110	Vertical Mixing (CVMix/KPP). K-Profile Parameterization boundary layer scheme.	Ocean	Intermediate
111	GOTM Vertical Mixing. k - ϵ / Mellor-Yamada turbulence closures.	Ocean	Physics-based
112	Pressure Gradient Force. Hydrostatic pressure gradient including baroclinic density effects.	Ocean	Physics-based

#	Process	Family	Fidelity
113	Coriolis & Horizontal Advection. Earth rotation and horizontal momentum advection.	Ocean	Physics-based
114	Surface Wind Stress. Atmospheric momentum flux to ocean surface.	Ocean	Physics-based
115	Explicit Bottom Drag. Quadratic bottom friction.	Ocean	Empirical
116	Topographic Wave Drag. Internal-wave dissipation over rough topography.	Ocean	Intermediate
117	Tidal Potential & Self-Attraction Loading. Astronomical tidal forcing with elastic Earth response.	Ocean	Physics-based
118	Tidal Forcing (constituents). M_2 , S_2 , K_1 , O_1 etc. at boundaries.	Ocean	Physics-based
119	ALE Vertical Coordinate / Thickness Advection. z -star, σ , hybrid vertical remapping.	Ocean	Physics-based
120	Tracer Advection (FCT/Monotone). Flux-corrected monotonic tracer transport.	Ocean	Physics-based
121	Shortwave Absorption (Jerlov/Chlorophyll). Penetrating solar radiation depth profiles.	Radiation	Empirical
122	Surface Bulk Forcing. Air-sea heat, freshwater, momentum bulk formulas.	Ocean	Intermediate
123	Frazil Ice Formation. Latent heat release when $T < \text{freezing}$ $\rightarrow$ ice nucleation.	Cryosphere	Physics-based
124	Stokes Drift / Langmuir Turbulence. Wave-driven transport enhancement of upper-ocean mixing.	Ocean	Physics-based
125	Wetting & Drying. Dynamic coastal cell wetting/drying.	Ocean	Physics-based
126	Ecosys Marine BGC. Multi-nutrient (N, P, Si, Fe) ecosystem: phytoplankton, zooplankton, DOM/POM.	Biogeochem	Intermediate
127	DMS Ocean Tracer. Biogenic dimethylsulfide production and sea-air flux.	Biogeochem	Intermediate
128	CFC Tracers (CFC-11/CFC-12). Transient ventilation tracers for circulation validation.	Biogeochem	Physics-based
129	Ideal Age Tracer. Passive tracer for ocean ventilation age.	Ocean	Physics-based
130	MacroMolecules Tracer. Dissolved organic macromolecule cycling.	Biogeochem	Intermediate
131	Transit Time Distribution (TTD). Ventilation transit time diagnostics.	Ocean	Physics-based

#	Process	Family	Fidelity
132	Land Ice–Ocean Fluxes. Sub-ice-shelf melt heat/freshwater exchange.	Cryosphere	Physics-based
133	Lagrangian Particle Tracking. Online water mass trajectory integration.	Ocean	Physics-based

Table 17: Ocean processes from MOM6 / MPAS-Ocean (32 entries).

D.6 Sea Ice — CICE / MPAS-Sea Ice

#	Process	Family	Fidelity
134	Ice Thermodynamics (Vertical). Multi-layer heat conduction with brine pockets; top/bottom growth/melt.	Cryosphere	Physics-based
135	Ice Thickness Distribution (ITD). Sub-grid multi-category thickness tracking.	Cryosphere	Intermediate
136	Elastic-Viscous-Plastic Dynamics (EVP). EVP rheology: stress–strain–rate–strength.	Cryosphere	Physics-based
137	Ice Transport (Incremental Remapping). Geometrically-based shape-preserving advection.	Cryosphere	Physics-based
138	Mechanical Redistribution (Ridging/Rafting). Pressure ridge and keel formation under convergence.	Cryosphere	Intermediate
139	Ice–Atmosphere Fluxes. Turbulent heat/moisture/momentum exchange.	Cryosphere	Intermediate
140	Ice–Ocean Fluxes. Heat, salt, freshwater exchange at ice bottom.	Cryosphere	Physics-based
141	Shortwave Radiation in Ice. Delta-Eddington multiple-scattering through snow/ice layers.	Radiation	Physics-based
142	Melt Ponds. Formation, drainage, refreezing; albedo feedback.	Cryosphere	Intermediate
143	Sea Ice Age Tracer. Multi-year vs. first-year ice extent tracking.	Cryosphere	Empirical
144	First-Year Ice Fraction. FY vs. MY ice partition for optical/biological properties.	Cryosphere	Empirical
145	Level Ice Fraction. Undeformed vs. ridged ice tracking.	Cryosphere	Empirical
146	Ice Aerosol Deposition. BC/dust on/in sea ice; albedo response.	Cryosphere	Intermediate
147	VP/EVP Constitutive Relations (MPAS). Variational and weak-ice rheology formulations.	Cryosphere	Physics-based

#	Process	Family	Fidelity
148	Icepack Thermodynamics (MPAS). Columnar thermodynamics via Icepack library.	Cryosphere	Physics-based

Table 18: Sea ice processes (15 entries).

D.7 Land Ice — CISM / MALI

#	Process	Family	Fidelity
149	Shallow Ice Approximation (SIA). Vertical-shear-dominated interior ice flow.	Cryosphere	Intermediate
150	Higher-Order Ice Dynamics (Blatter-Pattyn). First-order Stokes with membrane stresses.	Cryosphere	Physics-based
151	Full Stokes Dynamics (Albany). 3D Stokes via Trilinos/Albany for complex ice dynamics.	Cryosphere	Physics-based
152	Ice Sheet Thermodynamics. Advection-diffusion with strain heating and geothermal flux.	Cryosphere	Physics-based
153	Surface Mass Balance. Snowfall, melt (energy balance/PDD), rain, refreezing $\rightarrow$ net SMB.	Cryosphere	Intermediate
154	Basal Sliding. Weertman-type and Coulomb friction sliding laws.	Cryosphere	Intermediate
155	Calving. Flotation, thickness-based, eigen-calving, von Mises stress criteria.	Cryosphere	Empirical
156	Ice-Shelf Basal Melt. Ocean thermal forcing-driven sub-shelf melt rates.	Cryosphere	Intermediate
157	Subglacial Hydrology. Sheet/channel drainage controlling effective pressure and sliding.	Hydrology	Physics-based
158	Isostatic Bedrock Adjustment (GIA). Viscoelastic bedrock deformation under ice loads.	Geology	Intermediate
159	Sea Level Model. Gravitational, rotational, deformational sea level change.	Ocean	Physics-based

Table 19: Land ice processes from CISM / MALI (11 entries).

D.8 River Routing & Waves

#	Process	Family	Fidelity
160	RTM. Linear transport routing on global river network.	Hydrology	Empirical
161	MOSART Kinematic/Diffusion Wave Routing. Hillslope $\rightarrow$ sub-network $\rightarrow$ main channel cascade.	Hydrology	Intermediate

#	Process	Family	Fidelity
162	MOSART Reservoir Operations (WRM). Dam storage, regulated releases, water withdrawals.	Human Sys	Intermediate
163	MOSART Inundation. Floodplain wetting when discharge exceeds channel capacity.	Hydrology	Intermediate
164	MOSART Sediment Transport. Suspended/bed-load sediment routing through river network.	Geomorphology	Intermediate
165	MOSART River Heat. Thermal energy transport and river temperature.	Hydrology	Intermediate
166	MOSART Biogeochemistry. Dissolved C, N, and constituent transport.	Biogeochem	Intermediate
167	mizuRoute. IRF/KWT network-based streamflow routing on vector networks.	Hydrology	Intermediate
168	WW3 Wave Spectral Evolution. Wave action balance equation in frequency-direction space.	Ocean	Physics-based
169	WW3 Wind Input. Miles/Janssen wind-wave generation.	Ocean	Physics-based
170	WW3 Nonlinear Wave-Wave Interactions (DIA). Four-wave resonant energy redistribution.	Ocean	Physics-based
171	WW3 Wave Dissipation (Whitecapping). Steepness/spectral-saturation dissipation.	Ocean	Intermediate
172	WW3 Wave-Bottom Interaction. Bottom friction, refraction, shoaling.	Ocean	Intermediate
173	WW3 Wave-Ice Interaction. Wave attenuation in ice; ice breakup from flexural stress.	Cryosphere	Intermediate
174	WW3 Wave-Current Interaction. Radiation stress, Stokes drift coupling.	Ocean	Physics-based

Table 20: River routing and surface wave processes (15 entries).

D.9 Coupler & Data Models

#	Process	Family	Fidelity
175	CMEPS Air-Sea Flux. COARE-based turbulent flux computation.	Atmosphere	Intermediate
176	CMEPS Field Merging. Area-weighted surface field combination from ocean/ice/land.	Atmosphere	Physics-based
177	CMEPS Conservative Regridding. ESMF-based mapping between component grids.	Atmosphere	Physics-based

#	Process	Family	Fidelity
178	CMEPS Runoff Mapping. Freshwater routing from river mouths to coastal ocean cells.	Hydrology	Intermediate
179	Slab Ocean Model (DOCN%SOM). Mixed-layer SST evolution from fluxes + prescribed OHT.	Ocean	Empirical
180	Data Atmosphere (DATM). Prescribed reanalysis/observed forcing.	Atmosphere	Empirical
181	Data Sea Ice (DICE). Prescribed ice concentration with semi-prognostic fluxes.	Cryosphere	Empirical
182	Solar Variability. Spectral solar irradiance variations for radiation.	Radiation	Empirical

Table 21: Coupler and data model processes (8 entries).

E E3SM — Unique Processes Beyond CESM Heritage

E3SM shares substantial heritage with CESM (CAM→EAM, CLM→ELM, CICE→MPAS-Sea Ice). The following lists processes unique to or substantially modified in E3SM.

#	Process	Family	Fidelity
1	Theta-L Non-Hydrostatic Dynamical Core. Vertical Lagrangian θ - l formulation in HOMME.	Atmosphere	Physics-based
2	EAMxx/SCREAM C++ Physics Port. GPU-accelerated SHOC + P3 + RRTMGP + MAM4 + ZM on Kokkos.	Atmosphere	Physics-based
3	Neural-Network Cloud Fraction (EAMxx). ML-based cloud cover alternative to diagnostic scheme.	Atmosphere	Intermediate
4	Cloud-J Photolysis (UCI). Updated photolysis with cloud-J spectral binning.	Atmosphere	Physics-based
5	CRM SGS-TKE Turbulence (MMF). Subgrid TKE diffusion within embedded cloud-resolving model.	Atmosphere	Physics-based
6	BeTR (Biogeochemical Transport). Vertically resolved multi-phase reactive transport in soil.	Biogeochem	Physics-based
7	EMI (External Model Interface). Generic coupling to external BGC models (PFLOTRAN).	Biogeochem	Intermediate
8	FATES (Cohort Vegetation Demography). Size-structured competition, disturbance, recruitment.	Ecology	Physics-based
9	Plant-Microbe ECA Kinetics. Enzyme-mediated nutrient competition with Michaelis-Menten.	Biogeochem	Physics-based
10	MPAS-Ocean Unstructured Mesh Dynamics. Voronoi finite-volume split-explicit PE solver.	Ocean	Physics-based
11	MPAS-Sea Ice on Unstructured Mesh. VP/EVP + Icepack on SCVT meshes.	Cryosphere	Physics-based
12	MALI Ice Sheet (Albany). Unstructured-mesh Blatter-Pattyn + FCT + thermal solver.	Cryosphere	Physics-based
13	GCAM (Human Earth System). Integrated assessment model coupling for energy/land/water scenarios.	Human Sys	Empirical
14	MOAB Coupler. Online conservative remapping via MOAB mesh library.	Atmosphere	Physics-based
15	Ocean Sediment Transport / Flux Index. Coastal sediment transport analysis.	Geomorphology	Empirical

Table 22: E3SM-unique processes (15 entries).

F FATES — Functionally Assembled Terrestrial Ecosystem Simulator

FATES is a cohort-based vegetation demography model that replaces the static PFT framework in CLM/ELM with size-structured cohorts competing for light, water, and nutrients. It is hosted by E3SM, CESM, and CTSM, funded by DOE NGEE-Tropics. The detailed catalog below draws from 29 source files across 6 source directories (biogeochem, biogeophys, fire, main, parteh, radiation).

F.1 Radiation

#	Process	Family	Fidelity
1	Norman Multi-Layer Canopy Radiation. Radiative transfer through vertically-resolved canopy (Norman 1979); absorbed PAR/NIR per layer for sunlit/shaded leaves.	Radiation	Physics-based
2	Two-Stream Multi-Layer Plant Element (MLPE) Solver. Upward/downward diffuse + direct beam RT through multiple canopy elements with scattering/absorption.	Radiation	Physics-based
3	Canopy Radiation Driver. Top-level driver selecting Norman or Two-Stream; normalized canopy radiation profiles and sun/shade leaf area fractions.	Radiation	Physics-based
4	FATES Two-Stream Utilities. FATES-specific wrapper for Two-Stream: constructs radiation elements from cohort structure, maps absorbed radiation back to cohorts.	Radiation	Physics-based

Table 23: FATES radiation processes (4 entries).

F.2 Photosynthesis & Leaf Biophysics

#	Process	Family	Fidelity
5	Rubisco-Limited Photosynthesis (C3). Farquhar-von Caemmerer-Berry Rubisco carboxylation-limited assimilation; Michaelis-Menten kinetics with temperature-dependent V_{cmax} .	Biogeochem	Physics-based
6	RuBP-Limited Photosynthesis (C3). FvCB electron transport (light)-limited assimilation; uses J from PAR.	Biogeochem	Physics-based
7	RuBP-Limited Photosynthesis (C4). Light-limited assimilation for C4; quantum yield formulation.	Biogeochem	Physics-based
8	PEP Carboxylase-Limited Photosynthesis (C4). PEP carboxylase enzyme-limited assimilation for C4.	Biogeochem	Physics-based
9	Electron Transport Rate (J). Computes electron transport from PAR; FvCB hyperbolic and Johnson-Berry formulations.	Biogeochem	Physics-based

#	Process	Family	Fidelity
10	Stomatal Conductance — Medlyn Model. Optimal stomatal conductance (Medlyn et al. 2011) based on marginal water use efficiency; relates g_s to assimilation, VPD, CO_2 .	Biogeochem	Physics-based
11	Stomatal Conductance — Ball-Berry Model. Empirical model (Ball, Woodrow & Berry 1987); g_s from assimilation, relative humidity, leaf-surface CO_2 .	Biogeochem	Empirical
12	Intercellular CO_2 (Ci) Iteration. Bisection solver coupling photosynthetic demand with stomatal supply.	Biogeochem	Physics-based
13	Leaf Layer Photosynthesis. Integrates leaf biochemistry, stomatal conductance, and boundary layer conductance per canopy layer; net assimilation for sunlit/shaded fractions.	Biogeochem	Physics-based

Table 24: FATES photosynthesis and leaf biophysics (9 entries).

F.3 Plant Respiration

#	Process	Family	Fidelity
14	Leaf (Maintenance) Respiration. Dark respiration scaled from $V_{\text{cmax}25}$ with Atkin temperature acclimation.	Biogeochem	Intermediate
15	Fine Root Respiration. Temperature-dependent maintenance respiration distributed across soil layers; based on root nitrogen.	Biogeochem	Intermediate
16	Live Wood (Sapwood + Storage) Respiration. Temperature-dependent maintenance respiration of sapwood and storage tissues.	Biogeochem	Intermediate
17	Symbiotic N Fixation. Root-layer nitrogen fixation scaled by root biomass and soil conditions.	Biogeochem	Empirical

Table 25: FATES respiration and N fixation (4 entries).

F.4 Plant Hydraulics

#	Process	Family	Fidelity
18	Plant Hydraulics Driver (TFS-Hydro). Resistance-capacitance model: soil $\rightarrow$ root $\rightarrow$ stem $\rightarrow$ leaf water potential; Richards equation-based flow; tracks cavitation.	Hydrology	Physics-based
19	Van Genuchten Water Retention & Conductivity. θ vs ψ retention curve and unsaturated hydraulic conductivity (van Genuchten 1980).	Hydrology	Physics-based

#	Process	Family	Fidelity
20	Clapp-Hornberger-Campbell Water Retention & Conductivity. Alternative power-law formulation (Clapp & Hornberger 1978).	Hydrology	Physics-based
21	Smoothed CCH Water Transfer Functions. Smoothed variant with continuous derivatives for numerical stability.	Hydrology	Physics-based
22	TFS Pedotransfer Functions. Soil texture $\rightarrow$ van Genuchten hydraulic parameters via Tomasella-Hodnett.	Hydrology	Empirical
23	BTRAN Transpiration Wetness Factor. Soil moisture limitation on transpiration; weighted plant-available water across root-occupied layers.	Hydrology	Intermediate
24	Root Water Uptake Distribution. Distributes transpiration demand across soil layers proportional to root fraction and water availability.	Hydrology	Intermediate
25	Salinity Stress. Reduces BTRAN based on soil salinity; osmotic potential effect on water uptake.	Hydrology	Empirical

Table 26: FATES plant hydraulics (8 entries).

F.5 Fire (SPITFIRE)

#	Process	Family	Fidelity
26	Fire Weather Index (Nesterov Index). Cumulative fire weather index; resets with precipitation.	Fire	Empirical
27	Fire Danger Index (FDI). Ignition probability from fire weather; exponential Nesterov transformation (Venevsky et al. 2002).	Fire	Empirical
28	Ignition Sources. Lightning + anthropogenic ignition; anthropogenic via Li et al. (2012) population scaling.	Fire	Empirical
29	Effective Wind Speed. Wind modified by vegetation cover (tree/grass fractions).	Fire	Intermediate
30	Fuel Loading. Aggregates dead leaf litter, 4 CWD size classes, live grass into fuel classes.	Fire	Intermediate
31	Fuel Moisture. Moisture content from Nesterov index and drying ratio; class-specific extinction.	Fire	Empirical
32	Fuel Bulk Density & SAV. Average bulk density and surface-area-to-volume ratio; drives packing ratio.	Fire	Intermediate
33	Fuel Class Definitions. 6 classes: twigs (1h), small branches (10h), large branches (100h), trunks (1000h), dead leaves, live grass.	Fire	Empirical

#	Process	Family	Fidelity
34	Rothermel Reaction Intensity. Net energy release rate per unit fuel bed area (Rothermel 1972).	Fire	Physics-based
35	Optimum Packing Ratio. Packing ratio maximizing reaction intensity; function of SAV.	Fire	Physics-based
36	Forward Rate of Spread. Rothermel (1972) quasi-steady surface fire spread with wind and slope.	Fire	Physics-based
37	Backward Rate of Spread. Backing fire spread; empirical function of forward ROS and wind.	Fire	Empirical
38	Fire Duration & Size. Elliptical fire shape; length-to-breadth ratio; total area from forward/backward ROS.	Fire	Intermediate
39	Fireline Intensity. Byram’s fireline intensity (kW/m) from fuel consumed and ROS.	Fire	Physics-based
40	Scorch Height. Flame scorch height from fireline intensity (Van Wagner).	Fire	Empirical
41	Bark Thickness. Allometric bark thickness for fire resistance; PFT-specific.	Fire	Empirical
42	Cambial Mortality. Probability of cambial kill from fire residence time vs. critical bark thickness.	Fire	Intermediate
43	Crown Fire Mortality. Fraction of crown volume scorched and resulting crown fire mortality.	Fire	Intermediate
44	Total Fire Mortality. Combined cambial + crown scorch mortality per cohort.	Fire	Intermediate
45	Fuel Consumption & Residence Time. Fraction of each fuel class consumed; fire residence time for cambial damage.	Fire	Intermediate
46	Prescribed/Managed Fire Decision. Decision logic for prescribed burns based on weather window and fuel load thresholds.	Fire	Empirical
47	Prescribed Fire Burn Window. Evaluates meteorological permissibility for prescribed burn.	Fire	Empirical

Table 27: FATES/SPITFIRE fire processes (22 entries).

F.6 Mortality

#	Process	Family	Fidelity
48	Background Mortality. Constant PFT-specific rate representing unresolved stochastic mortality.	Ecology	Empirical

#	Process	Family	Fidelity
49	Carbon Starvation Mortality. Mortality from negative C balance when storage falls below threshold; linear ramp and exponential decay forms.	Ecology	Intermediate
50	Hydraulic Failure Mortality. Triggered when BTRAN or fractional conductivity loss exceeds critical threshold.	Ecology	Intermediate
51	Freezing Stress Mortality. Cold-induced mortality from accumulated freezing degree-days below PFT threshold.	Ecology	Empirical
52	Size-Dependent Senescence Mortality. Logistic increase with stem diameter; ontogenetic senescence.	Ecology	Empirical
53	Age-Dependent Senescence Mortality. Logistic increase with age beyond PFT-specific threshold.	Ecology	Empirical
54	Damage-Dependent Mortality. Additional mortality for cohorts with crown damage (wind); logistic function of damage class.	Ecology	Empirical
55	Fire Mortality. Combined cambial + crown fire kill applied to cohorts post-fire event.	Fire	Intermediate
56	Logging — Direct Harvest. Fraction of commercial-sized trees felled; DBH-bounded.	Human Sys	Empirical
57	Logging — Collateral Damage. Understory mortality from falling harvested trees.	Human Sys	Empirical
58	Logging — Mechanical/Infrastructure. Mortality from skid trails and logging roads.	Human Sys	Empirical

Table 28: FATES mortality processes (11 entries).

F.7 Phenology, Allometry, & Growth

#	Process	Family	Fidelity
59	GDD Accumulation. Growing degree-days above base temperature; drives cold-deciduous onset.	Ecology	Intermediate
60	Cold Deciduous Phenology. Leaf-on by GDD threshold; leaf-off by chilling day or daylength.	Ecology	Intermediate
61	Drought Deciduous Phenology. Soil moisture-driven leaf flush/senescence.	Ecology	Intermediate
62	Satellite/Prescribed Phenology. Externally prescribed LAI; bypasses prognostic phenology.	Ecology	Empirical
63	Phenology Leaf Flush. Allocates stored C/N/P to new leaves during onset; respects allometric targets.	Ecology	Intermediate

#	Process	Family	Fidelity
64	Deciduous Leaf Turnover. Retranslocates nutrients before abscission; transfers to litter.	Ecology	Intermediate
65	Height–Diameter Allometry. Multiple forms: power, O’Brien, Martínez-Cano, Chave.	Ecology	Empirical
66	Above-Ground Biomass Allometry. Total AGB from DBH; Saldarriaga, 2-parameter power, Chave.	Ecology	Empirical
67	Maximum Leaf Biomass Allometry. Max leaf biomass from DBH via crown area and SLA with trimming.	Ecology	Empirical
68	Crown Area Allometry. Crown area from DBH; power-law with canopy-position spread factor.	Ecology	Empirical
69	Sapwood Biomass Allometry. Sapwood cross-section and biomass from DBH and height via pipe model theory.	Ecology	Intermediate
70	Fine Root Biomass Allometry. Target fine root biomass from leaf area and nutrient acquisition needs.	Ecology	Intermediate
71	Storage Biomass Allometry. Target labile carbon pool as fraction of leaf biomass.	Ecology	Empirical
72	Below-Ground Woody Biomass. Coarse root biomass from AGB using fixed ratio.	Ecology	Empirical
73	Dead (Structural) Wood Biomass. Heartwood as residual after subtracting sapwood from total.	Ecology	Intermediate
74	LAI & SAI from Crown Geometry. Leaf/stem area index per tree from crown area, depth, biomass.	Ecology	Intermediate
75	C-Only Allometric Growth (PARTEH). Daily net C allocation to organs maintaining allometric targets.	Biogeochem	Intermediate
76	Carbon Pool Growth Derivatives. Instantaneous growth rates per C pool from NPP and allometric constraints.	Biogeochem	Intermediate
77	CNP Allometric Growth (PARTEH). C-N-P coupled allocation with nutrient limitation; fine root PID controller.	Biogeochem	Physics-based
78	Fine Root C:N:P Target Adjustment. PID-controller-based adjustment of fine root targets for nutrient acquisition.	Biogeochem	Intermediate
79	Prioritized Pool Replacement. Allocates C/N/P to deficit pools in priority order (leaf > fine root > sapwood > storage > structural).	Biogeochem	Intermediate
80	Maintenance Turnover. Continuous background turnover of leaves, fine roots; transfers to litter with retranslocation.	Biogeochem	Intermediate

#	Process	Family	Fidelity
81	Reproductive Allocation. C (and nutrients) to reproductive pool; seeds released at species-specific rate.	Biogeochem	Intermediate
82	Fire Biomass Losses. Removes fire-damaged organs; transfers to atmosphere (combustion) or litter.	Biogeochem	Intermediate
83	Crown Damage Biomass Losses. Removes organs by crown damage class; transfers to litter.	Biogeochem	Intermediate
84	Herbivory/Grazing Losses. Removes leaf biomass by PFT palatability and land use label.	Biogeochem	Empirical

Table 29: FATES phenology, allometry, and carbon allocation (26 entries).

F.8 Recruitment, Cohort & Patch Dynamics

#	Process	Family	Fidelity
85	Seed Production. C flux from reproductive allocation to site-level seed bank.	Ecology	Intermediate
86	Seed Germination. Fraction of seed bank germinating per time step; PFT-specific.	Ecology	Empirical
87	Seed Decay. Background mortality/decomposition of seeds.	Ecology	Empirical
88	Seed Dispersal Kernels. Spatial dispersal: exponential, exponential-power, log-sech PDFs.	Ecology	Intermediate
89	Recruitment (New Cohort Creation). Creates cohorts from germinated seeds; allometrically-consistent initialization.	Ecology	Intermediate
90	Cohort Fusion. Merges similar cohorts (PFT, size, damage class); weighted averaging of state variables.	Ecology	Intermediate
91	Cohort Termination. Removes cohorts below minimum density or size thresholds; transfers biomass to litter.	Ecology	Intermediate
92	Cohort Damage Recovery. Transitions damaged cohorts to lower damage classes over time.	Ecology	Empirical
93	Disturbance Rate Calculation. Patch-level disturbance rates from fire, treefall, logging, LUC.	Ecology	Intermediate
94	Patch Spawning (Disturbance). Creates new patches from disturbance events; partitions cohorts.	Ecology	Intermediate
95	Patch Fusion. Merges similar-aged patches; weighted averaging of all state variables.	Ecology	Intermediate
96	Patch Litter Redistribution. Distributes litter between newly created and existing patches.	Ecology	Intermediate

#	Process	Family	Fidelity
97	Canopy Layer Structure. Assigns cohorts to overstory/understory by height ranking and crown area.	Ecology	Intermediate
98	Canopy Promotion/Demotion. Moves cohorts between canopy layers when layer areas change.	Ecology	Intermediate
99	Canopy Spread Factor. Diameter-dependent crown area spread differentiating open-grown vs. closed-canopy.	Ecology	Empirical
100	Crown Damage Scheduling. Event-based crown damage application; assigns damage classes via lookup.	Ecology	Empirical

Table 30: FATES recruitment, cohort, and patch dynamics (16 entries).

F.9 Litter, Soil Interface, & Human Systems

#	Process	Family	Fidelity
101	Litter Pool Dynamics. Leaf litter (by PFT decomposition class), 4 CWD size classes, seed bank per patch; C/N/P tracking.	Biogeochem	Intermediate
102	CWD Size-Class Adjustment. Adjusts CWD fractions based on contributing tree DBH distribution (SPITFIRE-linked).	Biogeochem	Empirical
103	Fragmentation/Decomposition Scalar. CENTURY-model temperature + moisture scalar for litter fragmentation.	Biogeochem	Intermediate
104	Litter-to-Soil C & Nutrient Flux. Fragmented litter C/N/P to host soil BGC with exponential depth partitioning.	Biogeochem	Intermediate
105	Plant Nutrient Acquisition (N, P). Michaelis-Menten or ECA (Equilibrium Chemistry Approximation) uptake kinetics.	Biogeochem	Intermediate
106	Exudation / Root Efflux. Excess C exuded from roots into soil; labile C input to decomposition.	Biogeochem	Empirical
107	CH ₄ Boundary Conditions. Methane module boundary conditions: root fraction, photosynthetic flux, soil decomposition inputs.	Biogeochem	Empirical
108	Selective Logging / Harvest (Area-Based). Removes trees within DBH bounds at specified harvest rate per unit area.	Human Sys	Empirical
109	Harvest by Carbon Target. Alternative harvest mode targeting specified C removal.	Human Sys	Empirical
110	Harvest Debt Tracking. Tracks unfulfilled harvest demand; debt carried forward.	Human Sys	Empirical

#	Process	Family	Fidelity
111	Logging Litter Fluxes. Partitions logged tree biomass into commercial products (exported) and slash/litter (on-site CWD).	Human Sys	Empirical
112	Land Use Transition Matrices. LUH2 transition rates between primary/secondary forest, cropland, pasture, rangeland.	Human Sys	Empirical
113	Grazing / Browsing. Leaf biomass removal by land use label, PFT palatability, and plant height.	Human Sys	Empirical
114	Prescribed Fire Management. Decision logic and execution of managed burns within weather windows.	Human Sys	Empirical

Table 31: FATES litter/soil interface and human systems (14 entries).

F.10 Integration & Numerical Methods

#	Process	Family	Fidelity
115	Ecosystem Dynamics Driver. Top-level daily driver: phenology → allocation → mortality → disturbance → recruitment → structure.	Ecology	—
116	State Variable Integration. Integrates cohort-level growth rates (DBH, biomass pools) through daily time step.	Ecology	—
117	Mass/Energy Balance Checking. Comprehensive balance checks for C, N, P, and water; reports imbalances.	Ecology	—
118	Flux Accumulation. Accumulates sub-daily fluxes (photosynthesis, respiration, transpiration) to daily totals.	Ecology	—
119	Runge-Kutta-Fehlberg (RKF45) Integrator. Adaptive-step 4th/5th order ODE integrator with error control.	Ecology	—
120	Euler Integrator. Simple forward Euler as fallback for stiff/simple systems.	Ecology	—

Table 32: FATES integration and numerical methods (6 entries).

The expanded FATES catalog totals **90 process entries** across 7 process families (Radiation 4, Biogeochem 30, Hydrology 8, Fire 23, Ecology 35, Human Systems 14, plus 6 integration/numerical entries), drawn from 29 source files — more than doubling the original 42-entry catalog.

G iLand — Individual-Based Forest Landscape Model

#	Process	Family	Fidelity
1	Individual Tree Growth (3-PG variant). Light-use efficiency GPP with VPD/temp/soil water modifiers per tree.	Ecology	Intermediate
2	Light Competition. Beer-Lambert light interception on 2m height cells across the landscape.	Radiation	Intermediate
3	Establishment / Regeneration. Seed dispersal kernels, seedling light/moisture filtering, browsing.	Ecology	Intermediate
4	Mortality (Carbon Starvation). Death when C reserves deplete below threshold.	Ecology	Intermediate
5	Mortality (Stress/Intrinsic). Age-dependent and stress-based background mortality.	Ecology	Empirical
6	Phenology. Day-length and temperature-controlled leaf onset/offset.	Ecology	Empirical
7	Decomposition / Soil C-N (ICBM/2N). Cohort-based SOM decomposition with N cycling.	Biogeochem	Intermediate
8	Snag Dynamics. Standing dead tree decay, fall probability, CWD generation.	Biogeochem	Intermediate
9	Sapling Growth Module. Height-based sapling cohorts before individual tree transition.	Ecology	Intermediate
10	Water Cycle. Canopy interception, snowmelt, evapotranspiration, percolation, groundwater recharge.	Hydrology	Intermediate
11	Fire Disturbance Module. Stochastic fire ignition, spread probability, severity per species.	Fire	Empirical
12	Wind Disturbance. Wind damage probability based on tree height, species, and exposure.	Ecology	Empirical
13	Bark Beetle Disturbance. Beetle population dynamics, host selection, outbreak triggers.	Ecology	Intermediate
14	Forest Management. Thinning, clear-cut, shelter-wood prescriptions with scheduling.	Human Sys	Intermediate
15	Seed Dispersal. Species-specific dispersal kernels (exponential, fat-tailed).	Ecology	Intermediate
16	Browsing / Ungulate Herbivory. Height-dependent browse damage on seedlings/saplings.	Ecology	Empirical
17	Soil Water Bucket Model. Multi-layer simplified soil water balance.	Hydrology	Empirical

#	Process	Family	Fidelity
18	Permafrost / Soil Temperature. Simplified soil thermal regime for boreal forests.	Cryosphere	Empirical
19	Carbon Allocation. NPP partitioning to foliage, wood, fine roots, reserves.	Biogeochem	Intermediate
20	Species-Specific Allometry. Height-diameter, crown dimension, and biomass scaling.	Ecology	Empirical
21	Landscape Connectivity. Seed rain from neighboring cells; gene flow proxy.	Ecology	Intermediate

Table 33: iLand process catalog (21 entries). Primary families: Ecology (14), Biogeochem (4), Hydrology (2), Fire (1).

H Landlab — Earth Surface Dynamics Toolkit

Landlab provides a modular Python toolkit with 55+ components for geomorphology, hydrology, and landscape evolution research.

#	Process	Family	Fidelity
1	FlowAccumulator. Multi-algorithm flow routing and drainage area accumulation (D8, D4, Dinf).	Hydrology	Empirical
2	FlowDirectorSteepest / D8. Single-direction steepest-descent flow routing.	Hydrology	Empirical
3	FlowDirectorDinf. Divergent flow routing (Tarboton’s D-infinity).	Hydrology	Intermediate
4	FlowDirectorMFD. Multiple-flow-direction proportional routing.	Hydrology	Intermediate
5	DepressionFinderAndRouter. Pit filling and lake-over-flow routing.	Hydrology	Empirical
6	LakeMapperBarnes. Efficient lake delineation and fill (Barnes et al.).	Hydrology	Empirical
7	StreamPowerEroder. Detachment-limited stream power incision: $E = KA^m S^n$.	Geomorphology	Empirical
8	FastscapeEroder. Implicit $O(n)$ stream power (Braun & Willett 2013).	Geomorphology	Empirical
9	SedDepEroder. Transport-limited fluvial erosion with deposition.	Geomorphology	Empirical
10	ErosionDeposition. Simultaneous erosion and deposition (Davy & Lague 2009).	Geomorphology	Intermediate
11	Space (Stream Power with Alluvium Conservation). Bedrock + alluvium dual-layer incision.	Geomorphology	Intermediate
12	SpaceLargeScaleEroder. Landscape-scale SPACE variant.	Geomorphology	Intermediate
13	LinearDiffuser. $\partial z / \partial t = D \nabla^2 z$ hillslope creep.	Geomorphology	Empirical
14	TaylorNonLinearDiffuser. Critical-slope non-linear diffusion.	Geomorphology	Empirical
15	DepthDependentDiffuser. Soil-thickness-dependent creep transport.	Geomorphology	Intermediate
16	DepthDependentTaylorDiffuser. Combined depth-dependent + non-linear diffusion.	Geomorphology	Intermediate

#	Process	Family	Fidelity
17	TransportLengthHillslopeDiffuser. Detachment-transport hillslope model.	Geomorphology	Intermediate
18	LandslideComponent. Shallow landslide initiation via infinite slope stability.	Geomorphology	Intermediate
19	LandslideProbability. Probabilistic landslide hazard from soil/topo/water.	Geomorphology	Intermediate
20	BedrockLandslider. Deep-seated bedrock landslide model.	Geomorphology	Intermediate
21	OverlandFlow (de Almeida / Bates). 2D shallow water equations for surface flow.	Hydrology	Physics-based
22	KinwaveOverlandFlowModel. Kinematic wave overland flow routing.	Hydrology	Intermediate
23	GroundwaterDupuitPercolator. 2D Boussinesq unconfined aquifer flow.	Hydrology	Physics-based
24	SoilInfiltrationGreenAmpt. Green-Ampt infiltration model.	Hydrology	Physics-based
25	Radiation. Topographic solar radiation (slope, aspect, shadowing).	Radiation	Physics-based
26	PotentialEvapotranspiration. ET estimation from radiation and meteorology.	Hydrology	Intermediate
27	SoilMoisture. 1D soil water balance with leaky-bucket dynamics.	Hydrology	Empirical
28	Vegetation (cellular automaton). Stochastic vegetation dynamics on gridded landscapes.	Ecology	Empirical
29	VegCA. Cellular automaton for dryland vegetation pattern formation.	Ecology	Empirical
30	SpeciesEvolver. Tracking lineage diversification and phylogeography on evolving landscapes.	Evolution	Intermediate
31	Flexure. 2D lithospheric flexure under surface loads.	Geology	Physics-based
32	TectonicFiniteSketch. Tectonic deformation and faulting.	Geology	Intermediate
33	NormalFault. Normal fault displacement and scarp generation.	Geology	Intermediate
34	ListricKinematicExtender. Listric fault deformation.	Geology	Intermediate
35	SinkFiller. Iterative depression filling to create drainage-enforced DEMs.	Hydrology	Empirical

#	Process	Family	Fidelity
36	ChannelProfiler. Longitudinal channel profile extraction and analysis.	Geomorphology	Empirical
37	ChiFinder. Chi-elevation analysis for divide migration.	Geomorphology	Empirical
38	DrainageDensity. Drainage network density computation.	Geomorphology	Empirical
39	SteepnessFinder. Channel steepness index (k_{sn}) computation.	Geomorphology	Empirical
40	Weathering (Exponential). Soil production from bedrock as $f(H)$.	Geomorphology	Empirical
41	ExponentialWeathererIntegrated. Integrated soil production over time.	Geomorphology	Empirical
42	AreaSlopeTransporter. Area-slope sediment transport law.	Geomorphology	Empirical
43	GravelBedrockEroder. Gravel abrasion on bedrock channels.	Geomorphology	Intermediate
44	GravelRiverTransporter. Gravel transport capacity and downstream fining.	Geomorphology	Intermediate
45	NetworkSedimentTransporter. Parcel-based sediment routing on channel networks.	Geomorphology	Intermediate
46	PriorityFloodFlowRouter. Priority-flood-based ε -fill and route.	Hydrology	Empirical
47	HackCalculator. Hack’s law stream length-area analysis.	Geomorphology	Empirical
48	Profiler / TrickleDownProfiler. Channel profile analysis tools.	Geomorphology	Empirical
49	ErosionModel (Terrainbento). Configurable multi-process landscape evolution driver.	Geomorphology	Intermediate

Table 34: Landlab component catalog (49 entries). Primary families: Geomorphology (29), Hydrology (16), Geology (4), Ecology (2), Radiation (1), Evolution (1).

I LPJ-GUESS — Dynamic Global Vegetation Model

#	Process	Family	Fidelity
1	Photosynthesis (Haxeltine & Prentice / Collatz). Coupled photosynthesis–stomatal conductance.	Biogeochem	Physics-based
2	Autotrophic Respiration. Maintenance + growth respiration from N content and temperature.	Biogeochem	Intermediate
3	Vegetation Dynamics. Establishment, mortality, competition for light by cohorts.	Ecology	Intermediate
4	Bioclimatic Limits. PFT survival and establishment bioclimatic envelopes.	Ecology	Empirical
5	Phenology. Summergreen/raingreen/evergreen phenological strategies.	Ecology	Empirical
6	Carbon Allocation. NPP partitioning across leaves, sapwood, heartwood, fine roots.	Biogeochem	Intermediate
7	Fire (BLAZE). Process-based fire model with fuel moisture and fire weather.	Fire	Intermediate
8	Fire (SIMFIRE). Statistical global fire model based on productivity and climate.	Fire	Empirical
9	Soil Organic Matter (Century-based). Multi-pool decomposition with C-N coupling.	Biogeochem	Intermediate
10	Nitrogen Cycling. N mineralization, plant N uptake, N limitation of growth, symbiotic fixation.	Biogeochem	Intermediate
11	Methane Emissions (Wetlands). CH ₄ production and emission from wetland soils.	Biogeochem	Intermediate
12	Hydrology (2-Layer Bucket). Percolation, runoff, snow, evapotranspiration.	Hydrology	Empirical
13	Crop Module (CFT). Managed crop PFTs with sowing, harvest, fertilization for 15 crop types.	Human Sys	Intermediate
14	Managed Grassland / Pasture. Mowing and grazing management.	Human Sys	Empirical
15	Peatland Module. Peat accumulation, decomposition, and methane dynamics.	Biogeochem	Intermediate
16	Land Use Change. Deforestation, afforestation, crop expansion/abandonment transitions.	Human Sys	Intermediate
17	BVOC Emissions. Isoprene and monoterpene emissions from vegetation.	Biogeochem	Empirical

#	Process	Family	Fidelity
18	Soil Thermal (Permafrost). Multi-layer soil temperature and freeze-thaw dynamics for permafrost regions.	Cryosphere	Intermediate

Table 35: LPJ-GUESS process catalog (18 entries). Primary families: Biogeochem (8), Ecology (4), Human Sys (3), Fire (2), Hydrology (1).

J Noah-MP — Multi-Physics Land Surface Model

Noah-MP’s defining feature is offering multiple scheme options for each physical process, enabling systematic evaluation of structural uncertainty.

#	Process	Family	Fidelity
1	Vegetation Canopy Radiative Transfer (2-stream $\times$ gap). Modified two-stream with canopy gap probability.	Radiation	Physics-based
2	Snow Albedo — CLASS. Canadian Land Surface Scheme snow albedo (aging-based decay).	Radiation	Empirical
3	Snow Albedo — BATS. Biosphere-Atmosphere Transfer Scheme grain-size-dependent albedo.	Radiation	Empirical
4	Surface Exchange (M-O $\times$ Noah $\times$ Chen97). Monin-Obukhov with multiple roughness formulations.	Atmosphere	Intermediate
5	Stomatal Conductance — Ball-Berry. Humidity-based empirical stomatal model.	Biogeochem	Empirical
6	Stomatal Conductance — Jarvis. Multiplicative environmental stress factors on g_{max} .	Biogeochem	Empirical
7	Photosynthesis (Farquhar C3/C4). Farquhar biochemistry for carbon assimilation.	Biogeochem	Physics-based
8	Multi-Layer Snow Model. Up to 3 snow layers with compaction, melt, refreezing.	Cryosphere	Physics-based
9	Snow Liquid Water — Gravitational. Simple gravitational drainage.	Cryosphere	Empirical
10	Snow Liquid Water — SNTHERM. Snow thermal model with grain-size metamorphism.	Cryosphere	Physics-based
11	Soil Moisture — Richards Equation. Multi-layer unsaturated/saturated flow.	Hydrology	Physics-based
12	Runoff — TOPMODEL. Saturation-excess runoff with topographic index.	Hydrology	Intermediate
13	Runoff — SIMGM. Simple groundwater model with water table.	Hydrology	Intermediate
14	Runoff — XAJ (Xinjiang). Saturation-excess based on variable infiltration capacity.	Hydrology	Empirical
15	Runoff — Dynamic VIC. Variable infiltration capacity with dynamic parameterization.	Hydrology	Intermediate
16	Runoff — Free Drainage. Gravity-only lower boundary condition.	Hydrology	Empirical

#	Process	Family	Fidelity
17	Surface Resistance — Sakaguchi/Zeng. Soil evaporation resistance parameterization.	Hydrology	Empirical
18	Surface Resistance — Sellers. Vegetation-dependent surface resistance.	Hydrology	Empirical
19	Infiltration — Philip. Philip’s infiltration equation.	Hydrology	Physics-based
20	Infiltration — Green-Ampt. Wetting front-based infiltration.	Hydrology	Physics-based
21	Infiltration — Smith-Parlange. Smith-Parlange flux-based infiltration.	Hydrology	Physics-based
22	Tile Drainage. Agricultural subsurface drainage parameterization.	Hydrology	Intermediate
23	Canopy Interception. Precipitation interception, throughfall, drip from vegetation.	Hydrology	Intermediate
24	Soil Temperature. Multi-layer heat diffusion with phase change.	Hydrology	Physics-based
25	Frozen Soil Permeability. Ice-content-dependent hydraulic conductivity reduction.	Cryosphere	Intermediate
26	Glacier Model. Simple glacier ice accumulation and melt.	Cryosphere	Empirical
27	Carbon Model (Simple). Prognostic vegetation and soil carbon pools.	Biogeochem	Intermediate
28	Crop Module. Planting, phenology, harvest modules for agricultural PFTs.	Human Sys	Intermediate
29	Irrigation. Demand-based irrigation from soil moisture deficit.	Human Sys	Empirical
30	Urban Model. Simple single-layer urban canopy energy balance.	Human Sys	Empirical
31	Groundwater / Aquifer. Unconfined aquifer storage and exchange with soil column.	Hydrology	Intermediate
32	LAI Prescription / Phenology. Satellite-based or model-predicted LAI seasonal cycle.	Ecology	Empirical

Table 36: Noah-MP process catalog (32 entries). Multi-physics options yield 70 distinct parameterization combinations.

K ParFlow — Integrated Hydrological Model

ParFlow solves 3D variably-saturated groundwater flow (Richards equation) coupled to overland flow (kinematic/diffusion wave) on structured grids with terrain-following coordinates.

#	Process	Family	Fidelity
1	3D Richards Equation Solver. Variably-saturated subsurface flow with Newton-Krylov nonlinear solvers.	Hydrology	Physics-based
2	Van Genuchten Retention. Soil water retention and relative permeability curves.	Hydrology	Physics-based
3	Haverkamp Retention. Alternative water retention model.	Hydrology	Physics-based
4	Brooks-Corey Retention. Power-law retention curves.	Hydrology	Empirical
5	Overland Flow (Kinematic Wave). Manning’s equation surface runoff routing.	Hydrology	Intermediate
6	Overland Flow (Diffusive Wave). Diffusion-wave approximation for 2D surface flow.	Hydrology	Physics-based
7	Terrain-Following Grid (TFG). Topography-conformal mesh for hillslope/valley geometry.	Hydrology	Physics-based
8	CLM Coupling (Land Surface). Surface energy balance and ET via coupled CLM.	Hydrology	Physics-based
9	Wells (Injection/Extraction). Point source/sink well boundary conditions.	Hydrology	Physics-based
10	Reservoir / Lake Boundary. Surface water body storage boundary conditions.	Hydrology	Intermediate
11	Heterogeneous Permeability Fields. Geostatistical (turning bands) random field generation.	Geology	Intermediate
12	Indicator Geostatistics. Categorical geological facies simulation.	Geology	Intermediate
13	Constant / PFB Flux Boundaries. Specified flux and pressure boundary conditions.	Hydrology	Physics-based
14	Evapotranspiration (via CLM). Richards-CLM computed ET from root water uptake.	Hydrology	Physics-based
15	Snow Model (via CLM). Energy-balance multi-layer snow coupled to subsurface.	Cryosphere	Physics-based
16	Multi-GPU / CUDA Acceleration. GPU-accelerated Richards solver.	Hydrology	Physics-based

Table 37: ParFlow process catalog (16 entries). Primary family: Hydrology (14), Geology (2).

L WRF/WRF-SFIRE — Weather & Wildfire Modeling

WRF-SFIRE couples the Weather Research and Forecasting model with a fire spread model, enabling two-way atmosphere–wildfire interaction.

L.1 WRF Atmospheric Physics

#	Process	Family	Fidelity
1	ARW Dynamical Core. Fully compressible non-hydrostatic Euler equations on Arakawa C-grid, Runge-Kutta 3rd order.	Atmosphere	Physics-based
2	PBL — YSU. Yonsei University non-local K-profile boundary layer scheme.	Atmosphere	Intermediate
3	PBL — MYJ. Mellor-Yamada-Janjić 2.5-level local TKE closure.	Atmosphere	Intermediate
4	PBL — MYNN. Mellor-Yamada-Nakanishi-Niino 2.5/3.0 level closure.	Atmosphere	Intermediate
5	PBL — ACM2. Asymmetric Convective Model with non-local + local mixing.	Atmosphere	Intermediate
6	PBL — TEMF. Total Energy Mass Flux eddy-diffusivity mass-flux scheme.	Atmosphere	Intermediate
7	Microphysics — WSM3/5/6. WRF Single-Moment 3/5/6-class schemes (ice, snow, graupel).	Atmosphere	Intermediate
8	Microphysics — WDM5/6. WRF Double-Moment 5/6-class schemes.	Atmosphere	Physics-based
9	Microphysics — Thompson. Aerosol-aware mixed-phase bulk microphysics.	Atmosphere	Physics-based
10	Microphysics — Morrison 2-Moment. Full 2-moment for cloud, rain, ice, snow, graupel.	Atmosphere	Physics-based
11	Microphysics — Lin-Purdue. Lin et al. ice-phase microphysics.	Atmosphere	Intermediate
12	Microphysics — Kessler. Warm-rain autoconversion scheme.	Atmosphere	Empirical
13	Radiation — RRTM/RRTMG. Correlated-k longwave radiation for WRF.	Radiation	Physics-based
14	Radiation — Dudhia Shortwave. Simple downward integration shortwave scheme.	Radiation	Empirical
15	Radiation — Goddard Shortwave. Multi-band shortwave with aerosol/cloud interaction.	Radiation	Intermediate
16	Radiation — CAM. CAM radiation package within WRF.	Radiation	Physics-based

#	Process	Family	Fidelity
17	Cumulus — Kain-Fritsch. Mass-flux convective parameterization with CAPE removal closure.	Atmosphere	Intermediate
18	Cumulus — Betts-Miller-Janjić. Convective adjustment toward observed profiles.	Atmosphere	Empirical
19	Cumulus — Grell-Devenyi/Grell-Freitas. Ensemble/stochastic mass-flux convection.	Atmosphere	Intermediate
20	Cumulus — Tiedtke. Mass-flux with organized entrainment/detrainment.	Atmosphere	Intermediate
21	Land Surface — Noah. 4-layer soil T/moisture, urban, vegetation, snow.	Hydrology	Intermediate
22	Land Surface — Noah-MP. Multi-physics land surface (see Section J).	Hydrology	Intermediate
23	Land Surface — RUC. Rapid Update Cycle 6-layer soil/hydro/thermal model.	Hydrology	Intermediate
24	Land Surface — CLM. Community Land Model coupling within WRF.	Hydrology	Physics-based
25	Surface Layer — Revised MM5. Monin-Obukhov surface flux parameterization.	Atmosphere	Intermediate
26	Surface Layer — Eta. Janjić Eta model similarity theory.	Atmosphere	Intermediate

Table 38: WRF atmospheric physics (26 entries).

L.2 Fire & Chemistry

#	Process	Family	Fidelity
27	Level-Set Fire Spread. Narrow-band level set tracking of fire front with sub-grid refinement.	Fire	Physics-based
28	Rothermel Fire Spread Rate. Semi-empirical surface fire rate-of-spread from fuel, moisture, wind, slope.	Fire	Intermediate
29	Balbi Fire Spread Rate. Radiation-convection fire spread model (alternative to Rothermel).	Fire	Physics-based
30	13 Anderson Fuel Categories. NFDRS fuel model classification for fire behavior.	Fire	Empirical
31	Canopy Fire / Crown Fire. Vertical fire propagation from surface to crown.	Fire	Intermediate
32	Fire Emissions & Smoke Transport. Trace gas and particulate emissions from combustion; smoke plume advection.	Fire	Intermediate

#	Process	Family	Fidelity
33	Fire–Atmosphere Coupling (Two-Way). Sensible/latent heat flux from fire $\rightarrow$ atmosphere; wind $\rightarrow$ fire spread rate feedback.	Fire	Physics-based
34	WRF-Chem Gas-Phase Chemistry. RADM2, RACM, CB05 chemical mechanisms for tropospheric O ₃ , VOC, NO _x .	Atmosphere	Physics-based
35	WRF-Chem Aerosol (MADE/GOCART/MOSAIC). Modal/sectional aerosol microphysics/chemistry.	Atmosphere	Intermediate
36	WRF-Chem Photolysis (Fast-J/TUV). Online photolysis rate computation.	Atmosphere	Physics-based

Table 39: WRF-SFIRE fire and chemistry processes (10 entries).

L.3 WRF-Hydro / National Water Model

#	Process	Family	Fidelity
37	Overland Flow Routing. Diffusive wave routing on high-resolution terrain grids.	Hydrology	Physics-based
38	Channel Routing (Muskingum-Cunge). Kinematic-diffusive channel routing for river networks.	Hydrology	Intermediate
39	Subsurface Lateral Flow. Saturated lateral flow down topographic gradients.	Hydrology	Intermediate
40	Reservoir Routing. Level-pool reservoir routing with operating rules.	Human Sys	Intermediate
41	Lake Routing. Conceptual lake storage/outflow model.	Hydrology	Empirical
42	Bucket Model Groundwater. Conceptual exponential-decay groundwater baseflow.	Hydrology	Empirical
43	UDMAP Disaggregation. Upscale Noah-MP/CLM land surface fields to high-resolution routing grid.	Hydrology	Intermediate

Table 40: WRF-Hydro / NWM processes (7 entries).

M ORCHIDEE — Land Surface Model

ORCHIDEE 2.0 couples surface energy balance (SECHIBA) with vegetation dynamics (STOMATE), derived from LPJ.

M.1 SECHIBA (Surface Energy Balance)

#	Process	Family	Fidelity
1	Surface Energy Balance. Coupled radiation/turbulent flux solution for canopy and soil.	Atmosphere	Physics-based
2	11-Layer Richards Soil Hydrology (CWRR). Van Genuchten parameterized Richards equation solver.	Hydrology	Physics-based
3	Choisnel 2-Layer Bucket Hydrology. Simple bucket model alternative.	Hydrology	Empirical
4	Multi-Layer Snow Model. Explicit snow with compaction and thermal conduction.	Cryosphere	Physics-based
5	Soil Thermodynamics. Multi-layer soil temperature with freeze/thaw.	Hydrology	Physics-based
6	River Routing with Floodplains. Global routing scheme with floodplain inundation.	Hydrology	Intermediate
7	Canopy Interception & Throughfall. Precipitation partitioning by vegetation.	Hydrology	Intermediate
8	Albedo / Roughness. Surface optical/aerodynamic properties from PFT and soil/snow.	Radiation	Intermediate
9	Penman-Monteith Evapotranspiration. Physical ET computation with stomatal control.	Hydrology	Physics-based
10	BVOC Chemistry. Biogenic volatile organic compound emissions.	Biogeochem	Empirical
11	Diffuse Radiation Partitioning. Splitting of direct/diffuse radiation for canopy absorption.	Radiation	Intermediate

Table 41: ORCHIDEE SECHIBA processes (11 entries).

M.2 STOMATE (Vegetation Dynamics & Carbon)

#	Process	Family	Fidelity
12	Photosynthesis (Farquhar C3 / Collatz C4). Leaf-level carbon assimilation.	Biogeochem	Physics-based
13	Stomatal Conductance (Ball-Berry). Humidity-based stomatal regulation.	Biogeochem	Empirical

#	Process	Family	Fidelity
14	Autotrophic Respiration. Maintenance + growth respiration.	Biogeochem	Intermediate
15	Carbon Allocation. NPP partitioning to plant compartments.	Biogeochem	Intermediate
16	Phenology — Leaf Onset/Senescence. Temperature, moisture, and light triggers.	Ecology	Empirical
17	Vegetation Establishment. PFT-based establishment from climate suitability.	Ecology	Intermediate
18	Vegetation Mortality. Self-thinning, heat, and drought-driven mortality.	Ecology	Intermediate
19	Light Competition. Fractional coverage competition among PFTs.	Ecology	Intermediate
20	Fire (LPJ-derived). Litter-moisture and climate-dependent fire occurrence/spread.	Fire	Empirical
21	Litter & Soil Carbon (Century-based). Multi-pool SOM decomposition.	Biogeochem	Intermediate
22	Nitrogen Cycling. N mineralization, immobilization, plant uptake.	Biogeochem	Intermediate
23	Land Cover Change. PFT area transitions from deforestation/afforestation.	Human Sys	Intermediate
24	Wood Harvest. Harvested wood product pools.	Human Sys	Empirical
25	Grassland Management. Mowing and grazing.	Human Sys	Empirical
26	Crop Module. Agricultural PFT phenology and harvest.	Human Sys	Intermediate

Table 42: ORCHIDEE STOMATE processes (15 entries).

N VIC — Variable Infiltration Capacity Model

#	Process	Family	Fidelity
1	Variable Infiltration Capacity Runoff. Nonlinear soil moisture-storage-capacity distribution.	Hydrology	Intermediate
2	Multi-Layer Soil Temperature. Heat conduction with freeze/thaw through 3+ soil layers.	Hydrology	Physics-based
3	Frozen Soil Permafrost. Ice lens formation and thermal conductivity modification.	Cryosphere	Physics-based
4	Energy-Balance Snow Model (2-Layer). Mass and energy balance with liquid water retention.	Cryosphere	Physics-based

#	Process	Family	Fidelity
5	Blowing Snow Sublimation. Wind-transported snow sublimation loss.	Cryosphere	Intermediate
6	Penman-Monteith ET. Physically-based evapotranspiration.	Hydrology	Physics-based
7	Canopy Interception. Canopy water storage and throughfall partitioning.	Hydrology	Intermediate
8	Farquhar/Collatz Photosynthesis. Optional biochemical carbon assimilation.	Biogeochem	Physics-based
9	Soil Carbon Balance. Simple 3-pool soil carbon model with temperature/moisture controls.	Biogeochem	Intermediate
10	Lake Energy Balance with Ice. 1-D lake thermal model with ice formation.	Hydrology	Intermediate
11	Wetland Model. Saturated fraction and wetland water balance.	Hydrology	Empirical
12	Baseflow (ARNO). Non-linear baseflow recession from lower soil zone.	Hydrology	Empirical
13	Snow Albedo (Age-Dependent). Exponential decay of snow albedo with age.	Radiation	Empirical
14	Vegetation Library (12 Classes). Land cover properties for each grid cell tile.	Ecology	Empirical
15	Sub-Grid Elevation Bands. Snow/temperature processing by elevation band for mountainous terrain.	Cryosphere	Intermediate
16	Canopy Energy Balance. Full canopy energy budget with longwave/shortwave/turbulent fluxes.	Atmosphere	Physics-based
17	MTCLIM Meteorological Disaggregation. Daily-to-subdaily met forcing disaggregation.	Atmosphere	Empirical

Table 43: VIC process catalog (17 entries). Primary families: Hydrology (7), Cryosphere (5), Biogeochem (2), Atmosphere (2).

O NOAA-GFDL ESM4 — Component Overview

GFDL ESM4 submodules were not initialized in the reference clone; the catalog below is derived from `.gitmodules` and published model descriptions.

#	Process	Family	Fidelity
1	FV3 Dynamical Core. Finite-volume cubed-sphere non-hydrostatic atmosphere.	Atmosphere	Physics-based

#	Process	Family	Fidelity
2	GFDL Atmospheric Physics. Radiation, convection, cloud microphysics, PBL, gravity waves.	Atmosphere	Physics-based
3	AM4 Aerosol/Chemistry. Interactive tropospheric/stratospheric chemistry with aerosol.	Atmosphere	Physics-based
4	MOM6 Ocean. Modular Ocean Model v6 (see Table 17).	Ocean	Physics-based
5	SIS2 Sea Ice. GFDL sea ice model with multi-category thermodynamics and dynamics.	Cryosphere	Physics-based
6	LM4 Land Model. Cohort-based land with soil C-N, plant hydraulics, and river routing.	Ecology	Physics-based
7	COBALT Ocean BGC. Carbon-Ocean Biogeochemistry and Lower Trophics marine ecosystem.	Biogeochem	Physics-based
8	FMS Infrastructure. Flexible Modeling System I/O, diagnostics, domain decomposition.	Atmosphere	N/A
9	Ice Parameter Module. Sea-ice parameterization library.	Cryosphere	Intermediate

Table 44: GFDL ESM4 components (9 entries; submodules not populated).

P PEcAn — Predictive Ecosystem Analyzer

PEcAn is a scientific workflow system that wraps 20+ external ecosystem models with Bayesian calibration, uncertainty quantification, and benchmarking.

#	Process	Family	Fidelity
1	ED2 Model Wrapper. Ecosystem Demography model integration.	Ecology	Physics-based
2	SIPNET Wrapper. Simplified photosynthesis + evapotranspiration model.	Biogeochem	Empirical
3	BioCro Wrapper. Bioenergy crop growth model (C4 grasses, Miscanthus).	Ecology	Intermediate
4	CLM4.5 Wrapper. Community Land Model integration.	Ecology	Physics-based
5	FATES Wrapper. FATES vegetation demography integration.	Ecology	Physics-based
6	JULES Wrapper. Joint UK Land Environment Simulator.	Ecology	Physics-based
7	LPJ-GUESS Wrapper. Dynamic vegetation model integration.	Ecology	Intermediate
8	DALEC Wrapper. Data Assimilation Linked Ecosystem Carbon model.	Biogeochem	Empirical

#	Process	Family	Fidelity
9	LINKAGES Wrapper. Forest succession model.	Ecology	Intermediate
10	MAESPA Wrapper. Multilayer tree canopy radiation + photosynthesis + water balance.	Ecology	Physics-based
11	RothC Wrapper. Rothamsted soil carbon turnover model.	Biogeochem	Empirical
12	Bayesian Photosynthesis Calibration. Hierarchical Bayes inversion for V_{cmax} , J_{max} from leaf gas exchange.	Biogeochem	Physics-based
13	Radiative Transfer Model Inversion. PROSPECT/SAIL inversion for leaf/canopy traits from spectral data.	Radiation	Physics-based
14	Allometry Module. Tree diameter→height/biomass/crown allometric relationships.	Ecology	Empirical
15	Met Processing (CF Standards). Meteorological data ingestion, gap-filling, and downscaling.	Atmosphere	Intermediate
16	Data Assimilation (State/Parameter). Ensemble Kalman filter and particle filter for state updating.	Ecology	Physics-based
17	Sensitivity Analysis (Morris/Sobol). Global sensitivity for model parameters.	Ecology	Physics-based
18	Uncertainty Quantification. Monte Carlo uncertainty propagation with Latin hypercube sampling.	Ecology	Physics-based
19	Benchmarking Module. Model-data comparison scoring against AmeriFlux, MODIS, forest inventory.	Ecology	Intermediate
20	Gaussian Process Emulator. Surrogate model construction for expensive simulations.	Ecology	Physics-based

Table 45: PEcAn workflow and model wrapper catalog (20 entries).

Q LANDIS-II Core / Core-Model-v6

#	Process	Family	Fidelity
1	Succession Extension Interface. Base class for forest succession state variables and dynamics.	Ecology	Intermediate
2	Species / Cohort Data Model. Species-level parameterization and cohort tracking per site.	Ecology	Intermediate
3	Ecoregion Framework. Spatially indexed ecological region properties.	Ecology	Empirical
4	Disturbance Extension Interface. Plugin architecture for fire, wind, insects, harvest.	Ecology	Intermediate

#	Process	Family	Fidelity
5	Seed Dispersal. Species-specific effective and maximum dispersal distances.	Ecology	Intermediate
6	Site-Level Establishment. Probabilistic establishment filtering per species.	Ecology	Intermediate

Table 46: LANDIS-II Core framework (6 entries). Extensions (NECN, PnET, SCRPPLE, etc.) provide full process models.

R JULES — Joint UK Land Environment Simulator

JULES (Best et al., 2011; Clark et al., 2011) is the community land surface model of the UK Met Office, used as the land component of UKESM1 and HadGEM3. It simulates energy balance, hydrology, snow, vegetation dynamics, carbon and nitrogen cycling, fire, urban surfaces, and crop growth on a regular grid. JULES is notable for its multi-layer snow scheme, TRIFFID dynamic vegetation, and INFERNO fire model.

#	Process	Family	Fidelity
1	Surface Energy Balance. Coupled solution of sensible, latent, ground, and radiative fluxes via Penman–Monteith on tiles.	Atmosphere	Physics-based
2	Tile-Based Land Surface. Sub-grid heterogeneity via 9 surface types (broadleaf, needleleaf, C3/C4 grass, shrub, urban, lake, soil, ice) with independent energy/water budgets.	Hydrology	Intermediate
3	Multi-Layer Snow Scheme. Up to 3-layer prognostic snow with compaction, grain growth, liquid water retention, and albedo aging.	Cryosphere	Physics-based
4	Zero-Layer Snow Scheme. Single-layer diagnostic snow for computational efficiency.	Cryosphere	Empirical
5	Canopy Radiation. Multi-layer two-stream canopy radiation with separate direct/diffuse, sunlit/shaded leaf partitioning.	Radiation	Physics-based
6	Photosynthesis (C3/C4). Leaf-level Farquhar (C3) and Collatz (C4) photosynthesis with enzyme kinetics and co-limitation.	Biogeochem	Physics-based
7	Stomatal Conductance. Jacobs (1994) or Medlyn (2011) stomatal models coupled to photosynthesis.	Biogeochem	Intermediate
8	Plant Respiration. Maintenance respiration (Q10 temperature dependence) and growth respiration as fixed fraction of NPP.	Biogeochem	Intermediate
9	TRIFFID Dynamic Vegetation. Lotka–Volterra competition among 5 PFTs for fractional cover based on NPP and height-dependent dominance.	Ecology	Intermediate
10	Phenology. Leaf onset/senescence driven by temperature and moisture thresholds with growing degree day accumulation.	Ecology	Empirical

#	Process	Family	Fidelity
11	Soil Hydrology (Richards Equation). Multi-layer (4-layer default) Richards equation with van Genuchten or Brooks–Corey hydraulic functions.	Hydrology	Physics-based
12	Soil Thermodynamics. Multi-layer heat diffusion with phase change (freeze–thaw), temperature-dependent thermal conductivity.	Hydrology	Physics-based
13	PDM Runoff. Probability Distributed Model for surface runoff generation based on sub-grid soil moisture distribution.	Hydrology	Intermediate
14	TOPMODEL Runoff. Topography-based saturation-excess runoff using topographic wetness index.	Hydrology	Intermediate
15	Soil Carbon (RothC). 4-pool soil carbon decomposition (DPM, RPM, BIO, HUM) with temperature and moisture rate modifiers.	Biogeochem	Intermediate
16	Soil Nitrogen Cycling. Ammonium/nitrate pools with mineralization, nitrification, denitrification, leaching, and N ₂ O emissions.	Biogeochem	Intermediate
17	Methane Emissions. Wetland CH ₄ emissions from substrate availability, temperature, and water table depth.	Biogeochem	Intermediate
18	INFERNO Fire Model. Interactive fire with human/lightning ignition, fuel moisture/load thresholds, burnt area, and fire emissions.	Fire	Intermediate
19	River Routing (RFM). 1-D kinematic wave river flow model for global river discharge.	Hydrology	Intermediate
20	Irrigation Demand. Crop-specific irrigation scheduling from soil moisture deficit relative to field capacity.	Human Systems	Empirical
21	Crop Model (JULES-crop). Phenology-driven crop growth with leaf/stem/root/storage organ partitioning for wheat, rice, maize, soybean.	Human Systems	Intermediate
22	Urban Energy Balance (MORUSES). 2-tile urban canyon model (roof, road+wall) with radiative trapping, anthropogenic heat, and thermal inertia.	Atmosphere	Intermediate
23	Lake Model (FLake). 2-layer lake thermodynamics with mixed layer, thermocline, and ice cover.	Hydrology	Intermediate
24	Permafrost / Soil Freeze–Thaw. Deep soil column (up to 10 m) with phase change, supercooled liquid water, and bedrock heat capacity.	Cryosphere	Physics-based

#	Process	Family	Fidelity
25	Dust Emission. Wind-erosion dust flux from bare soil tiles using Woodward (2001) scheme.	Atmosphere	Intermediate

Table 47: JULES process catalog (25 entries).

S CABLE — Community Atmosphere Biosphere Land Exchange

CABLE (Kowalczyk et al., 2006; Wang et al., 2011) is the Australian land surface model used in ACCESS-ESM and ACCESS-CM2 (CMIP6). It features a two-leaf canopy model, a six-layer soil scheme, the CASA-CNP biogeochemical model, and the POP demographic model for forest dynamics.

#	Process	Family	Fidelity
1	Surface Energy Balance. Coupled solution of radiation, sensible heat, latent heat, and ground heat flux on vegetated and bare soil tiles.	Atmosphere	Physics-based
2	Two-Leaf Canopy Model. Separate sunlit/shaded leaf photosynthesis and conductance, distinguishing direct and diffuse radiation interception.	Radiation	Physics-based
3	Photosynthesis. Farquhar–von Caemmerer (C3) and Collatz (C4) leaf-level biochemistry with nitrogen-dependent V_{cmax} .	Biogeochem	Physics-based
4	Stomatal Conductance (Leuning). Ball–Berry–Leuning model coupling stomatal aperture to photosynthesis, humidity deficit, and CO_2 .	Biogeochem	Intermediate
5	Aerodynamic Conductance. Roughness sublayer parameterization with Raupach (1994) within-canopy and above-canopy resistances.	Atmosphere	Physics-based
6	Soil Hydrology (6-Layer Richards). Six-layer soil moisture with Richards equation, Clapp–Hornberger hydraulic functions, and gravitational drainage.	Hydrology	Physics-based
7	Soil Thermodynamics (6-Layer). Six-layer heat diffusion with freeze–thaw and layered thermal conductivity.	Hydrology	Physics-based
8	Snow Model (3-Layer). Prognostic snow with compaction, liquid water retention, and grain metamorphism.	Cryosphere	Intermediate
9	Canopy Interception. Precipitation interception and throughfall/stemflow partitioning on LAI-dependent storage.	Hydrology	Empirical
10	CASA-CNP Biogeochemistry. Carbon, nitrogen, and phosphorus cycling through plant/litter/soil pools with 3 litter and 3 soil pools.	Biogeochem	Intermediate
11	Leaf Litter Decomposition. Temperature and moisture-dependent decomposition of metabolic, structural, and CWD litter.	Biogeochem	Intermediate

#	Process	Family	Fidelity
12	Nitrogen Limitation. Down-regulation of photosynthesis when soil N supply cannot match demand; biological N fixation.	Biogeochem	Intermediate
13	Phosphorus Limitation. Weathering-based P supply, sorption/desorption, and biochemical mineralization unique to Australian P-limited soils.	Biogeochem	Intermediate
14	POP (Patch of Patches). Demographic forest model with age-class patches, self-thinning, disturbance gaps, and landscape-level averaging.	Ecology	Intermediate
15	Phenology. Prescribed or climate-driven LAI with drought-deciduous and cold-deciduous types.	Ecology	Empirical
16	Root Water Uptake. Layer-weighted root extraction with exponential root density profile.	Hydrology	Empirical
17	Tile-Based Heterogeneity. Sub-grid tiling by PFT and bare soil with independent energy/water/carbon budgets.	Hydrology	Intermediate
18	SLI (Soil–Litter–Isotope). Advanced soil model coupling litter layer energy balance with soil moisture and heat.	Hydrology	Intermediate

Table 48: CABLE process catalog (18 entries).

T CLASSIC — Canadian Land Surface Scheme Including Biogeochemical Cycles

CLASSIC (Melton et al., 2020) unifies the Canadian Land Surface Scheme (CLASS; Versegny, 1991) with the Canadian Terrestrial Ecosystem Model (CTEM; Arora, 2003). It is the land component of CanESM5 (CMIP6) and provides coupled energy, water, carbon, and nitrogen cycling with dynamic vegetation and prognostic fire.

#	Process	Family	Fidelity
1	Surface Energy Balance. Implicit coupled solution of surface temperature, sensible/latent heat, and ground heat flux over 4 sub-areas (needleleaf, broadleaf, crops, grass/bare).	Atmosphere	Physics-based
2	Multi-Layer Soil (3-Layer). Prognostic temperature and liquid/frozen moisture in 3 soil layers with thermal/hydraulic coupling.	Hydrology	Intermediate
3	Snow Physics (1-Layer). Single-layer snow with aging albedo, densification, liquid water retention, and heat capacity.	Cryosphere	Intermediate
4	Canopy Conductance. Jarvis-type stomatal conductance with PAR, temperature, VPD, and soil moisture stress functions.	Biogeochem	Intermediate
5	Photosynthesis. Farquhar (C3) and Collatz (C4) leaf-level models integrated over sunlit/shaded fractions with nitrogen scaling.	Biogeochem	Physics-based
6	Autotrophic Respiration. Maintenance respiration (stem, root, leaf) with Q10 temperature dependence plus growth respiration.	Biogeochem	Intermediate
7	Heterotrophic Respiration. Decomposition of 2 litter pools (structural, metabolic) and 1 soil carbon pool; temperature and moisture rate modifiers.	Biogeochem	Intermediate
8	Dynamic Vegetation (CTEM). Competition among 9 PFTs driven by NPP-based colonization, mortality (heat stress, drought, disturbance), and bioclimatic limits.	Ecology	Intermediate
9	Phenology. Cold-deciduous and drought-deciduous leaf onset/offset with carbon cost of leaf deployment.	Ecology	Empirical
10	Allocation. NPP allocation to leaf, stem, root with allometric constraints varying by PFT and light competition.	Ecology	Intermediate

#	Process	Family	Fidelity
11	Fire (CTEM-Fire). Area burned from probability of fire occurrence (lightning + human ignition) $\times$ fire spread rate $\times$ fire duration; biomass combustion and post-fire mortality.	Fire	Intermediate
12	Land Use Change. Deforestation, afforestation, and cropland expansion with carbon pool transfers.	Human Systems	Empirical
13	Wetland Methane. Heterotrophic respiration-based substrate with temperature-dependent methanogenesis and oxidation.	Biogeochem	Intermediate
14	Permafrost Carbon. Deep soil carbon cycling in permafrost layers with thaw-dependent decomposition rates.	Biogeochem	Intermediate
15	Nitrogen Cycle. Plant N uptake, biological N fixation, litter/soil N mineralization with C:N ratio constraints on decomposition.	Biogeochem	Intermediate
16	River Routing. Linear reservoir-based river flow routing to ocean grid cells.	Hydrology	Empirical

Table 49: CLASSIC process catalog (16 entries).

U SUMMA — Structure for Unifying Multiple Modeling Alternatives

SUMMA (Clark et al., 2015a,b) is a hydrological modeling framework developed at NCAR that enables controlled experimentation with alternative process representations within a single code. Its explicit representation of modeling decisions — which process parameterization to use at each modeling layer — makes it uniquely relevant to MAESMA’s mission of autonomous process selection.

#	Process	Family	Fidelity
1	Radiation Transfer (Multiple). Two-stream, Beer–Lambert, or Noah-style canopy radiation with switchable formulations.	Radiation	Physics-based
2	Stomatal Resistance (Multiple). Ball–Berry, Jarvis, or simple-resistance formulations selectable at runtime.	Biogeochem	Intermediate
3	Canopy Shortwave. Separate direct/diffuse partitioning with variable canopy scattering.	Radiation	Physics-based
4	Canopy Longwave. Multi-layer or single-layer longwave radiation within the canopy.	Radiation	Intermediate
5	Throughfall / Canopy Interception. Maximum storage capacity model vs. Rutter model for canopy water budget.	Hydrology	Empirical
6	Snow Layering. Dynamic layer splitting/merging (CLM-style, VIC-style, or Anderson-style) with variable max layers.	Cryosphere	Intermediate
7	Snow Compaction. Anderson (1976) or empirical overburden compaction models.	Cryosphere	Intermediate
8	Snow Albedo. BATS decay, CLASS aging, or lookup-based snow albedo parameterizations.	Cryosphere	Intermediate
9	Snow Liquid Water. Bucket or Richards-equation percolation within snowpack layers.	Cryosphere	Intermediate
10	Surface Energy Balance. Implicit solution coupling surface temperature, turbulent fluxes, and ground conduction.	Atmosphere	Physics-based
11	Soil Hydrology (Richards). Multi-layer mixed form Richards equation with switchable hydraulic functions (Campbell or van Genuchten).	Hydrology	Physics-based
12	Groundwater. Water table depth as prognostic variable with aquifer recharge/discharge coupling to soil column.	Hydrology	Intermediate
13	Soil Thermodynamics. Multi-layer Fourier heat conduction with freeze–thaw and de Vries thermal conductivity.	Hydrology	Physics-based

#	Process	Family	Fidelity
14	Baseflow. TOPMODEL-based, linear reservoir, or non-linear power-law baseflow parameterizations.	Hydrology	Intermediate
15	Surface Runoff. Saturation-excess (TOPMODEL), infiltration-excess (Green-Ampt), or combined.	Hydrology	Intermediate
16	Evapotranspiration. Penman–Monteith or resistance-based approaches with separate soil/canopy/snow evaporation.	Hydrology	Physics-based
17	Lateral Subsurface Flow. Hillslope-scale lateral redistribution of soil moisture via Darcy’s law.	Hydrology	Physics-based
18	Upper Boundary Condition (Multiple). Dirichlet, Neumann, or flux-based upper boundary for heat/moisture.	Hydrology	Intermediate
19	Lower Boundary Condition (Multiple). Free drainage, prescribed head, or zero-flux lower boundary.	Hydrology	Intermediate
20	Numerical Solver. Coupled or split solution of energy/mass conservation via backward Euler with Newton–Raphson iteration.	Hydrology	Physics-based

Table 50: SUMMA process catalog (20 entries). SUMMA is uniquely structured around switchable process representations, reflecting 20+ modeling decisions.

V ED2 — Ecosystem Demography Model 2

ED2 (Medvigy et al., 2009; Longo et al., 2019) is a cohort-based terrestrial biosphere model developed at Harvard that resolves individual-level vegetation dynamics through size- and age-structured cohorts within patches. As a precursor to FATES, ED2 provides independent implementations of canopy radiation, photosynthesis, allocation, disturbance, and demographic dynamics with distinct algorithmic choices.

#	Process	Family	Fidelity
1	Canopy Radiation (Multi-Layer). Exact solution of multi-layer two-stream radiation within vertically resolved canopy strata.	Radiation	Physics-based
2	Photosynthesis. Farquhar–Leuning coupled leaf-level model with Arrhenius temperature dependencies and co-limitation.	Biogeochem	Physics-based
3	Stomatal Conductance. Leuning (1995) stomatal model coupled iteratively to photosynthesis and leaf energy balance.	Biogeochem	Intermediate
4	Leaf Energy Balance. Individual-leaf Penman–Monteith with iterative convergence of surface temperature and fluxes.	Atmosphere	Physics-based
5	Stem Respiration. Diameter-based maintenance respiration with Q10 temperature scaling.	Biogeochem	Intermediate
6	Root Respiration. Biomass-dependent root maintenance respiration.	Biogeochem	Intermediate
7	Growth Respiration. Fixed fraction (0.33) of positive NPP allocated to construction costs.	Biogeochem	Empirical
8	Allocation. Allometric height–diameter–crown–root partitioning with obligate carbon balance closure.	Ecology	Intermediate
9	Cohort Dynamics. Size-structured cohort fusion/fission for computational efficiency while preserving demographic resolution.	Ecology	Intermediate
10	Patch Dynamics. Age-structured patch mosaic representing disturbance history; patches fuse when structurally similar.	Ecology	Intermediate
11	Gap-Phase Disturbance. Treefall gap creation with size-dependent mortality rates creating new patches.	Ecology	Intermediate
12	Mortality (5 Mechanisms). Carbon starvation, cold, disturbance, senescence, and density-dependent self-thinning mortality.	Ecology	Intermediate
13	Phenology (4 Schemes). Drought-deciduous, cold-deciduous, evergreen, and light-controlled leaf flushing/abscission.	Ecology	Empirical

#	Process	Family	Fidelity
14	Soil Carbon Decomposition. Fast/slow/structural soil carbon pools with heterotrophic respiration driven by temperature and moisture.	Biogeochem	Intermediate
15	Soil Hydrology. Multi-layer soil moisture with infiltration, percolation, and root water extraction.	Hydrology	Intermediate
16	Fire Disturbance. Prescribed or stochastic fire occurrence with combustion completeness dependent on fuel moisture.	Fire	Empirical
17	Seed Dispersal / Recruitment. Seed rain, germination, and sapling establishment into existing patches.	Ecology	Empirical
18	Allometry. DBH-based allometric equations for height, crown area, biomass, and LAI following Chave-type tropical/temperate relations.	Ecology	Intermediate

Table 51: ED2 process catalog (18 entries).

W PFLOTRAN — Subsurface Reactive Transport

PFLOTRAN (Lichtner et al., 2015) is a DOE-funded massively parallel code for simulating subsurface multiphase flow, multicomponent reactive transport, and geomechanics. It operates on structured and unstructured grids, supports coupling to CLM/ELM, and fills essential gaps in deep subsurface biogeochemistry, groundwater contamination, and nuclear waste repository modeling.

#	Process	Family	Fidelity
1	Variably-Saturated Flow (Richards). Richards equation with Mualem–van Genuchten or Brooks–Corey constitutive relations.	Hydrology	Physics-based
2	Multiphase Flow. Two-phase (liquid–gas) or three-phase (liquid–gas–NAPL) Darcy flow with capillary pressure.	Hydrology	Physics-based
3	Reactive Transport. Advection-dispersion-reaction equation for multicomponent solute transport with operator splitting.	Biogeochem	Physics-based
4	Aqueous Speciation. Thermodynamic equilibrium speciation using law of mass action with extended Debye–Hückel activity coefficients.	Biogeochem	Physics-based
5	Mineral Precipitation/Dissolution. Transition State Theory (TST) rate laws with surface-area-dependent kinetics.	Geology	Physics-based
6	Sorption. Linear K_d , Langmuir, Freundlich, ion exchange, and surface complexation adsorption models.	Biogeochem	Intermediate
7	Microbial Degradation. Monod kinetics with multiple electron donor/acceptor pairs, biomass growth, and inhibition terms.	Biogeochem	Intermediate
8	Colloid Transport. Facilitated transport of contaminants with filtration, attachment/detachment kinetics.	Hydrology	Physics-based
9	Heat Transport. Energy conservation with conduction, advection, and latent heat of phase change in porous media.	Hydrology	Physics-based
10	Geomechanics. Linear poroelasticity coupling pore pressure to solid displacement via Biot’s equations.	Geology	Physics-based
11	CLM–PFLOTRAN Coupling. Interface replacing CLM subsurface hydrology/biogeochemistry with PFLOTRAN reactive transport.	Hydrology	Physics-based
12	Fracture Flow. Discrete fracture network (DFN) or dual-continuum flow in fractured porous media.	Hydrology	Physics-based

#	Process	Family	Fidelity
13	Radionuclide Decay. Multi-species radioactive decay chains with ingrowth and branching ratios.	Biogeochem	Physics-based
14	Structural Grid. Structured Cartesian or unstructured polyhedral meshes with PETSc-based parallel solvers.	Hydrology	Physics-based
15	CO ₂ Sequestration. Supercritical CO ₂ injection, residual/solubility/mineral trapping in saline aquifers.	Human Systems	Physics-based

Table 52: PFLOTRAN process catalog (15 entries).

X ROMS — Regional Ocean Modeling System

ROMS (Shchepetkin & McWilliams, 2005) is a widely-used free-surface, terrain-following, primitive-equation regional ocean model. It is used globally for coastal, estuarine, and open-ocean research with biogeochemical and sediment transport extensions. ROMS fills the regional/coastal ocean gap complementing the global models (MOM6, MPAS-Ocean) already cataloged.

#	Process	Family	Fidelity
1	3D Primitive Equations. Hydrostatic Boussinesq momentum, continuity, temperature, and salinity equations with free surface.	Ocean	Physics-based
2	Free Surface. Split-explicit barotropic/baroclinic time stepping for fast external gravity waves.	Ocean	Physics-based
3	Advection (3rd/4th-Order). Third-order upstream or fourth-order centered horizontal/vertical advection schemes.	Ocean	Physics-based
4	Horizontal Mixing. Laplacian or biharmonic viscosity/diffusivity along geopotential, sigma, or isopycnal surfaces.	Ocean	Physics-based
5	Vertical Mixing (KPP). K-Profile Parameterization boundary layer scheme with interior Richardson-number mixing.	Ocean	Physics-based
6	Vertical Mixing (GLS). Generic Length Scale turbulence closure (k- ϵ , k- ω , k-kl, gen) via Umlauf-Burchard (2003).	Ocean	Physics-based
7	Tidal Forcing. Barotropic tidal constituents at open boundaries with tidal potential body force.	Ocean	Physics-based
8	Open Boundary Conditions. Radiation, Flather, Chapman, clamped, or gradient conditions for tracers and momentum.	Ocean	Intermediate
9	Bottom Boundary Layer. Logarithmic or quadratic drag with wave-current bottom stress interaction.	Ocean	Physics-based
10	Sediment Transport. Non-cohesive bed load (Meyer-Peter-Müller) and suspended load with multiple grain classes.	Geomorphology	Physics-based
11	Bed Morphodynamics. Bed evolution from erosion/deposition with active-layer stratigraphy tracking.	Geomorphology	Physics-based
12	NPZD Biogeochemistry. Nutrient-Phytoplankton-Zooplankton-Detritus pelagic ecosystem with light-dependent primary production.	Biogeochem	Intermediate

#	Process	Family	Fidelity
13	Fennel Biogeochemistry. Extended NPZD with chlorophyll-to-carbon ratios, oxygen cycling, and multiple nutrient limitation (N, P, Si).	Biogeochem	Intermediate
14	Air–Sea Flux. Bulk formulae (COARE or Fairall) for momentum, heat, and freshwater exchange with atmosphere.	Atmosphere	Intermediate
15	Wetting/Drying. Intertidal wetting and drying for estuarine and tidal flat simulations.	Ocean	Intermediate
16	Ice Shelf Thermodynamics. Sub-ice-shelf melt via three-equation thermodynamic formulation at ice–ocean interface.	Cryosphere	Physics-based
17	Sea Ice (Budgell). Elastic-viscous-plastic sea ice dynamics and thermodynamics (optional coupling).	Cryosphere	Intermediate
18	Lagrangian Floats / Drifters. Online particle tracking with optional behavior (diurnal vertical migration, settling).	Ocean	Intermediate
19	Nesting / Multi-Grid. One-way or two-way grid refinement with AGRIF or composition coupling.	Ocean	Intermediate
20	4D-Var Data Assimilation. Tangent linear and adjoint models for incremental strong-constraint 4D-Var.	Ocean	Physics-based

Table 53: ROMS process catalog (20 entries).

Y SWAT — Soil and Water Assessment Tool

SWAT (Arnold et al., 1998; Neitsch et al., 2011) is the world’s most widely-used watershed model, simulating hydrology, sediment, nutrient, and pesticide transport at the basin scale. SWAT operates on a semi-distributed HRU (Hydrologic Response Unit) framework and is extensively applied for agricultural water quality, climate impact assessment, and TMDL (Total Maximum Daily Load) studies.

#	Process	Family	Fidelity
1	Surface Runoff (SCS-CN). Modified SCS Curve Number method with antecedent moisture and slope adjustment.	Hydrology	Empirical
2	Surface Runoff (Green-Ampt). Infiltration-excess runoff via Green-Ampt Mein-Larson equation.	Hydrology	Physics-based
3	Evapotranspiration (Multiple). Penman-Monteith, Priestley-Taylor, or Hargreaves methods selectable.	Hydrology	Intermediate
4	Soil Percolation. Layer-based water movement with storage routing and percolation to shallow aquifer.	Hydrology	Intermediate
5	Lateral Subsurface Flow. Kinematic storage model for lateral soil-water redistribution.	Hydrology	Intermediate
6	Groundwater (Shallow Aquifer). Unconfined aquifer recharge, baseflow, revap to soil, and pumping.	Hydrology	Intermediate
7	Groundwater (Deep Aquifer). Confined aquifer recharge as fraction of percolation; deep aquifer loss.	Hydrology	Empirical
8	Channel Routing. Variable storage or Muskingum routing of streamflow through channel network.	Hydrology	Intermediate
9	Reservoir Routing. Target release, monthly/daily outflow tables, or measured discharge for impoundments.	Human Systems	Empirical
10	Snow Melt. Temperature-index melt with elevation band sub-daily temperature distribution.	Cryosphere	Empirical
11	Sediment Yield (MUSLE). Modified Universal Soil Loss Equation driven by runoff energy rather than rainfall.	Geomorphology	Empirical
12	Channel Sediment Routing. Stream power-based degradation and Bagnold-equation deposition in channels.	Geomorphology	Empirical
13	Nitrogen Cycle. Five soil N pools (NO_3 , NH_4 , active/stable organic, fresh organic) with mineralization, nitrification, denitrification, plant uptake, volatilization, and leaching.	Biogeochem	Intermediate

#	Process	Family	Fidelity
14	Phosphorus Cycle. Six soil P pools (labile, active/stable mineral, organic, fresh organic, solution) with sorption, mineralization, and transport.	Biogeochem	Intermediate
15	Pesticide Fate. Degradation, sorption, washoff, leaching, and volatilization for user-defined pesticide compounds.	Human Systems	Intermediate
16	Crop Growth (Simplified EPIC). Heat-unit-driven phenology, radiation-use-efficiency biomass accumulation, harvest index, and nutrient/water stress.	Human Systems	Intermediate
17	Irrigation / Management. Auto-irrigation, fertilizer, tillage, and grazing operations on HRU schedule.	Human Systems	Empirical
18	Urban Runoff. Build-up/wash-off model for impervious fraction with pollutant loading.	Human Systems	Empirical

Table 54: SWAT process catalog (18 entries).

Z GEOS-Chem — Atmospheric Chemistry Transport Model

GEOS-Chem (Bey et al., 2001; <https://geos-chem.org>) is a global 3-D model of atmospheric composition driven by NASA GEOS meteorological fields. It simulates tropospheric and stratospheric chemistry, aerosol microphysics, emissions, and deposition for over 200 species, and is used by hundreds of research groups worldwide. GEOS-Chem fills a critical gap in the MAESMA knowledgebase: detailed atmospheric gas-phase and heterogeneous chemistry.

#	Process	Family	Fidelity
1	Tropospheric Chemistry (Full). 350 reaction mechanism with HOx, NOx, Ox, VOC, halogen chemistry and JPL/IUPAC kinetics.	Atmosphere	Physics-based
2	Stratospheric Chemistry. 120 additional reactions including Ox–NOx–HOx–ClOx–BrOx catalytic cycles and heterogeneous PSC chemistry.	Atmosphere	Physics-based
3	Photolysis (FAST-JX). Online photolysis rate calculation accounting for overhead ozone, aerosol, and cloud optical depth.	Radiation	Physics-based
4	Sulfate Aerosol. SO ₂ oxidation (gas-phase OH, aqueous H ₂ O ₂ /O ₃), sulfate nucleation, condensation, and wet/dry removal.	Atmosphere	Intermediate
5	Carbonaceous Aerosol. Primary OC/BC emissions, SOA formation via 2-product or VBS scheme, and hygroscopic aging.	Atmosphere	Intermediate
6	Mineral Dust. Size-resolved dust emission (DEAD/Zender), gravitational settling, and heterogeneous uptake.	Atmosphere	Intermediate
7	Sea Salt Aerosol. Wind-speed-dependent emission, deliquescence, and size-dependent deposition.	Atmosphere	Intermediate
8	Nitrate / Ammonium Aerosol. ISORROPIA-II thermodynamic equilibrium for HNO ₃ –NH ₃ –H ₂ SO ₄ system.	Atmosphere	Physics-based
9	Wet Deposition. In-cloud and below-cloud scavenging with Henry’s law solubility and retention efficiency.	Atmosphere	Intermediate
10	Dry Deposition (Wesely). Resistance-in-series model with aerodynamic, quasi-laminar, and surface resistances.	Atmosphere	Intermediate
11	Anthropogenic Emissions. Gridded emission inventories (CEDS, EDGAR, EPA NEI) with sector-specific temporal profiles.	Human Systems	Empirical
12	Biogenic Emissions (MEGAN). Model of Emissions of Gases and Aerosols from Nature — isoprene, monoter-	Biogeochem	Intermediate

#	Process	Family	Fidelity
	penes, sesquiterpenes, and oVOCs from canopy-scale parameterization.		
13	Biomass Burning Emissions. GFED, FINN, or QFED fire emission inventories with injection height parameterization.	Fire	Empirical
14	Transport (Advection). Lin–Rood flux-form semi-Lagrangian advection on cubed-sphere or latitude–longitude grids.	Atmosphere	Physics-based
15	Convective Transport. Archived convective mass fluxes from GEOS for deep and shallow convective tracer redistribution.	Atmosphere	Intermediate
16	Boundary Layer Mixing. Non-local PBL mixing using archived GEOS turbulent diffusivities and mixed-layer depths.	Atmosphere	Intermediate
17	Mercury Cycle. Elemental/reactive/particulate Hg emissions, oxidation (Br, OH), deposition, and re-emission with land/ocean reservoirs.	Biogeochem	Intermediate
18	CO ₂ / CH ₄ Simulation. Tagged tracers for fossil fuel, biosphere, fire, and wetland sources with atmospheric transport for inverse modeling.	Biogeochem	Intermediate
19	Aerosol–Radiation Interaction. Online aerosol optical depth calculation for radiative transfer feedback (when coupled to GCM).	Radiation	Intermediate
20	Halogen Chemistry (Cl/Br/I). Reactive halogen cycling from sea salt, biogenic emissions, and heterogeneous recycling on aerosols.	Atmosphere	Physics-based

Table 55: GEOS-Chem process catalog (20 entries).

AA Delft3D — Coastal and Estuarine Hydrodynamics

Delft3D (Lesser et al., 2004; Deltares) is a widely-used integrated modeling suite for coastal, estuarine, riverine, and lake hydrodynamics. It couples flow, waves, sediment transport, morphodynamics, water quality, and ecology. Delft3D fills gaps in nearshore processes, estuarine mixing, tidal dynamics, and coastal morphology that global ocean models do not resolve.

#	Process	Family	Fidelity
1	Shallow Water Equations. 2D depth-averaged or 3D sigma-layer hydrostatic Navier–Stokes with Coriolis, pressure gradient, and wind stress.	Ocean	Physics-based
2	Turbulence Closure. k- ϵ model, k-L, or algebraic mixing-length for vertical eddy viscosity/diffusivity.	Ocean	Physics-based
3	Tidal Propagation. Astronomic tidal constituents, self-attraction and loading, and tidal flats wetting/drying.	Ocean	Physics-based
4	Wave–Current Interaction (SWAN). Online coupling with SWAN spectral wave model for radiation stress, wave setup, and longshore currents.	Ocean	Physics-based
5	Wind-Driven Circulation. Spatially varying wind stress with Charnock or Smith–Banke drag coefficients.	Atmosphere	Intermediate
6	Salinity Transport. Advection–diffusion of salinity with baroclinic density gradients driving estuarine circulation.	Ocean	Physics-based
7	Temperature Transport. Heat budget with shortwave absorption (Beer–Lambert), longwave, sensible/latent fluxes.	Ocean	Physics-based
8	Sediment Transport (Cohesive). Krone/Partheniades erosion/deposition for mud with flocculation and consolidation.	Geomorphology	Physics-based
9	Sediment Transport (Non-Cohesive). van Rijn (1993) bed load and suspended load with reference concentration.	Geomorphology	Physics-based
10	Bed Morphodynamics. Bed level update from sediment mass balance with bank erosion, dredging, and nourishment.	Geomorphology	Physics-based
11	Water Quality (DELWAQ). Process library of 100+ water quality processes: BOD/DO, nutrients, algae growth, toxics, heavy metals.	Biogeochem	Intermediate

#	Process	Family	Fidelity
12	Algal Bloom Dynamics. Multi-species algal growth with Monod nutrient limitation, self-shading, grazing, and settling.	Ecology	Intermediate
13	Secondary Flow. Spiral motion in river bends for lateral sediment transport and point-bar morphology.	Ocean	Physics-based
14	Dam Break / Flood Propagation. Riemann-solver-based flood wave propagation over complex bathymetry.	Hydrology	Physics-based
15	Oil Spill Tracking. Lagrangian particle tracking with weathering (evaporation, dispersion, emulsification).	Human Systems	Intermediate
16	Flexible Mesh (D-Flow FM). Unstructured mesh solver for arbitrary coastline and channel geometry.	Ocean	Physics-based

Table 56: Delft3D process catalog (16 entries).

AB LISFLOOD-FP — Large-Scale Flood Inundation Model

LISFLOOD-FP (Bates & De Roo, 2000; Neal et al., 2012) is a raster-based flood inundation model designed for flood hazard mapping at scales from rivers to continents. It provides multiple solver options from simplified diffusive wave to full shallow water equations, making it a key reference for multi-fidelity hydrological modeling.

#	Process	Family	Fidelity
1	Shallow Water Equations (2D). Full 2D shallow water solver with Roe-type approximate Riemann scheme and explicit time stepping.	Hydrology	Physics-based
2	Diffusive Wave (ACC). Acceleration-limited diffusive wave approximation (Neal et al., 2012) for efficient continental-scale routing.	Hydrology	Intermediate
3	Inertial / Subgrid Channel. 1D subgrid channel model with bankfull transition to floodplain using Newton–Raphson water level convergence.	Hydrology	Physics-based
4	Manning Friction. Spatially distributed Manning’s n for floodplain and channel roughness.	Hydrology	Empirical
5	Wetting / Drying. Thin-film or depth-threshold wetting and drying with mass conservation.	Hydrology	Intermediate
6	Rainfall Input. Direct rainfall-runoff on DEM cells with simple infiltration loss (Green–Ampt or constant rate).	Hydrology	Intermediate
7	Evaporation Loss. Simple potential-evaporation removal from surface water.	Hydrology	Empirical
8	Boundary Conditions. Free-slip wall, stage hydrograph, flow hydrograph, or tidal harmonic open boundaries.	Hydrology	Intermediate
9	DEM Pre-Processing. Hydrological conditioning, flat-area routing, and sub-grid topography statistics.	Geomorphology	Empirical
10	Adaptive Time Stepping. CFL-based adaptive Δt for explicit solver stability.	Hydrology	Physics-based

Table 57: LISFLOOD-FP process catalog (10 entries).

AC LM3-PPA / BiomeE — Perfect Plasticity Approximation Vegetation Models

The Perfect Plasticity Approximation (PPA; Purves et al., 2008; Strigul et al., 2008) replaces the computational complexity of explicit crown geometry with a single analytical closure: at demographic equilibrium, canopy packing is “perfectly plastic,” so the total crown area of trees above the canopy closure height z^* equals ground area. This yields a sharp canopy–understory partition solvable in $O(N)$ cohorts rather than $O(N^2)$ crown overlap tests. ED2-PPA (Weng et al., 2015) embedded PPA inside the ED2 framework. LM3-PPA extended it into GFDL’s land model (Shevliakova et al., 2009). BiomeE (Weng et al., 2019; 2022) is the latest evolution, coupling PPA-based vegetation demography to full biogeochemistry within GFDL ESM4.1. Together these models define an alternative canopy-closure paradigm to the multi-layer two-stream schemes used in FATES and ED2.

#	Process	Family	Fidelity
1	Perfect Plasticity Canopy Closure. Analytical z^* height partitioning: total crown area above z^* equals ground area, yielding a binary canopy/understory light environment.	Ecology	Physics-based
2	Crown Area Allometry. Species-specific crown area–diameter power law ($A_c = a \cdot D^b$) controlling competitive light capture.	Ecology	Intermediate
3	Canopy Light Extinction (Beer–Lambert). Exponential light attenuation through canopy and understory strata with PFT-specific extinction coefficients.	Radiation	Intermediate
4	Photosynthesis (Farquhar / LUE). Farquhar C3 biochemistry (LM3-PPA) or light-use-efficiency model (BiomeE) integrated over sunlit/shaded fractions.	Biogeochem	Physics-based
5	Stomatal Conductance. Medlyn-type optimal stomatal model coupling carbon gain to water loss.	Biogeochem	Intermediate
6	Autotrophic Respiration. Leaf, sapwood, and fine-root maintenance respiration with Q10 temperature scaling plus growth respiration.	Biogeochem	Intermediate
7	Allocation (Allometric). Carbon allocation to leaf, sapwood, heartwood, fine root, and reproduction constrained by allometric invariants.	Ecology	Intermediate
8	Height Growth. Diameter-increment-driven height growth via species-specific H–D allometry; asymptotic maximum height.	Ecology	Intermediate
9	Diameter Growth. Annual carbon balance converted to diameter increment via pipe-model sapwood-area constraint.	Ecology	Intermediate

#	Process	Family	Fidelity
10	Reproduction / Seed Rain. Fraction of NPP allocated to reproduction; seed dispersal kernel; density-dependent sapling establishment.	Ecology	Intermediate
11	Mortality (Carbon Starvation). Death rate increases exponentially when stored carbon falls below a critical reserve.	Ecology	Intermediate
12	Mortality (Density-Dependent). Self-thinning mortality following Yoda's $-\frac{3}{2}$ power law at high stem density.	Ecology	Intermediate
13	Mortality (Background / Senescence). Age- or size-dependent background mortality rate.	Ecology	Empirical
14	Cohort Dynamics. Size-structured cohort tracking with annual growth, mortality, and recruitment; cohort merging for efficiency.	Ecology	Intermediate
15	Patch Age Structure. Disturbance-driven patch age distribution; young patches transition through successional stages.	Ecology	Intermediate
16	Soil Biogeochemistry (CENTURY-Type). Fast/slow/passive soil carbon pools with temperature and moisture decomposition modifiers; BiomeE adds N cycle.	Biogeochem	Intermediate
17	Nitrogen Cycle (BiomeE). Plant N uptake, resorption, litter N, soil mineralization, nitrification, and N-limitation of photosynthesis.	Biogeochem	Intermediate
18	Litter Dynamics. Leaf, fine-root, and woody litter input to metabolic/structural pools with lignin-fraction quality control.	Biogeochem	Intermediate
19	Disturbance Regime. Prescribed or stochastic disturbance (fire, windthrow) that resets patches to age zero with partial biomass kill.	Fire	Empirical
20	Trait-Based Competition. PFT competition mediated by height growth rate, shade tolerance (via Amax, Rdark), and wood density trade-offs.	Ecology	Intermediate
21	Water Stress (BiomeE). Soil moisture limitation on stomatal conductance and photosynthesis with PFT-specific wilting point.	Hydrology	Intermediate
22	Phenology. Prescribed or climate-driven (GDD, photoperiod) leaf onset/senescence for deciduous PFTs.	Ecology	Empirical

Table 58: LM3-PPA / BiomeE process catalog (22 entries). Covers ED2-PPA (Weng et al., 2015), LM3-PPA, and BiomeE (Weng et al., 2019).

AD SORTIE-ND — Spatially Explicit Individual-Based Forest Model

SORTIE (Pacala et al., 1993, 1996) is the canonical individual-based forest gap model. Unlike cohort models (ED2, FATES), SORTIE tracks every individual tree’s spatial (x, y) position, enabling explicit neighborhood interactions for light, seed dispersal, and competition. SORTIE-ND (Neighborhood Dynamics; Murphy, 2008) is the modern open-source descendant. It serves as the reference for full spatial resolution against which PPA and cohort approximations are benchmarked.

#	Process	Family	Fidelity
1	Spatially Explicit Light (GLI). Global Light Index computation: for each sapling, a hemisphere is divided into sectors; neighboring tree crowns project shadows to compute fractional sky visibility.	Radiation	Physics-based
2	Crown Geometry. Explicit allometric crown radius, crown depth, and shape (ellipsoid, cone) per species for shadow casting.	Ecology	Intermediate
3	Growth (NCI). Neighborhood Competition Index: diameter growth = $f(\text{size, light, sum of distance-weighted neighbor basal areas})$.	Ecology	Intermediate
4	Growth (Relative Height). Height growth relative to local neighborhood max height with species-specific shade tolerance.	Ecology	Intermediate
5	Mortality (Stochastic). Size- and growth-rate-dependent annual mortality probability; slow-growing suppressed trees die faster.	Ecology	Intermediate
6	Mortality (Senescence). Maximum-age-dependent background death rate.	Ecology	Empirical
7	Seed Dispersal (Kernel). Spatially explicit seed rain from parent trees with species-specific 2D dispersal kernels (Weibull, lognormal, 2Dt).	Ecology	Intermediate
8	Substrate-Dependent Establishment. Sapling establishment probability conditioned on substrate type (tip-up mound, fresh CWD, forest floor, scarified).	Ecology	Intermediate
9	Clonal Reproduction. Vegetative sprouting from stumps or roots with species-specific probability and spatial offset.	Ecology	Intermediate
10	Harvest / Silviculture. Individual-tree selection, diameter-limit, or area-based harvest prescriptions with residual stand specification.	Human Systems	Empirical

#	Process	Family	Fidelity
11	Storm Damage. Wind-speed- and size-dependent blow-down probability with domino-effect neighborhood damage.	Ecology	Intermediate
12	Snag Dynamics. Standing dead tree decay through snag height classes with substrate generation for regeneration.	Ecology	Intermediate
13	Allometry. Species-specific DBH-to-height, DBH-to-crown-radius, and DBH-to-biomass allometric equations.	Ecology	Intermediate
14	Neighborhood Density Effects. Conspecific and heterospecific neighbor density effects on growth and mortality via Janzen–Connell mechanisms.	Ecology	Intermediate

Table 59: SORTIE-ND process catalog (14 entries). The canonical spatially explicit individual-based forest model.

AE JABOWA / FORET — Classic Gap Models

JABOWA (Botkin et al., 1972) was the first forest gap model; FORET (Shugart, 1984) generalized it. Their core innovation — simulating patch-scale (0.1 ha) forest dynamics with individual trees competing for light within gaps — spawned an entire model lineage (ZELIG, LINKAGES, FORCLIM, UVAFME). These models remain foundational references for the gap-dynamics paradigm.

#	Process	Family	Fidelity
1	Gap Light Regime. Available light for each tree computed from the leaf area above it in a single vertical column; Beer–Lambert extinction within the gap (0.1 ha patch).	Radiation	Intermediate
2	Diameter Growth (Potential Curve). Species-specific growth equation: $\Delta D = G \cdot D \cdot (1 - DH/D_{\max}H_{\max})/(274 + 3b_2D - 4b_3D^2)$ modified by environmental scalars.	Ecology	Intermediate
3	Environmental Growth Modifiers. Multiplicative scalars (0–1) for (i) available light, (ii) growing degree-days, (iii) soil moisture (drought days), and (iv) soil nitrogen.	Ecology	Empirical
4	Shade Tolerance Classification. Species assigned to tolerance classes (very intolerant → very tolerant) determining minimum light threshold for survival and growth.	Ecology	Empirical
5	Mortality (Stress). If diameter growth falls below a minimum threshold for several consecutive years, death probability increases steeply (intrinsic stress kill).	Ecology	Intermediate
6	Mortality (Random). Small annual age-independent background mortality (1–2% yr ^{−1}) ensuring most trees die before reaching maximum age.	Ecology	Empirical
7	Regeneration. Species establish in gaps when (i) growing-degree-day range is met, (ii) light exceeds tolerance threshold, and (iii) stochastic lottery selects from the species pool.	Ecology	Intermediate
8	Height–Diameter Allometry. $\text{Height} = 137 + b_2D - b_3D^2$; parabolic allometry with species-specific coefficients.	Ecology	Empirical
9	Leaf Area (LAI). Individual LAI computed from diameter and species-specific crown parameters; summed vertically for light competition.	Ecology	Empirical
10	Soil Moisture Budget (FORET). Simple bucket model: precipitation, PET (Thornthwaite), and available water capacity determine drought stress days.	Hydrology	Empirical
11	Decomposition / Nutrient Cycling (LINKAGES). Litter quality (lignin:N) drives decomposition rate; AET-scaled annual soil nitrogen availability (Pastor and Post, 1985).	Biogeochem	Intermediate

#	Process	Family	Fidelity
12	Biogeographic Envelope. Species presence filtered by minimum/maximum growing-degree-day and minimum January temperature thresholds.	Ecology	Empirical

Table 60: JABOWA / FORET / LINKAGES gap model catalog (12 entries). Foundational gap-dynamics models.

AF FORMIND — Individual-Based Tropical Forest Model

FORMIND (Köhler & Huth, 1998; Fischer et al., 2016) is an individual-based forest gap model designed for species-rich tropical forests. It operates on a grid of 20×20 m patches, tracks individual trees grouped into PFTs, and emphasizes carbon balance-driven growth, logging, and fragmentation. FORMIND bridges the gap between classic temperate gap models and tropical diversity, with particular strength in logging impact assessment.

#	Process	Family	Fidelity
1	Light Competition (3D). Vertical light profile in 20×20 m patches computed from crown geometry; multi-layer canopy with penumbra effects.	Radiation	Intermediate
2	Photosynthesis (Gross). Light-response curve per PFT (Michaelis–Menten or rectangular hyperbola); integrated over crown layers.	Biogeochem	Intermediate
3	Respiration. Maintenance respiration (leaf, stem, root) with allometric scaling plus growth respiration.	Biogeochem	Intermediate
4	Carbon Balance Growth. Net carbon gain allocated to diameter increment via allometric pipe model; negative carbon balance triggers mortality.	Ecology	Intermediate
5	Allometry. PFT-specific DBH–height (Chave-type), DBH–crown-diameter, and DBH–biomass relations.	Ecology	Intermediate
6	Recruitment. PFT-specific seedling ingrowth rate (stems ha ⁻¹ yr ⁻¹) into smallest size class filtered by light availability.	Ecology	Empirical
7	Seed Dispersal. Spatially explicit dispersal kernel between patches; wind- and animal-dispersed modes.	Ecology	Intermediate
8	Mortality (3 Modes). Background mortality, diameter-growth-dependent stress mortality, and crowding mortality from gap closure.	Ecology	Intermediate
9	Gap Formation. Large-tree fall creates gaps by damaging neighbors within the crown-fall zone.	Ecology	Intermediate
10	Logging Module. Selective logging by minimum diameter, species group, and cutting cycle; skid-trail collateral damage to residual stand.	Human Systems	Intermediate
11	Fragmentation. Edge effects on tree mortality, microclimate, and seed dispersal near forest–non-forest boundaries.	Ecology	Intermediate

#	Process	Family	Fidelity
12	Succession / PFT Competition. Pioneer vs. late-successional PFT trade-offs via maximum growth rate, shade tolerance, and wood density.	Ecology	Intermediate
13	Soil Water Bucket. Simple one-layer soil water balance with rainfall input, ET loss, and drought stress on growth.	Hydrology	Empirical
14	Land-Use Change. Deforestation, regrowth, and agricultural rotation scenarios on the patch grid.	Human Systems	Empirical

Table 61: FORMIND process catalog (14 entries).

AG LANDIS-II Extensions — Landscape Disturbance and Succession

LANDIS-II (Scheller et al., 2007) is a spatially explicit landscape model operating on raster cells (typically 100–250 m) over large domains (10^4 – 10^6 ha). The Core framework (cataloged separately) provides the extension interface; the extensions implement the actual process representations. Here we catalog the NECN (Net Ecosystem Carbon & Nitrogen), SCRPPLE (fire), and Dynamic Fuel System extensions that make LANDIS-II competitive with process-based DGVMs at landscape scales.

#	Process	Family	Fidelity
1	NECN Soil Carbon. CENTURY-based 4-pool soil organic matter (active, slow, passive, SOM) decomposition with climate and texture controls.	Biogeochem	Intermediate
2	NECN Soil Nitrogen. Mineralization, nitrification, denitrification, leaching, and volatile N loss coupled to soil C decomposition.	Biogeochem	Intermediate
3	NECN Aboveground NPP. Radiation-use-efficiency estimate with PFT-specific parameters and water/temperature stress modifiers.	Biogeochem	Intermediate
4	NECN Litter Decomposition. Structural vs. metabolic litter partitioned by lignin:N ratio; surface and soil litter pools.	Biogeochem	Intermediate
5	NECN Dead Wood. Coarse woody debris decay through 5 decay classes with species-specific decay constants.	Biogeochem	Empirical
6	Biomass Succession. Cohort-level growth based on maximum ANPP, age-dependent decline, and neighborhood shade competition.	Ecology	Intermediate
7	Age-Only Succession. Simplest succession: species presence/absence by age since disturbance and shade tolerance class.	Ecology	Empirical
8	PnET Succession. Photosynthesis–Evapotranspiration-based succession using Aber & Federer (1992) physiological model within LANDIS-II.	Ecology	Intermediate
9	SCRPPLE Fire. Stochastic fire ignition (lightning/human), weather-dependent spread on landscape grid, fire severity from burn conditions.	Fire	Intermediate
10	Dynamic Fuel System. Fuel type assignment from cohort composition, age, and previous disturbance; fuel moisture based on weather stream.	Fire	Intermediate

#	Process	Family	Fidelity
11	Base Fire. Simpler stochastic fire regime with mean fire return interval per fire region; size drawn from fire-size distribution.	Fire	Empirical
12	Wind Disturbance. Stochastic windthrow events with size and severity drawn from empirical distributions; directional wind damage.	Ecology	Intermediate
13	Biological Disturbance (BDA). Insect/pathogen outbreaks modeled as host-density-dependent epidemics with spatial dispersal.	Ecology	Intermediate
14	Harvest / Forest Management. Spatially explicit timber harvest prescriptions: clearcut, shelterwood, selection; management areas and rotations.	Human Systems	Intermediate
15	Seed Dispersal. Species-specific effective and maximum dispersal distances with probability of long-distance dispersal events.	Ecology	Intermediate
16	Establishment (Climate Filtering). Species establishment probability filtered by soil moisture, temperature, and light given overstory composition.	Ecology	Intermediate
17	Climate Change (Climate Library). Spatially explicit future climate inputs with monthly temperature and precipitation for all extensions.	Atmosphere	Empirical
18	Land-Use / Land-Cover Change. Prescribed conversion of forest cells to agriculture, urban, or restoration based on scenario maps.	Human Systems	Empirical

Table 62: LANDIS-II Extensions catalog (18 entries). Extends the Core framework (6 entries cataloged separately).

AH UVAFME — University of Virginia Forest Model Enhanced

UVAFME (Shuman et al., 2017; Foster et al., 2019) is a modern descendant of the JABOWA/FORST gap model lineage, enhanced for global-scale application. It simulates individual-tree dynamics on replicated 500 m² plots with stochastic climate variability, permafrost coupling, and fire, and has been applied from boreal to tropical forests. UVAFME bridges classic gap models to the scale of DGVMs while retaining individual-tree resolution.

#	Process	Family	Fidelity
1	Individual-Tree Growth. Species-specific potential diameter growth modified by light, temperature (GDD), drought, and nutrient scalars.	Ecology	Intermediate
2	Gap Light Model. Vertical leaf-area summation within 500 m ² plots; Beer–Lambert extinction for available light at each tree’s crown midpoint.	Radiation	Intermediate
3	Stochastic Climate Variability. Monthly temperature and precipitation drawn from species-specific distributions to capture inter-annual variability effects on growth/mortality.	Atmosphere	Intermediate
4	Growing-Degree-Day Filtering. Species establishment and growth constrained by GDD thresholds; cold-hardiness filtering by minimum winter temperature.	Ecology	Empirical
5	Drought Stress. Soil moisture budget (Thornthwaite AET/PET) with species-specific drought tolerance thresholds.	Hydrology	Empirical
6	Mortality (Stress / Background). Growth-rate-dependent stress mortality (slow-growing trees die faster) plus age-independent background rate.	Ecology	Intermediate
7	Fire Module. Stochastic fire occurrence with mean return interval; fire kills trees based on size (bark thickness) and species fire tolerance.	Fire	Intermediate
8	Permafrost Coupling. Active-layer depth limits rooting depth; permafrost thaw deepens the active layer, enabling succession transitions (boreal-to-temperate).	Cryosphere	Intermediate
9	Soil Decomposition. Annual litter input decomposed at AET-controlled rate; soil nitrogen availability feeds back on growth via nutrient scalar.	Biogeochem	Intermediate
10	Regeneration. Stochastic seedling establishment from species pool when light and climate conditions are met; seedling bank for some species.	Ecology	Intermediate
11	Height–Diameter–Crown Allometry. Species-specific parabolic H–D allometry and crown area–diameter scaling.	Ecology	Empirical

#	Process	Family	Fidelity
12	Tree-Level Biomass. Aboveground and belowground biomass from allometric equations; leaf area from specific leaf area $\times$ leaf biomass.	Ecology	Intermediate
13	Plot Replication. Hundreds of replicate plots averaged for landscape-level statistics; variance across plots captures spatial heterogeneity.	Ecology	Intermediate
14	Global Application. >100 species parameterized globally from trait databases; runs forced by ERA5 or CMIP climate grids.	Ecology	Intermediate

Table 63: UVAFME process catalog (14 entries).

AI FORCLIM — Forest Dynamics Under Climate Change

FORCLIM (Bugmann, 1996, 2001) is a widely-used gap model designed for climate change impact studies. Developed at ETH Zürich, it integrates a bioclimatic envelope approach with mechanistic carbon balance and has been applied extensively across European and North American forests. FORCLIM’s modular structure (environment, plant, and soil submodels) makes it a template for multi-fidelity process substitution.

#	Process	Family	Fidelity
1	Gap Light. Vertical leaf-area integration within $\frac{1}{12}$ ha patch; Beer–Lambert attenuation with species-specific light extinction.	Radiation	Intermediate
2	Diameter Growth. Maximum potential growth (parabolic function of D) reduced by light, temperature, and drought multipliers.	Ecology	Intermediate
3	Bioclimatic Limits. Minimum winter temperature, GDD requirements, and drought tolerance define species range; dynamic with changing climate.	Ecology	Empirical
4	Drought Index (Monthly). Monthly bucket hydrology: precipitation vs. PET (Thornthwaite); species drought tolerance maps to growth reduction.	Hydrology	Empirical
5	Mortality (Growth-Dependent). Probability of survival decreases when realized growth < 10% of maximum for consecutive years (Bigler & Bugmann, 2004).	Ecology	Intermediate
6	Mortality (Background). Small constant annual death probability calibrated to species longevity.	Ecology	Empirical
7	Regeneration (Seed-Based). Species-specific maximum seedling number modulated by light, temperature, and browsing pressure.	Ecology	Intermediate
8	Browsing Pressure. Ungulate browsing reduces regeneration of palatable species; browsing intensity as external driver.	Ecology	Empirical
9	Soil Decomposition (Bucket-N). Single-layer soil nitrogen availability from AET-driven litter decomposition (LINKAGES-type).	Biogeochem	Intermediate
10	Height–DBH Allometry. Species-specific parabolic allometry with maximum height/diameter constraints.	Ecology	Empirical
11	Stand Structure Output. Stem density, basal area, biomass, LAI, and species composition by diameter class; validation against NFI data.	Ecology	Intermediate

#	Process	Family	Fidelity
12	Management Module. Thinning and harvesting prescriptions (from below, from above, clearcut, shelterwood) at species/diameter-class level.	Human Systems	Intermediate

Table 64: FORCLIM process catalog (12 entries).

AJ SORTIE-NG (Erickson) — GPU-Native Spatially Explicit Individual-Based Forest Model

SORTIE-NG (Erickson, 2018) is a next-generation redesign of the SORTIE individual-based forest model, re-architected from the ground up for GPU execution. Where classical SORTIE processes every tree sequentially on CPU — resulting in $O(N^2)$ neighborhood interactions and hemisphere-based light computations that dominate runtime — SORTIE-NG maps all per-tree operations to massively parallel GPU kernels. The design philosophy treats the forest stand as a particle system: each tree is an independent data element processed by GPU threads in lockstep. Key innovations include CUDA-parallelized hemispherical light integration (replacing the $O(N^2)$ CPU shadow loop with a GPU rasterization pass over crown projections), batched seed dispersal via FFT-accelerated kernel convolution on device, and GPU-side allometric computation for diameter/height/crown radius updates. The model retains SORTIE’s biological realism — species-specific light response, spatially explicit competition, stochastic recruitment and mortality — while achieving 50–200× speedup on modern GPU hardware. This paradigm informs MAESMA’s GPU-native architecture for ecology process families.

#	Process / Capability	Family	Fidelity
1	GPU Hemispherical Light Model. Each tree’s light environment computed via GPU rasterization of neighboring crown projections onto a hemisphere discretized into sky sectors; replaces SORTIE’s sequential $O(N^2)$ ray-casting with a single parallel render pass per cohort batch. Global Light Index (GLI) per tree.	Radiation	Physics-based
2	Parallel Individual-Tree Processing. Every tree in the stand is a GPU thread: growth, mortality, and reproduction computed in parallel SIMT warps. Stand sizes of 10^5 – 10^6 individuals processed per kernel launch.	Ecology	Intermediate
3	GPU Neighborhood Interaction. Spatial competition indices (NCI: neighborhood crowding index) computed via GPU spatial hashing with fixed-radius neighbor queries, reducing $O(N^2)$ to $O(N)$ amortized.	Ecology	Intermediate
4	FFT-Accelerated Seed Dispersal. Species-specific dispersal kernels (Weibull, lognormal, 2Dt) convolved with parent-tree density maps on GPU using cuFFT; produces seedling establishment probability surfaces at sub-meter resolution.	Ecology	Physics-based
5	GPU Allometric Computation. Diameter–height, diameter–crown radius, and diameter–biomass allometries evaluated as vectorized GPU functions across all individuals simultaneously. Species-specific coefficient lookup via texture memory.	Ecology	Empirical

#	Process / Capability	Family	Fidelity
6	Stochastic Mortality (GPU RNG). Per-tree mortality drawn from species-specific probability functions using GPU-parallel cuRAND streams. Size-dependent, competition-dependent, and storm-event mortality modes.	Ecology	Intermediate
7	Substrate-Dependent Recruitment. Seedling establishment probability conditioned on substrate type (mineral soil, decayed wood, moss, litter), light availability (GLI threshold), and conspecific density. GPU-parallel evaluation across all microsites.	Ecology	Empirical
8	Crown Plasticity and Asymmetric Competition. Individual crowns deform toward light gaps; asymmetric competition where larger trees suppress smaller neighbors proportional to crown overlap area computed via GPU geometric intersection.	Ecology	Intermediate
9	Species-Specific Growth Response. Radial growth as a function of light (GLI), tree size (DBH), local competition (NCI), and climate envelopes. Nonlinear response surfaces evaluated on GPU with per-species parameter vectors.	Ecology	Intermediate
10	Spatial Structure Tracking. Full (x, y) coordinates maintained for every individual on GPU device memory. Pair correlation functions, Ripley's K, and mark correlation computed as GPU reduction kernels for real-time spatial pattern analysis.	Ecology	Intermediate
11	Disturbance Integration (Wind/Fire/Harvest). Spatially explicit windthrow (topographic exposure + crown drag), surface fire (flame length + bark thickness survival), and selective harvest (marking rules applied as GPU predicates).	Fire	Intermediate
12	Gap Dynamics and Canopy Turnover. Tree death creates gaps; gap light regimes computed by GPU hemisphere rasterization of surviving neighbors. Successional trajectory emerges from species-specific gap-phase regeneration.	Ecology	Intermediate
13	Multi-Species Coexistence Tracking. Shannon diversity, species importance values, size-class distributions, and compositional turnover tracked as GPU-parallel reductions per timestep. Enables real-time biodiversity monitoring.	Ecology	Empirical
14	Real-Time 3D Canopy Visualization. GPU-generated crown geometry (ellipsoidal or allometric) rendered via compute-to-render pipeline; live visualization of stand structure, light environment, and species composition during simulation.	Ecology	Intermediate

#	Process / Capability	Family	Fidelity
15	Climate Envelope Forcing. Temperature, precipitation, and growing-degree-day inputs drive species recruitment probability, growth rate modifiers, and drought-induced mortality thresholds. GPU-side interpolation from gridded climate data.	Atmosphere	Empirical
16	Soil Resource Competition. Below-ground resource partitioning via root-zone overlap computed from species-specific rooting depth profiles and spatial proximity; modulates growth alongside light competition.	Biogeochem	Intermediate

Table 65: SORTIE-NG (Erickson) process catalog (16 entries). GPU-native spatially explicit forest model.

AK DeepLand (Erickson) — Deep Learning Land Surface Model

DeepLand (Erickson) is a neural network-based land surface model that replaces traditional physics parameterizations with learned operators while enforcing physical constraints through architecture design and physics-informed loss functions. Rather than hand-coding stomatal conductance equations (Ball–Berry, Medlyn), soil hydraulic functions (van Genuchten, Brooks–Corey), or snow albedo decay curves, DeepLand learns these mappings directly from high-resolution process model output and observational data using the NVIDIA Modulus framework for physics-informed neural operator training. The model operates as a hybrid system: a process-based backbone handles prognostic state evolution (mass and energy conservation are structurally guaranteed by the architecture), while neural networks learn the constitutive relations, closure terms, and subgrid parameterizations that drive the largest uncertainties in conventional LSMs. Key Modulus principles embedded in the design include PDE residual loss for Richards equation and heat diffusion, Fourier Neural Operators for resolution-invariant spatial operators, and transfer learning from high-resolution simulations to coarser operational grids. DeepLand informs MAESMA’s approach to physics–ML hybrid representations and neural operator-based process discovery.

#	Process / Capability	Family	Fidelity
1	Neural Stomatal Conductance. Replaces Ball–Berry/Medlyn with a neural network mapping $(R_{\text{net}}, \text{VPD}, T_{\text{leaf}}, C_a, \text{SWC}) \rightarrow g_s$. Trained on eddy covariance observations; physics-informed loss enforces non-negative conductance, correct VPD response sign, and water-use efficiency bounds.	Ecology	Physics-based
2	FNO-Based Richards Equation Solver. Fourier Neural Operator learns the solution operator for variably saturated soil water flow. Trained on paired numerical solutions across soil texture classes; resolution-invariant formulation transfers from 1 cm training grid to 10 cm operational grid. PDE residual loss enforces $\partial\theta/\partial t = \nabla \cdot (K(\theta)\nabla\psi(\theta))$ without paired simulation data via PINO.	Hydrology	Physics-based
3	Neural Soil Thermal Conductivity. DeepONet learns the conductivity-moisture-texture-density mapping; branch network encodes soil properties, trunk network encodes depth–time coordinates. Replaces Johansen/Côté–Konrad empirical fits.	Hydrology	Physics-based
4	Learned Radiative Transfer (Canopy). Neural network canopy radiative transfer replacing two-stream approximation: maps $(LAI, \text{leaf angle, solar zenith, spectral albedos}) \rightarrow (\text{absorbed PAR, net radiation})$. Trained against Monte	Radiation	Physics-based

#	Process / Capability	Family	Fidelity
	Carlo photon transport simulations; enforces energy conservation via architecture (output sums to incoming).		
5	Neural Snow Albedo and Metamorphism. Learns snow albedo decay, grain growth, and melt–refreeze dynamics from SNICAR offline simulations and satellite retrievals. Physics-informed loss enforces albedo $\in [0, 1]$ and monotonic aging in absence of new snowfall.	Cryosphere	Physics-based
6	Hybrid Photosynthesis Model. Process-based Farquhar–von Caemmerer–Berry backbone (Rubisco/RuBP limitation structure preserved); neural network learns temperature acclimation of $V_{c,max}$, J_{max} and mesophyll conductance as residual corrections to standard Q_{10} functions. GPU-accelerated batch evaluation across all grid cells.	Ecology	Physics-based
7	MeshGraphNet for Groundwater–Surface Water Exchange. Graph neural network operating on unstructured watershed mesh; learns lateral groundwater flow, seepage, and river–aquifer exchange fluxes. Message-passing architecture preserves mass conservation per mesh element.	Hydrology	Physics-based
8	Neural Turbulent Flux Parameterization. Replaces Monin–Obukhov similarity theory for surface layer turbulent fluxes (H, LE) : neural network maps (wind speed, ΔT , Δq , z_0 , stability) $\rightarrow (C_H, C_E)$. Trained on eddy covariance tower data; physics-informed loss enforces correct stability function limits and flux–gradient relationship sign.	Atmosphere	Physics-based
9	Transfer Learning: High-Res to Coarse Grid. Models trained at high resolution (e.g., 1 km CLM) are transferred to coarser grids (e.g., 100 km ESM) using Modulus-style domain adaptation. Learned subgrid distributions replace tile-fraction approaches; FNO resolution invariance enables direct weight transfer without retraining.	Ecology	Intermediate
10	Conservation-Enforcing Architecture. Network architecture structurally guarantees conservation: water balance ($P = ET + R + \Delta S$) and energy balance ($R_{net} = H + LE + G$) enforced as hard constraints via output normalization layers, not soft loss terms. Closure errors are identically zero by construction.	Hydrology	Physics-based
11	Subgrid Heterogeneity Learning. Neural network learns subgrid probability distributions of soil moisture, LAI, and snow cover from high-resolution training data. Replaces	Hydrology	Intermediate

#	Process / Capability	Family	Fidelity
	tiling approaches (CLM PCT, JULES tiles) with learned continuous distributions that capture spatial correlations.		
12	Neural Phenology. Learns budburst, senescence, and leaf area dynamics from satellite LAI time series (MODIS, Sentinel-2) conditioned on temperature, photoperiod, and soil moisture history. Recurrent architecture (LSTM/GRU) captures memory effects; physics-informed loss enforces non-negative LAI and seasonal carbon budget closure.	Ecology	Intermediate
13	GPU-Native End-to-End Training. Full model (backbone + neural components) trains end-to-end on GPU with automatic differentiation through both physics and ML components. Mixed-precision training (BF16 backbone, FP32 accumulation) on multi-GPU via PyTorch FSDP or JAX pjit. Training data: eddy covariance, FLUXNET, SMAP, GRACE, tower meteorology.	Ecology	Intermediate
14	Uncertainty Quantification via Deep Ensembles. Each neural component is an ensemble of M networks ($M = 5-10$); epistemic uncertainty estimated from ensemble spread. Aleatoric uncertainty captured by heteroscedastic output layers. Total predictive uncertainty propagated through the process chain and reported per grid cell per timestep.	Ecology	Intermediate
15	Symbolic Distillation of Learned Operators. After training, PySR extracts interpretable symbolic expressions from neural components (e.g., discovering a power-law soil hydraulic conductivity from the FNO Richards solver). Distilled expressions are deposited as high-interpretability knowledgebase entries alongside their neural parents.	Ecology	Intermediate
16	Online Adaptation and Continual Learning. Neural components update their weights online as new observational data streams arrive (SMAP, GRACE, AmeriFlux), using elastic weight consolidation to prevent catastrophic forgetting. Drift-detection triggers full retraining when distribution shift exceeds threshold.	Ecology	Intermediate
17	Benchmarking Against Traditional LSMs. Automated head-to-head comparison against CLM, Noah-MP, JULES, and CABLE across PLUMBER-2 tower sites, FLUXNET-CH4 wetlands, and SnowMIP sites. Skill metrics (KGE, RMSE, bias, timing errors) tracked per process component and globally.	Ecology	Intermediate

#	Process / Capability	Family	Fidelity
18	Carbon Cycle Neural Closure. Neural network learns heterotrophic respiration, litter decomposition, and soil carbon turnover rates from FLUXNET NEE partitioning and soil incubation data. Physics-informed loss enforces non-negative pools and mass conservation ($\Delta C = \text{NPP} - R_h - \text{DOC}_{\text{export}}$).	Biogeochem	Physics-based

Table 66: DeepLand (Erickson) process catalog (18 entries). Deep learning land surface model integrating NVIDIA Modulus principles.

AL NVIDIA Earth-2 / earth2studio — Foundation Weather Model Platform

NVIDIA Earth-2 (earth2studio) is an open-source Python framework providing a unified interface to pre-trained AI foundation weather and climate models, ensemble generation, diagnostic workflows, and GPU-accelerated data pipelines. earth2studio treats weather prediction as a composable inference pipeline: data sources (GFS, ERA5, HRRR, IFS, CDS) feed into pre-trained models, whose outputs flow through diagnostic transforms and perturbation methods. MAESMA registers these foundation models as Atmosphere R0/R1 rungs in the knowledgebase, enabling the agent swarm to select AI-driven atmospheric forcing when weather dynamics are not the dominant source of error. The Foundation Model Agent manages model versioning, benchmarking against reanalysis, and automatic registration of new architectures as they are released.

#	Capability	Family	Fidelity
1	FourCastNet (SFNO). Spherical Fourier Neural Operator on 0.25° equiangular grid; 73 prognostic channels (surface + pressure-level T, u, v, z, q); 6-hourly autoregressive; 10-day global forecast in 2 seconds on single GPU. Resolution-invariant spectral architecture.	Atmosphere	Physics-based
2	Pangu-Weather. 3D transformer on 13 pressure levels + surface; separate 1h/3h/6h/24h lead-time models; trained on 39 years of ERA5. Pressure-level prognostics enable vertical profile prediction.	Atmosphere	Physics-based
3	GraphCast. Graph neural network on icosahedral mesh (0.25°); message-passing over multi-mesh hierarchy; 10-day forecasts competitive with HRES; autoregressive with 6h steps.	Atmosphere	Physics-based
4	DLWP-CS. Deep Learning Weather Prediction on cubed-sphere grid; UNet encoder-decoder; 6h timestep; trained on ERA5; lightweight architecture suitable for large ensemble generation.	Atmosphere	Intermediate
5	GenCast. Diffusion-based probabilistic weather model; generates calibrated ensemble members from a single forward pass; captures multimodal forecast distributions for extreme events; 12-day probabilistic forecasts.	Atmosphere	Physics-based
6	CorrDiff (Super-Resolution). Score-based diffusion model for downscaling coarse forecasts ($25\text{ km} \rightarrow 2\text{ km}$); trained on HRRR; preserves extreme value statistics and spatial coherence. Bridges global foundation models to regional process-model grids.	Atmosphere	Physics-based
7	Ensemble Perturbation Methods. Bred Vectors, Gaussian, Spherical Harmonics, and Laplacian perturbation methods	Atmosphere	Intermediate

#	Capability	Family	Fidelity
	for generating calibrated ensemble spread from deterministic models. GPU-parallel perturbation generation.		
8	Diagnostic Pipelines. GPU-accelerated computation of derived quantities: accumulated cyclone energy (ACE), atmospheric river detection (IVT), fire weather indices (FWI, FFMCI, DMC, DC, ISI, BUI), heat wave indices, precipitation extremes. Direct feed into MAESMA benchmarking agent.	Atmosphere	Intermediate
9	Data Source Abstraction. Unified interface to GFS (NOAA), ERA5 (ECMWF/CDS), HRRR (NOAA), IFS (ECMWF), and satellite-derived analysis products. Lazy loading, coordinate normalization, and on-the-fly regridding via xarray + GPU interpolation.	Atmosphere	Empirical
10	Autoregressive Rollout Engine. GPU-optimized autoregressive inference loop for multi-day forecasts: state caching, mixed-precision (FP16/BF16), batch parallelism across ensemble members, and checkpoint-restart for long rollouts. Supports model chaining (e.g., FourCastNet → CorrDiff downscaling).	Atmosphere	Intermediate
11	Model Zoo Management. Versioned model registry with ONNX/PyTorch weight storage, automatic download, hardware-aware model selection (GPU memory constraints), and benchmark comparison against operational NWP (GFS, IFS HRES). New model releases auto-registered as knowledge-base candidates.	Atmosphere	Intermediate
12	Precipitation and Cloud Parameterization Priors. Foundation model outputs provide learned priors for convective and stratiform precipitation, cloud fraction, and radiative flux that downstream process discovery can use as training targets or baseline comparisons.	Atmosphere	Intermediate
13	Tropical Cyclone Tracking and Intensity. Post-processing pipeline detecting, tracking, and classifying tropical cyclones from foundation model outputs; intensity estimation from wind field maxima and pressure minima; enables rapid catastrophe assessment.	Atmosphere	Intermediate
14	Seamless Prediction (Weather-S2S-Climate). Multiscale inference bridging weather (1–15 days), subseasonal-to-seasonal (S2S, 2–8 weeks), and decadal climate states via model chaining and bias correction. Covers the prediction gap between NWP and climate projection.	Atmosphere	Intermediate

Table 67: NVIDIA Earth-2 / earth2studio capabilities catalog (14 entries). Foundation weather model platform providing Atmosphere R0/R1 rungs.

AM ESA PhiSat-2 — On-Board AI Earth Observation Principles

PhiSat-2 (ESA, 2022–) is the European Space Agency’s technology demonstrator for on-board artificial intelligence in Earth observation satellites. Deploying inference models on a low-power VPU (Intel Movidius Myriad 2) directly on the spacecraft, PhiSat-2 demonstrated that scientifically valuable data can be selected, classified, and prioritized *at the point of acquisition* — eliminating downlink bandwidth waste on cloud-contaminated or scientifically redundant imagery. While PhiSat-2 is a single satellite mission, its architectural principles define a paradigm that MAESMA generalizes: autonomous observation intelligence, edge inference for real-time process monitoring, and active observation tasking driven by model uncertainty. The Autonomous Observation Agent instantiates these principles across the full observation stack — from LEO satellites to field sensor networks.

#	Capability	Family	Fidelity
1	On-Board Cloud Detection and Filtering. CNN classifier (quantized INT8 for VPU deployment) applied to raw hyperspectral imagery; classifies cloud cover per-tile; discards frames exceeding cloud threshold before downlink. Reduces data volume by 30–70% without ground processing.	Atmosphere	Intermediate
2	Edge-AI Fire Detection. On-board detection of active fire pixels from thermal/SWIR bands; sub-pixel fire radiative power estimation; real-time alert generation with geolocation and confidence score. Latency: seconds from acquisition to alert.	Fire	Intermediate
3	Autonomous Flood Mapping. SAR/optical change detection deployed on-board; classifies inundation extent versus permanent water; generates flood mask products at point of acquisition for near-real-time disaster response.	Hydrology	Intermediate
4	Vegetation Stress Classification. Spectral index computation (NDVI, EVI, NDWI, CRI) and anomaly detection relative to phenological baseline; flags drought stress, pest damage, and defoliation events for priority downlink.	Ecology	Intermediate
5	Land Cover Change Detection. Multi-temporal comparison of classified scenes; detects deforestation, urbanization, agriculture expansion, and wetland loss between revisit passes. On-board differencing eliminates redundant stable-scene imagery.	Ecology	Empirical
6	Observation Value Scoring. On-board estimation of scientific value per scene: novelty (deviation from climatolog-	Human Systems	Intermediate

#	Capability	Family	Fidelity
	ical expectation), relevance (overlap with model uncertainty hotspots downlinked as target maps), and quality (atmospheric contamination, off-nadir angle). Prioritizes high-value scenes for limited downlink bandwidth.		
7	Compressed Feature Downlink. Instead of raw imagery, transmits extracted feature vectors (spectral indices, classified labels, detected events) when downlink bandwidth is severely constrained. Preserves information content at 10–100× compression ratio.	Human Systems	Intermediate
8	Active Tasking Interface. Uplink protocol for receiving observation requests from ground-based agents (MAESMA’s Autonomous Observation Agent). Adjustable pointing, spectral mode, spatial resolution, and priority scheduling based on model-driven information gain estimates.	Human Systems	Intermediate
9	Snow and Ice Extent Mapping. On-board classification of snow cover, sea ice extent, and glacier calving events from optical/thermal imagery; feeds cryosphere process models with near-real-time boundary conditions at high revisit frequency.	Cryosphere	Intermediate
10	Coastal and Water Quality Monitoring. On-board spectral analysis for chlorophyll-a, turbidity, harmful algal blooms, and sediment plume detection in coastal/inland waters. Flags anomalous events for priority processing.	Ocean	Intermediate
11	Multi-Sensor Data Fusion Readiness. On-board co-registration and fusion of co-manifested sensors (optical, SAR, thermal, lidar); produces analysis-ready fused products that downstream models can ingest directly without ground-based preprocessing.	Atmosphere	Intermediate
12	Model-in-the-Loop Deployment. Ground-to-orbit deployment of MAESMA-discovered classifiers: the process discovery pipeline trains a new edge-inference model (e.g., improved fire detector from residual analysis), quantizes it for VPU/NPU, and uploads it to the satellite for on-board execution. Closes the loop from knowledgebase to orbit.	Fire	Intermediate

Table 68: PhiSat-2 on-board AI capabilities catalog (12 entries). Autonomous observation intelligence principles.

AN Google Research FireBench — High-Fidelity Wildfire Simulation Dataset

FireBench (Google Research, 2024) is the largest high-fidelity wildfire simulation dataset, combining observational environmental data with coupled fire–atmosphere large-eddy simulations (Swirl-LM). While not itself a process model codebase, FireBench provides critical validation infrastructure for fire process representations in the MAESMA knowledgebase.

#	Capability	Family	Role
1	Coupled Fire–Atmosphere LES. High-fidelity fire–atmosphere coupled simulations (Swirl-LM) resolving turbulent multiphase combustion, plume dynamics, and fire–wind feedbacks at meter-scale resolution.	Fire	Benchmark
2	Ensemble Fire Evolution Scenarios. Generation of ensembles of fire evolution under varied environmental conditions (wind, terrain, fuel, moisture) for statistical validation.	Fire	Benchmark
3	Observational Environmental Data Integration. Real topography, vegetation, and weather data combined with simulation for realistic fire scenarios.	Fire	Benchmark
4	Multi-Variable Time-Series Statistics. Apache Beam pipeline for computing spatiotemporal statistics (min, max, mean) of all simulation variables across TB-scale datasets.	Fire	Benchmark
5	Fire Spread Rate Validation. Ground-truth ROS fields from coupled LES for validating Rothermel, CFS FBP, and physics-based fire spread models at R1–R3 fidelity.	Fire	Benchmark
6	Plume Dynamics Reference. Resolved buoyant plume structure, entrainment, and lofting for validating parameterized plume models at R2/R3 fidelity.	Atmosphere	Benchmark
7	Turbulent Fire–Wind Interaction. Resolved fire–atmosphere feedback including fire-induced winds, local vorticity, and convective column dynamics for calibrating coupled models.	Fire	Benchmark
8	ML Training Data for Fire Emulators. TB-scale simulation data suitable for training neural operator surrogates (FNO, DeepONet) for fast fire spread emulation.	Fire	Benchmark

Table 69: FireBench capabilities catalog (8 entries). FireBench serves as validation infrastructure rather than a process model; all entries are benchmarking resources.

AO MARBL — Marine Biogeochemistry Library

MARBL is the ocean biogeochemistry module used in CESM/POP2 and MOM6. It implements the Biogeochemical Elemental Cycling (BEC) model with explicit representation of multiple phytoplankton functional types, zooplankton, dissolved/particulate organic matter, and full nutrient cycling (N, P, Si, Fe). Standalone library that can be coupled to any ocean circulation model.

#	Process	Family	Fidelity
1	Phytoplankton Growth. Light- and nutrient-limited growth for multiple functional types (diatoms, small phyto, diazotrophs, coccolithophores).	Biogeochem	Intermediate
2	Nutrient Uptake (Michaelis-Menten). Uptake kinetics for NO_3 , NH_4 , PO_4 , SiO_3 , Fe with Michaelis-Menten half-saturation.	Biogeochem	Intermediate
3	Nitrogen Fixation. Diazotroph N_2 fixation controlled by temperature, light, Fe, P availability.	Biogeochem	Intermediate
4	Zooplankton Grazing. Parameterized zooplankton grazing on phytoplankton with prey-switching and sloppy feeding.	Biogeochem	Empirical
5	Phytoplankton Mortality & Aggregation. Non-grazing mortality; aggregation to particulate organic matter.	Biogeochem	Empirical
6	Dissolved Organic Matter Cycling. Production, bacterial remineralization, semi-labile/refractory DOM (C, N, P, Fe).	Biogeochem	Intermediate
7	Particulate Organic Matter Flux. Sinking POM (POC, POP, PON, bSi, CaCO_3) with ballast-effect mineral protection.	Biogeochem	Intermediate
8	Remineralization (Martin Curve). Depth-dependent remineralization of sinking particles following power-law or ballast model.	Biogeochem	Empirical
9	Nitrification. Oxidation of NH_4 to NO_3 as function of light inhibition and temperature.	Biogeochem	Intermediate
10	Denitrification. Water-column and sediment denitrification dependent on O_2 concentration.	Biogeochem	Intermediate
11	Iron Cycle. Dissolved Fe sources (sediment, dust, hydrothermal), scavenging by particles, ligand complexation.	Biogeochem	Intermediate
12	Silicon Cycle. Biogenic silica production by diatoms, dissolution in water column and sediments.	Biogeochem	Intermediate
13	CaCO_3 Production & Dissolution. Calcification by coccolithophores; dissolution below lysocline.	Biogeochem	Intermediate

#	Process	Family	Fidelity
14	Air-Sea Gas Exchange. CO ₂ and O ₂ flux across air-sea interface via wind-speed dependent piston velocity.	Biogeochem	Physics-based
15	Carbonate Chemistry. Full ocean carbon chemistry solver (DIC, alkalinity, pH, pCO ₂).	Biogeochem	Physics-based
16	Oxygen Dynamics. Dissolved O ₂ production (photosynthesis), consumption (respiration, nitrification), air-sea exchange.	Biogeochem	Intermediate
17	Chlorophyll-to-Carbon Ratio. Photoadaptive Chl:C ratio varying with light, nutrients, temperature.	Biogeochem	Intermediate
18	Sediment Flux Coupling. Bottom boundary conditions for nutrient return (Fe, PO ₄ , NO ₃) from sediments.	Biogeochem	Empirical
19	DMS Production. Dimethylsulfide production from phytoplankton, photolysis, bacterial consumption.	Biogeochem	Empirical
20	Light Attenuation. PAR propagation through water column with Chl-dependent attenuation (K_d).	Radiation	Intermediate

Table 70: MARBL ocean biogeochemistry catalog (20 entries). Primary family: Biogeochem (19), Radiation (1).

AP PISM — Parallel Ice Sheet Model

PISM is the most widely used open-source ice sheet model. It uses a hybrid shallow ice approximation (SIA) + shallow shelf approximation (SSA) stress balance, thermomechanically coupled, with a sophisticated subglacial hydrology model. Used in IPCC ISMIP6 projections.

#	Process	Family	Fidelity
1	Shallow Ice Approximation (SIA). Non-sliding internal deformation ice flow via Glen’s law on the horizontal velocity field.	Cryosphere	Physics-based
2	Shallow Shelf Approximation (SSA). Membrane stress balance for ice shelves and fast-flowing ice streams.	Cryosphere	Physics-based
3	Hybrid SIA+SSA Stress Balance. Superposition of SIA and SSA velocities for grounded ice with basal sliding.	Cryosphere	Physics-based
4	Blatter Higher-Order Stress. Optional Blatter-Pattyn first-order Stokes stress balance for steep topography.	Cryosphere	Physics-based
5	Ice Thermodynamics. Enthalpy-based 3D temperature/moisture field in polythermal ice (advection-diffusion).	Cryosphere	Physics-based
6	Basal Sliding (Pseudo-plastic). Pseudo-plastic or Coulomb sliding laws relating basal shear stress to sliding velocity.	Cryosphere	Physics-based
7	Subglacial Hydrology (Distributed). Distributed subglacial water transport (till-layer and linked-cavity models).	Hydrology	Intermediate
8	Subglacial Hydrology (Routing). Subglacial water routing along hydraulic potential gradient.	Hydrology	Physics-based
9	Basal Yield Stress. Till yield stress from subglacial water pressure via Mohr-Coulomb criterion.	Cryosphere	Physics-based
10	Calving (Eigen-Calving). Calving rate proportional to product of principal strain rates at ice front.	Cryosphere	Intermediate
11	Calving (von Mises). Calving based on tensile stress exceeding threshold.	Cryosphere	Intermediate
12	Calving (Thickness). Calving where ice thickness drops below threshold at terminus.	Cryosphere	Empirical
13	Grounding Line Dynamics. Sub-grid grounding line position parameterization; flotation criterion.	Cryosphere	Physics-based
14	Marine Ice Sheet Instability. Flux-driven instability on retrograde bed slopes with buttressing.	Cryosphere	Physics-based
15	Surface Mass Balance (PDD). Positive degree-day temperature-index surface mass balance.	Cryosphere	Empirical

#	Process	Family	Fidelity
16	Surface Mass Balance (dEBM). Diurnal energy balance melt model for surface mass balance.	Cryosphere	Intermediate
17	Ocean Thermal Forcing. Sub-shelf melt from ocean temperature/salinity (quadratic, PICO, Beckmann-Goosse).	Ocean	Intermediate
18	PICO Ocean Box Model. Potsdam Ice-shelf Cavity mOdel: box model of sub-ice-shelf ocean circulation and melting.	Ocean	Intermediate
19	Ice Rheology (Glen's Flow Law). Temperature-dependent viscosity via Glen's power law ($n = 3$) with enhancement factor.	Cryosphere	Physics-based
20	Bed Deformation (Elastic Lithosphere). Glacial isostatic adjustment: elastic lithosphere, relaxing asthenosphere (ELRA/Lingle-Clark).	Geology	Physics-based
21	Fracture Mechanics. Fracture density field advected with ice flow; damage mechanics for crevassing.	Cryosphere	Intermediate
22	Age Tracking. Advection of ice age as a tracer through the ice body.	Cryosphere	Physics-based
23	Geothermal Heat Flux. Spatially variable geothermal heat flux as basal boundary condition.	Geology	Empirical

Table 71: PISM ice sheet dynamics catalog (23 entries). Primary families: Cryosphere (17), Hydrology (2), Ocean (2), Geology (2).

AQ OGGM — Open Global Glacier Model

OGGM is a modular glacier evolution model designed for global-scale applications. It uses flowline dynamics, temperature-index mass balance, and automated glacier inventory processing. Used in GlacierMIP and IPCC AR6.

#	Process	Family	Fidelity
1	Glacier Outline Processing. Automated extraction of glacier outlines, topography, and flowlines from RGI inventory and DEMs.	Cryosphere	Intermediate
2	Centerline/Flowline Extraction. Geometric extraction of glacier centerlines and flowlines from surface topography.	Cryosphere	Intermediate
3	Mass Balance (Temperature-Index). Monthly temperature-index mass balance model with precipitation correction and temperature sensitivity.	Cryosphere	Empirical
4	Mass Balance (PyGEM). Coupling to PyGEM energy-balance mass balance model for higher fidelity.	Cryosphere	Intermediate
5	Flowline Ice Dynamics (SIA). Shallow ice approximation along flowlines with trapezoidal cross-section.	Cryosphere	Physics-based
6	Ice Thickness Inversion. Inversion of surface velocity/slope for ice thickness using mass conservation (Farinotti et al.).	Cryosphere	Intermediate
7	Bed Topography Estimation. Estimation of glacier bed from surface topography and ice thickness inversion.	Cryosphere	Intermediate
8	Glacier Length/Area/Volume Evolution. Forward time integration of glacier geometry responding to climate forcing.	Cryosphere	Intermediate
9	Frontal Ablation (Tidewater). Calving flux for tidewater glaciers as function of water depth and ice thickness.	Cryosphere	Empirical
10	Climate Downscaling. Temperature lapse rate correction and precipitation scaling for glacier elevation.	Atmosphere	Empirical
11	Dynamical Spinup. Iterative spinup to match observed glacier state at reference date.	Cryosphere	Intermediate
12	Glacier Retreat/Advance. Terminus position changes and area/volume evolution under transient climate.	Cryosphere	Intermediate
13	Equilibrium Line Altitude. Computation of ELA from mass balance profile.	Cryosphere	Empirical
14	Surface Velocity. Surface velocity estimation from SIA flow law.	Cryosphere	Physics-based
15	Accumulation/Ablation Zones. Spatial delineation of accumulation and ablation zones from mass balance profile.	Cryosphere	Empirical

Table 72: OGGM glacier dynamics catalog (15 entries). Primary families: Cryosphere (14), Atmosphere (1).

AR APSIM — Agricultural Production Systems Simulator

APSIM is the premier farming-systems simulator, modeling the soil–plant–atmosphere continuum with 30+ crop species. It simulates crop growth, soil water/nutrient dynamics, residue management, erosion, and farm economics. Widely used in agricultural research globally.

#	Process	Family	Fidelity
1	Crop Phenology. Temperature-driven (thermal time / GDD) phenological development with photoperiod and vernalization.	Ecology	Intermediate
2	Leaf Area Development. Leaf area expansion based on thermal time, leaf appearance rate, node number, and water/N stress.	Ecology	Intermediate
3	Canopy Photosynthesis (RUE). Radiation use efficiency: biomass = intercepted PAR $\times$ RUE, with stress modifiers.	Biogeochem	Empirical
4	Canopy Photosynthesis (DCaPST). Detailed coupled canopy photosynthesis-stomatal conductance (Farquhar + Ball-Berry).	Biogeochem	Physics-based
5	Biomass Partitioning. Daily allocation of assimilate to roots, stems, leaves, grain according to growth stage.	Ecology	Intermediate
6	Grain Filling. Grain yield from source (assimilate supply) and sink (grain number $\times$ potential weight).	Ecology	Intermediate
7	Root Growth. Root depth advance with thermal time; root length density distribution with depth.	Ecology	Intermediate
8	Crop Water Stress. Ratio of actual to potential transpiration modifying RUE, leaf expansion, phenology.	Hydrology	Intermediate
9	Crop Nitrogen Demand/Uptake. N demand from critical N concentration functions; uptake by mass flow + diffusion.	Biogeochem	Intermediate
10	SoilWat (Water Balance). Cascading-bucket soil water: infiltration (SCS curve number or Green-Ampt), redistribution, drainage, evaporation.	Hydrology	Intermediate
11	SWIM (Richards). Mechanistic Richards equation 1-D soil water balance (optional).	Hydrology	Physics-based
12	Soil Evaporation. Two-stage evaporation from soil surface (Ritchie approach).	Hydrology	Empirical
13	Soil Organic Matter (SoilN). Multi-pool SOM model (FBIOM/FINERT + active/humic) with C:N ratios.	Biogeochem	Intermediate

#	Process	Family	Fidelity
14	Soil Nitrogen Transformations. Mineralization, nitrification, denitrification, urea hydrolysis, NH_3 volatilization.	Biogeochem	Intermediate
15	Soil Phosphorus. Labile/sorbed/organic P pools and transformations.	Biogeochem	Intermediate
16	Surface Organic Matter (Residues). Crop residue decomposition on soil surface; C/N release; mulch effects on evaporation.	Biogeochem	Intermediate
17	Erosion (USLE). Soil erosion via Universal Soil Loss Equation variants.	Geomorphology	Empirical
18	Soil Temperature. Soil temperature profile from air temperature using EPIC-based approach.	Hydrology	Empirical
19	Tillage. Tillage operations modifying soil structure, residue incorporation, bulk density.	Human Systems	Empirical
20	Fertilizer Application. Timing, rate, and placement of mineral fertilizers (N, P, K, S).	Human Systems	Empirical
21	Irrigation Scheduling. Rule-based and deficit-triggered irrigation management.	Human Systems	Empirical
22	Crop Rotation / Intercropping. Multi-crop sequences, relay planting, intercropping competition for light/water/N.	Ecology	Intermediate
23	Pasture/Grazing. Pasture growth, animal intake, trampling, excreta return.	Ecology	Intermediate
24	Microclimate. Surface energy balance, canopy temperature, humidity within crop canopy.	Atmosphere	Intermediate
25	Economic/Management Rules. Farm-level management rules, economics, decision-making scripts.	Human Systems	Empirical

Table 73: APSIM agricultural systems catalog (25 entries). Primary families: Ecology (7), Biogeochem (7), Hydrology (5), Human Systems (4), Geomorphology (1), Atmosphere (1).

AS DSSAT — Decision Support System for Agrotechnology Transfer

DSSAT is the classic crop simulation platform containing 30+ crop models (CERES-Maize, CERES-Wheat, CROPGRO-Soybean, etc.) with soil water/nitrogen/phosphorus modules. Heavily used in agricultural climate impact assessment.

#	Process	Family	Fidelity
1	CERES Crop Growth. Temperature-driven development (thermal time), light interception (Beer’s law), RUE-based biomass production.	Ecology	Intermediate
2	CROPGRO Engine. Mechanistic hourly canopy photosynthesis (Boote/Pickering), respiration, C balance for legumes/vegetables.	Biogeochem	Intermediate
3	CROPSIM Engine. Wheat/barley growth with detailed tiller dynamics and kernel number determination.	Ecology	Intermediate
4	Phenology (CERES). Phase-based development via thermal time, photoperiod sensitivity, vernalization.	Ecology	Intermediate
5	Leaf Area Growth. Leaf area expansion driven by temperature and leaf number; senescence from age/stress.	Ecology	Intermediate
6	Grain/Yield Formation. Source-sink grain filling: kernel number $\times$ kernel growth rate; harvest index.	Ecology	Intermediate
7	Root Growth & Water Uptake. Root elongation with depth; water extraction from soil layers via root length density.	Ecology	Intermediate
8	Nitrogen Demand & Uptake. Crop N demand from critical/minimum N concentrations; root uptake from soil NH_4/NO_3 .	Biogeochem	Intermediate
9	Phosphorus Uptake. Crop P demand and uptake from labile soil P pools.	Biogeochem	Intermediate
10	Carbon Balance. Plant-level carbon balance: photosynthesis, maintenance/growth respiration, senescence.	Biogeochem	Intermediate
11	Soil Water (Ritchie). Tipping-bucket: rainfall, infiltration (SCS CN), ET (Priestley-Taylor or FAO-56), drainage.	Hydrology	Intermediate
12	Soil Evaporation-Transpiration. Two-stage bare soil evaporation; transpiration limited by root distribution and soil water.	Hydrology	Intermediate
13	Soil Temperature. Soil temperature from surface energy balance (EPIC approach).	Hydrology	Empirical

#	Process	Family	Fidelity
14	Soil Organic C/N (CENTURY). CENTURY-based SOM model (metabolic/structural litter, active/slow/passive pools).	Biogeochem	Intermediate
15	Soil Organic C/N (Godwin). Original DSSAT 2-pool SOM decomposition model.	Biogeochem	Empirical
16	Nitrogen Transformations. Mineralization, immobilization, nitrification, denitrification, leaching, volatilization.	Biogeochem	Intermediate
17	Phosphorus Dynamics. Labile/active/stable inorganic P pools; organic P mineralization.	Biogeochem	Intermediate
18	Potassium Dynamics. K uptake, fixation, exchange, leaching.	Biogeochem	Intermediate
19	Surface Runoff. SCS Curve Number method for surface runoff.	Hydrology	Empirical
20	Flood Management (Rice). Ponded water balance for paddy rice (CERES-Rice).	Hydrology	Intermediate
21	Management Operations. Planting, harvest, fertilization, irrigation, tillage, organic amendments scheduling.	Human Systems	Empirical
22	Seasonal Analysis. Multi-year, multi-location analysis with historical/synthetic weather.	Human Systems	Empirical
23	Weather Generator (WGEN). Stochastic daily weather generation for scenario analysis.	Atmosphere	Empirical
24	Pest/Disease Coupling. Interface for coupling pest and disease models to crop growth.	Ecology	Empirical

Table 74: DSSAT crop modeling catalog (24 entries). Primary families: Ecology (7), Biogeochem (9), Hydrology (5), Human Systems (2), Atmosphere (1).

AT CryoGrid — Permafrost and Frozen Ground Dynamics

CryoGrid is a purpose-built permafrost model solving coupled heat transfer and water/ice dynamics in frozen/unfrozen ground. It excels at representing periglacial processes, ground ice dynamics, thermokarst, and snow-ground coupling with much higher vertical resolution and process detail than CLM or ORCHIDEE permafrost modules.

#	Process	Family	Fidelity
1	Heat Conduction (Fourier). 1D heat conduction through soil/rock column with depth-dependent thermal properties.	Cryosphere	Physics-based
2	Freeze-Thaw (Enthalpy). Phase change of water/ice via enthalpy formulation; latent heat effects on temperature.	Cryosphere	Physics-based
3	Unfrozen Water Content. Soil freezing characteristic curve: unfrozen water as function of temperature below 0 °C.	Cryosphere	Physics-based
4	Active Layer Dynamics. Seasonal thaw depth computation; interannual active layer thickness variability.	Cryosphere	Physics-based
5	Permafrost Table Migration. Long-term permafrost aggradation/degradation under changing climate.	Cryosphere	Physics-based
6	Excess Ice / Ground Ice. Representation of massive ground ice bodies, ice wedges, and their thaw-settlement.	Cryosphere	Intermediate
7	Thermokarst / Subsidence. Ground subsidence from thawing of ice-rich permafrost; talik formation.	Cryosphere	Intermediate
8	Snow (Multi-layer). Multi-layer snowpack: heat conduction, compaction, metamorphism, water percolation, refreezing.	Cryosphere	Physics-based
9	Snow-Ground Coupling. Thermal coupling between snowpack and ground; insulation effects.	Cryosphere	Physics-based
10	Soil Thermal Properties. Thermal conductivity and heat capacity as functions of mineral/organic content, water/ice saturation.	Cryosphere	Physics-based
11	Organic Soil Layers. Variable-thickness organic horizons (peat, moss) with distinct thermal/hydraulic properties.	Ecology	Intermediate
12	Soil Water (Richards). Variably saturated water flow via Richards equation (unfrozen zone).	Hydrology	Physics-based
13	Evapotranspiration. Surface energy balance-driven ET with soil evaporation resistance.	Hydrology	Intermediate
14	Surface Energy Balance. Full surface energy balance: short-wave, longwave, sensible/latent heat, ground heat flux.	Atmosphere	Physics-based
15	Lateral Heat Flow. 2D/3D lateral heat transport for thermal erosion, lake-permafrost interaction.	Cryosphere	Intermediate

#	Process	Family	Fidelity
16	Lake-Permafrost Interaction. Talik formation beneath thermokarst lakes; lake ice growth/decay.	Cryosphere	Intermediate
17	Salt Transport. Coupled salt/heat transport for saline permafrost (cryopegs).	Cryosphere	Intermediate
18	Bedrock Thermal. Deep geothermal gradient and bedrock heat conduction.	Geology	Physics-based

Table 75: CryoGrid permafrost catalog (18 entries). Primary families: Cryosphere (13), Hydrology (2), Ecology (1), Atmosphere (1), Geology (1).

AU XBeach — Coastal Morphodynamics

XBeach is the leading open-source nearshore morphodynamic model. It resolves short-wave groups, infragravity waves, wave-driven currents, sediment transport, and beach/dune erosion during storms. Used globally for coastal hazard assessment.

#	Process	Family	Fidelity
1	Short-wave Energy Propagation. Wave action balance equation: shoaling, refraction, breaking (Baldock/Roelvink), directional spreading.	Ocean	Physics-based
2	Infragravity Wave Generation. Bound long wave forcing by short-wave groups; free infragravity wave propagation.	Ocean	Physics-based
3	Wave Breaking. Depth-limited wave breaking dissipation (roller model).	Ocean	Physics-based
4	Roller Energy. Surface roller energy balance: production from breaking, dissipation to bore turbulence.	Ocean	Physics-based
5	Wave Setup / Setdown. Radiation stress gradients driving mean water level changes.	Ocean	Physics-based
6	Depth-Averaged Currents. 2DH shallow water equations with wave forcing (radiation stress + roller).	Ocean	Physics-based
7	Longshore Currents. Wave-driven longshore currents from oblique wave incidence.	Ocean	Physics-based
8	Cross-shore Currents (Undertow). Return flow / undertow from wave mass transport and roller.	Ocean	Physics-based
9	Bed Shear Stress. Combined wave-current bed shear stress (Soulsby-van Rijn, Van Thiel-Van Rijn).	Geomorphology	Physics-based
10	Sediment Transport (Bed Load). Bed-load transport from combined wave-current forcing.	Geomorphology	Intermediate
11	Sediment Transport (Suspended). Depth-averaged suspended sediment advection-diffusion with wave-stirring source.	Geomorphology	Physics-based
12	Sediment Avalanching. Dry/wet slope avalanching when bed slope exceeds critical angle of repose.	Geomorphology	Empirical
13	Bed Level Update (Exner). Morphological bed evolution via sediment continuity (Exner equation).	Geomorphology	Physics-based
14	Morphological Acceleration. Morfac time-scaling for accelerated morphological evolution.	Geomorphology	Empirical

#	Process	Family	Fidelity
15	Dune Erosion. Dune face retreat from wave impact, slumping, and avalanching.	Geomorphology	Intermediate
16	Overwash & Breaching. Barrier island overwash and breach formation/evolution.	Geomorphology	Intermediate
17	Groundwater Flow (1D). Simple groundwater table dynamics within beach/dune (affects infiltration/exfiltration).	Hydrology	Intermediate
18	Ship-wave Effects. Wake from passing vessels as boundary forcing.	Ocean	Intermediate
19	Vegetation Drag. Wave and flow attenuation by vegetation (mangroves, seagrass).	Ecology	Intermediate
20	Storm Surge Coupling. Boundary conditions from tidal/surge models (via ADCIRC, Delft3D).	Ocean	Physics-based

Table 76: XBeach coastal morphodynamics catalog (20 entries). Primary families: Ocean (10), Geomorphology (8), Hydrology (1), Ecology (1).

AV SURFEX/TEB — Urban Canopy and Land Surface

SURFEX is Météo-France’s externalized land surface platform containing TEB (Town Energy Balance), ISBA (land surface), and lake/ocean surface schemes. TEB is the leading physics-based urban canopy model resolving the energy balance of streets, walls, roofs, and gardens within urban canyons.

#	Process	Family	Fidelity
1	Urban Canyon Radiation. Multiple reflections of short-wave and longwave radiation between road, walls, and sky within street canyon.	Radiation	Physics-based
2	Wall Energy Balance. Conductive heat transfer through building walls (multi-layer); external/internal surface temperatures.	Atmosphere	Physics-based
3	Roof Energy Balance. Roof surface energy balance: solar absorption, longwave emission, sensible/latent heat, conduction.	Atmosphere	Physics-based
4	Road Energy Balance. Road surface energy balance with thermal storage in pavement layers.	Atmosphere	Physics-based
5	Canyon Air Temperature. Energy balance of canyon air space; mixing with above-roof air.	Atmosphere	Physics-based
6	Anthropogenic Heat. Waste heat from building HVAC, industrial activity, and traffic.	Human Systems	Intermediate
7	Building Energy Model (BEM). Indoor energy demand: heating, cooling, ventilation, internal heat gains.	Human Systems	Intermediate
8	Urban Hydrology. Rainfall interception on roofs, road runoff, stormwater drainage, garden infiltration.	Hydrology	Intermediate
9	Urban Vegetation (Garden). Garden soil-vegetation energy/water balance within urban tiles.	Ecology	Intermediate
10	Street Trees. Radiative and evapotranspirative effects of street trees in canyon geometry.	Ecology	Intermediate
11	Green/Cool Roofs. Modified roof properties for vegetated and high-albedo roofing.	Human Systems	Intermediate
12	Snow on Urban Surfaces. Snow accumulation and melt on roofs, roads, and walls with plowing management.	Cryosphere	Intermediate
13	Urban Heat Island. Emergent UHI from canyon trapping, reduced ventilation, thermal mass, and anthropogenic heat.	Atmosphere	Physics-based

#	Process	Family	Fidelity
14	ISBA Land Surface. ISBA soil-vegetation-atmosphere scheme: soil hydrology (diffusion/force-restore), photosynthesis, snow.	Hydrology	Physics-based
15	ISBA Carbon Cycle. Ecosystem respiration, NEE, soil C decomposition within ISBA component.	Biogeochem	Intermediate
16	ISBA Multi-Energy Balance. Separate energy balance for vegetation canopy, bare soil, snow under canopy.	Atmosphere	Physics-based
17	Flake Lake Model. 1-D lake thermal model (FLake) for inland water bodies.	Hydrology	Intermediate
18	Sea Surface (ECUME). Air-sea flux parameterization over ocean surfaces.	Ocean	Intermediate

Table 77: SURFEX/TEB urban and land surface catalog (18 entries). Primary families: Atmosphere (6), Human Systems (3), Hydrology (3), Ecology (2), Radiation (1), Biogeochem (1), Cryosphere (1), Ocean (1).

AW EwE — Ecopath with Ecosim

Ecopath with Ecosim is the world’s most used marine ecosystem modeling platform (applied in 200+ countries/ecosystems). It consists of Ecopath (mass-balance snapshot), Ecosim (temporal dynamics), and Ecospace (spatial). It represents full food webs from phytoplankton to apex predators, including fisheries and coral reef ecosystems.

#	Process	Family	Fidelity
1	Mass-Balance (Ecopath). Static mass-balance of ecosystem: production, consumption, respiration, egestion for all functional groups.	Ecology	Intermediate
2	Trophic Flow Network. Quantified trophic interactions (diet composition matrix) across all functional groups.	Ecology	Intermediate
3	Production/Biomass Ratios. Production rates from empirical relationships with body size, temperature, mortality.	Ecology	Empirical
4	Consumption (Foraging Arena). Ecosim predator-prey dynamics: foraging arena theory mediating prey vulnerability.	Ecology	Intermediate
5	Predator-Prey Dynamics. Time-dynamic biomass changes from differential equations: growth, predation, fishing mortality.	Ecology	Intermediate
6	Primary Production Forcing. Phytoplankton production driven by nutrient concentration and environmental forcing.	Biogeochem	Empirical
7	Fisheries Exploitation. Fishing mortality by fleet, gear type, effort dynamics, discards.	Human Systems	Intermediate
8	Marine Protected Areas. Spatial closures with differential habitat quality and movement rates.	Ecology	Intermediate
9	Spatial Dynamics (Ecospace). 2D grid-based spatial representation with habitat preferences, movement, dispersal.	Ecology	Intermediate
10	Habitat Capacity. Habitat-mediated carrying capacity: coral cover, seagrass, depth, temperature effects.	Ecology	Intermediate
11	Nutrient Loading. External nutrient forcing driving eutrophication and primary production changes.	Biogeochem	Empirical
12	Climate Forcing Functions. Temperature, pH, oxygen forcing driving physiological rates and distribution shifts.	Atmosphere	Empirical

#	Process	Family	Fidelity
13	Ontogenetic Diet Shifts. Size/age-based changes in diet composition as organisms grow.	Ecology	Intermediate
14	Multi-Stanza Groups. Life-stage (juvenile/adult) representation with recruitment dynamics.	Ecology	Intermediate
15	Detritus Pathways. Dead organic matter cycling: detritus production, microbial decomposition, detritivore consumption.	Biogeochem	Intermediate
16	Coral-Algae Competition. Competition dynamics between coral and macroalgae mediated by herbivory pressure.	Ecology	Intermediate
17	Network Analysis. Ecosystem network metrics: trophic level, omnivory, flow diversity, system maturity.	Ecology	Intermediate

Table 78: EwE marine ecosystem catalog (17 entries). Primary families: Ecology (12), Biogeochem (3), Human Systems (1), Atmosphere (1).

AX CARMA — Community Aerosol and Radiation Model for Atmospheres

CARMA is a bin-resolved (sectional) aerosol microphysics model that explicitly tracks the aerosol size distribution across 20–40 bins. Unlike modal approaches (MAM in CESM), CARMA resolves the full size distribution evolution from nucleation to sedimentation. Used for stratospheric aerosols, volcanic plumes, noctilucent clouds, wildfire smoke, and mineral dust.

#	Process	Family	Fidelity
1	Homogeneous Nucleation. Classical nucleation theory for new particle formation (H_2SO_4 - H_2O binary, ternary with NH_3).	Atmosphere	Physics-based
2	Heterogeneous Nucleation. Ice nucleation on existing particles (deposition, immersion, contact modes).	Atmosphere	Physics-based
3	Condensation/Evaporation. Mass transfer to/from particles: growth by condensation of supersaturated vapor.	Atmosphere	Physics-based
4	Coagulation. Particle-particle coagulation (Brownian, gravitational, turbulent collection) updating size distribution.	Atmosphere	Physics-based
5	Sedimentation. Gravitational settling velocity as function of particle size, density, altitude (Stokes-Cunningham).	Atmosphere	Physics-based
6	Brownian Diffusion. Vertical mixing of aerosol by Brownian diffusion (small particles).	Atmosphere	Physics-based
7	Aerosol Thermodynamics. Equilibrium partitioning of semi-volatile species between gas and aerosol phases.	Atmosphere	Physics-based
8	Particle Growth by Coagulation. Size distribution evolution from coagulation kernel integration over size bins.	Atmosphere	Physics-based
9	Sulfate Chemistry. H_2SO_4 production from SO_2 oxidation (gas-phase and aqueous); condensation onto particles.	Atmosphere	Physics-based
10	Meteoric Dust Ablation. Meteoric material injection into mesosphere; size-dependent ablation.	Atmosphere	Intermediate
11	Volcanic Injection. Sulfate aerosol formation from volcanic SO_2 injection at stratospheric altitudes.	Atmosphere	Intermediate
12	Noctilucent/PMC Clouds. Ice particle formation at mesopause from water vapor deposition on meteor smoke nuclei.	Atmosphere	Physics-based
13	Wildfire Smoke Aging. Brown carbon absorption evolution, coagulation of smoke particles in aging plumes.	Atmosphere	Intermediate
14	Aerosol Optical Properties. Mie theory: extinction, single scattering albedo, asymmetry parameter per bin.	Radiation	Physics-based

#	Process	Family	Fidelity
15	Radiative Heating Rates. Aerosol-induced shortwave/long-wave heating rate perturbations.	Radiation	Physics-based
16	Dry Deposition. Size-dependent dry deposition velocity to surface.	Atmosphere	Intermediate
17	Wet Scavenging. In-cloud and below-cloud scavenging of aerosol by precipitation.	Atmosphere	Intermediate
18	Mixed Composition Particles. Internal mixing of multiple aerosol components within size bins (sulfate + organics + dust).	Atmosphere	Intermediate

Table 79: CARMA aerosol microphysics catalog (18 entries). Primary families: Atmosphere (16), Radiation (2).

AY ATS — Advanced Terrestrial Simulator

ATS is a multiphysics code built on the Amanzi framework that solves coupled surface/subsurface thermal hydrology with a focus on permafrost environments. It integrates surface energy balance, snow physics, Richards equation in variably-saturated frozen soils, and overland flow. Specifically designed for Arctic/subarctic environments where soil thermal-hydrological coupling is critical.

#	Process	Family	Fidelity
1	Richards Equation (Variably Saturated). 3D variably-saturated flow via Richards equation with nonlinear permeability (van Genuchten).	Hydrology	Physics-based
2	Frozen Soil Hydraulics. Modified relative permeability for ice-bearing soils; impedance of water flow by ice.	Cryosphere	Physics-based
3	Coupled Energy Equation. 3D subsurface energy transport: conduction + advection by water; enthalpy with phase change.	Cryosphere	Physics-based
4	Freeze-Thaw Dynamics. Coupled thermal-hydrological freeze-thaw: cryosuction, ice lens formation, frost heave potential.	Cryosphere	Physics-based
5	Surface Energy Balance. Full radiation + turbulent + ground heat flux balance; skin temperature solver.	Atmosphere	Physics-based
6	Snow Physics. Multi-layer snow: accumulation, densification, wind redistribution, melt, sublimation.	Cryosphere	Physics-based
7	Overland Flow (Diffusion Wave). 2D diffusion wave overland flow coupled to subsurface via infiltration/exfiltration.	Hydrology	Physics-based
8	Microtopography / Ponding. Depression storage, rill connectivity, microtopographic flow routing.	Hydrology	Intermediate
9	Evapotranspiration (Priestley-Taylor). ET partitioning: transpiration, bare soil evaporation, canopy evaporation.	Hydrology	Intermediate
10	Canopy Water/Energy. Simple canopy interception, throughfall, canopy radiation budget.	Ecology	Intermediate
11	Soil Thermal Properties. Temperature- and moisture-dependent thermal conductivity (Kersten/Johansen) and heat capacity.	Hydrology	Physics-based
12	Organic Soil Properties. Peat/organic horizon hydraulic and thermal properties distinct from mineral soil.	Ecology	Intermediate
13	Deformable Soil Column. Soil column deformation from ice thaw/frost heave (subsidence/heave).	Cryosphere	Intermediate

#	Process	Family	Fidelity
14	Permafrost Distribution. Simulated spatial distribution of permafrost from coupled energy/water balance.	Cryosphere	Physics-based
15	Wetland Hydrology. Saturated/inundated conditions with water table dynamics above/below surface.	Hydrology	Physics-based
16	Biogeochemistry Coupling. Interface for coupling to PFLO-TRAN for reactive transport.	Biogeochem	Intermediate
17	Mesh Management (Unstructured). Flexible unstructured mesh for complex topography and watershed geometry.	Hydrology	Physics-based
18	Lateral Subsurface Flow. 3D lateral water movement through heterogeneous subsurface.	Hydrology	Physics-based

Table 80: ATS integrated hydrology catalog (18 entries). Primary families: Hydrology (8), Cryosphere (6), Ecology (2), Atmosphere (1), Biogeochem (1).

AZ BFM — Biogeochemical Flux Model

BFM is a comprehensive marine biogeochemistry model resolving lower trophic levels with explicit representation of multiple plankton functional types, bacteria, detritus, and full C/N/P/Si/Fe/O cycling. Distinguished from MARBL by its European heritage, NEMO coupling, and detailed microbial loop.

#	Process	Family	Fidelity
1	Phytoplankton Photosynthesis. Light-limited, nutrient-limited carbon fixation for 4 types (diatoms, flagellates, picophyto, dinoflagellates).	Biogeochem	Intermediate
2	Phytoplankton Nutrient Uptake. Multi-nutrient (N, P, Si, Fe) uptake with Droop quota dynamics (luxury uptake, internal quotas).	Biogeochem	Intermediate
3	Chlorophyll Synthesis. Photoacclimation: dynamic Chl:C ratio based on light and nutrient status.	Biogeochem	Intermediate
4	Zooplankton Grazing. Grazing by meso- and microzooplankton with prey preferences and threshold feeding.	Biogeochem	Intermediate
5	Bacterial Dynamics. Explicit heterotrophic bacteria: DOC uptake, remineralization, nutrient uptake/competition.	Biogeochem	Intermediate
6	Microbial Loop. Bacteria-DOM-flagellate interactions; viral lysis pathway (implicit).	Biogeochem	Intermediate
7	Dissolved Organic Matter. Semi-labile and refractory DOM pools (C, N, P); bacterial utilization and abiotic degradation.	Biogeochem	Intermediate
8	Particulate Organic Matter. Sinking POM: production from mortality/egestion; depth-dependent remineralization.	Biogeochem	Intermediate
9	Dissolved Inorganic Carbon. DIC dynamics: photosynthesis, respiration, air-sea exchange, carbonate chemistry.	Biogeochem	Physics-based
10	Alkalinity. Total alkalinity budget: calcification, CaCO_3 dissolution, nutrient uptake/release.	Biogeochem	Physics-based
11	Oxygen Dynamics. O_2 production/consumption; air-sea exchange; suboxic/anoxic transitions.	Biogeochem	Intermediate
12	Nitrogen Cycle (Full). NO_3 , NH_4 , NO_2 pools; nitrification, denitrification (water column + sediment), N_2 fixation.	Biogeochem	Intermediate
13	Phosphorus Cycle. PO_4 uptake/remineralization; luxury uptake with internal P quota.	Biogeochem	Intermediate
14	Silicon Cycle. Silicic acid uptake by diatoms; biogenic silica dissolution.	Biogeochem	Intermediate

#	Process	Family	Fidelity
15	Iron Cycle. Dissolved Fe: dust input, scavenging, biological uptake, ligand complexation.	Biogeochem	Intermediate
16	Calcification / CaCO_3 . Calcite/aragonite production, dissolution below saturation horizon, ballast effect.	Biogeochem	Intermediate
17	Benthic-Pelagic Coupling. Organic matter deposition to sediments; nutrient return flux to water column.	Biogeochem	Intermediate
18	Light Penetration. PAR attenuation through water column (self-shading by Chl + CDOM + detritus).	Radiation	Intermediate
19	Air-Sea CO_2 Exchange. CO_2 flux via wind-speed dependent piston velocity; surface pCO_2 from carbonate system.	Biogeochem	Physics-based
20	N_2O / DMS Production. Trace gas cycling: N_2O from nitrification/denitrification; DMS from DMSP cleavage.	Biogeochem	Empirical
21	Sea-Ice Biogeochemistry. Brine channel biology; ice algae production; nutrient release upon melt.	Cryosphere	Intermediate
22	River Input. Riverine dissolved/particulate nutrient and carbon inputs to coastal zones.	Biogeochem	Empirical

Table 81: BFM marine biogeochemistry catalog (22 entries). Primary families: Biogeochem (20), Radiation (1), Cryosphere (1).

BA MAESTRA/MAESTRO — 3D Individual-Tree Canopy Model

MAESTRA (Multi-Array Evapotranspiration Simulation of Trees and Radiation Absorption) is a 3D individual-tree canopy model computing radiation interception, photosynthesis, transpiration, and respiration for tree stands at half-hourly resolution. Originally developed by Ying-Ping Wang (CSIRO, 1996) and extended by Belinda Medlyn and the Jarvis group (Edinburgh, 1997–2001) for the EU ECOCRAFT project. Crowns are discretized into up to 720 volumetric grid points; ray paths trace through all neighbouring crowns with Beer–Lambert extinction. Norman (1979) iterative multi-scattering resolves PAR, NIR, and thermal fluxes. Farquhar–von Caemmerer C3 photosynthesis couples iteratively with Ball-Berry, Ball-Berry-Leuning, or Jarvis stomatal conductance; Penman-Monteith transpiration and iterative leaf energy balance close the surface flux loop. Six crown shapes (cone, half-ellipsoid, paraboloid, full-ellipsoid, cylinder, box) with beta-function leaf area density distributions and ellipsoidal leaf-angle distributions (Campbell 1986) are supported. Its successor MAESPA extends the model to soil water balance.

#	Process	Family	Fidelity
1	3D Crown Radiation Transfer. Beam/diffuse/thermal radiation through individual tree crowns; Beer–Lambert extinction with weighted pathlengths through neighbouring crowns (TREDST/WPATH); sunlit/shaded separation. 6 crown shapes.	Radiation	Physics-based
2	Solar Geometry. Full astronomical position (Barkstrom 1981): declination, equation of time, daylength, half-hourly zenith/azimuth, slope correction factors (Steven & Unsworth 1979/1980).	Radiation	Physics-based
3	Farquhar–von Caemmerer C3 Photosynthesis. Rubisco-limited (A_c) and electron transport-limited (A_j) assimilation; peaked Arrhenius $V_{\max}/J_{\max}$; ECOCRAFT parameterization; coupled iterative C_i solution.	Biogeochem	Physics-based
4	Stomatal Conductance Suite. Three models: Jarvis multiplicative (4 VPD options), Ball-Berry, Ball-Berry-Leuning. Soil moisture modifier (Granier & Loustau 1994).	Ecology	Intermediate
5	Penman-Monteith Transpiration & Energy Balance. Leaf-level and canopy-scale PM ET with iterative leaf temperature (Leuning et al. 1995). Radiation conductance, forced/free convection boundary layer.	Hydrology	Physics-based
6	Autotrophic Respiration. Q10 maintenance for 5 pools (foliage, stem, branch, fine root, coarse root) plus growth respiration from biomass increment $\times$ construction cost.	Biogeochem	Intermediate
7	Crown Geometry & Allometry. 6 crown shapes; beta-function LAD (vertical/horizontal, 3 age classes); ellipsoidal	Ecology	Intermediate

#	Process	Family	Fidelity
	leaf-angle distribution (Campbell 1986); simple 4-parameter phenology; allometric stem/branch/root biomass.		
8	Met Processing & Radiation Partitioning. Daily→half-hourly disaggregation: sinusoidal T cycle; Spitters (1986) beam/diffuse; Bristow-Campbell transmissivity; exponential canopy wind; humidity conversions; CO ₂ /OTC scenarios.	Atmosphere	Intermediate
9	Canopy Integration & Stand Aggregation. Summation of leaf-level fluxes to tree-level and stand-level (SUMTREES): CO ₂ , transpiration, sensible heat, respiration. Weighted by stocking density.	Ecology	Intermediate
10	Leaf Optics & Multi-Scattering. Norman (1979) iterative scheme on Equivalent Horizontal Canopy layers for PAR/NIR/thermal; absorptance/reflectance/transmittance per wavelength; soil reflectance.	Radiation	Physics-based

Table 82: MAESTRA/MAESTRO 3D canopy model catalog (10 entries). Primary families: Radiation (3), Ecology (3), Biogeochem (2), Hydrology (1), Atmosphere (1).

BB Knowledgebase Summary Statistics

Source Model	Entries	Primary Family	Physics-Based
Badlands	33	Geomorphology	1
CESM (CAM+CLM+MOM6+CICE+CISM+MO SART+WWA)	182	Atmosphere	65
E3SM (unique beyond CESM)	15	Atmosphere/Ocean	11
FATES	90	Ecology	18
iLand	21	Ecology	0
Landlab	49	Geomorphology	7
LPJ-GUESS	18	Biogeochem	1
Noah-MP	32	Hydrology	9
ParFlow	16	Hydrology	12
WRF / WRF-SFIRE / WRF-Hydro	43	Atmosphere	14
ORCHIDEE	26	Hydrology/BGC	5
VIC	17	Hydrology	5
GFDL ESM4	9	Atmosphere	6
PEcAn	20	Ecology	9
LANDIS-II Core	6	Ecology	0
JULES	25	Hydrology	7
CABLE	18	Hydrology/BGC	6
CLASSIC	16	Biogeochem	2
SUMMA	20	Hydrology	8
ED2	18	Ecology	3
PFLOTRAN	15	Geology/BGC	13
ROMS	20	Ocean	12
SWAT	18	Hydrology	1
GEOS-Chem	20	Atmosphere	6
Delft3D	16	Ocean	12

Source Model	Entries	Primary Family	Physics-Based
LISFLOOD-FP	10	Hydrology	3
LM3-PPA / BiomeE	22	Ecology	2
SORTIE-ND	14	Ecology	1
JABOWA / FORET / LINKAGES	12	Ecology	0
FORMIND	14	Ecology	0
LANDIS-II Extensions (NECN+)	18	Ecology/BGC	0
UVAFME	14	Ecology	0
FORCLIM	12	Ecology	0
SORTIE-NG (Erickson)	16	Ecology	2
DeepLand (Erickson)	18	Hydrology/Ecology	10
Earth-2 / earth2studio (NVIDIA)	14	Atmosphere	5
PhiSat-2 principles (ESA)	12	Multi-family	0
FireBench (benchmark)	8	Fire	—
MARBL	20	Biogeochem	2
PISM	23	Cryosphere	13
OGGM	15	Cryosphere	2
APSIM	25	Ecology/BGC	2
DSSAT	24	Ecology/BGC	0
CryoGrid	18	Cryosphere	11
XBeach	20	Ocean/Geomorphology	12
SURFEX/TEB	18	Atmosphere	8
EwE	17	Ecology	0
CARMA	18	Atmosphere	12
ATS	18	Hydrology/Cryo	12
BFM	22	Biogeochem	3
MAESTRA	10	Radiation/Ecology	4

Source Model	Entries	Primary Family	Physics-Based
Total	1195		336

Table 83: Summary of initial Process Knowledgebase entries by source model/platform (51 models + 1 benchmark suite). FireBench entries are benchmarking resources, not process implementations.

Process Family	Count	Top Contributors
Ecology	218	FATES, EwE, APSIM, DSSAT, ED2, SORTIE-ND, SORTIE-NG, DeepLand, PPA/BiomeE, JABOWA/FORET, FORMIND, iLand, LANDIS-II, UVAFME, FORCLIM, PhiSat-2, MAESTRA
Hydrology	188	CLM, ParFlow, SUMMA, ATS, APSIM, DSSAT, SWAT, JULES, CABLE, Noah-MP, LISFLOOD-FP, ROMS, CryoGrid, DeepLand, PhiSat-2, MAESTRA
Biogeochem	184	MARBL, BFM, CLM, E3SM, FATES, GEOS-Chem, PFLOTRAN, APSIM, DSSAT, LANDIS-II NECN, PPA/BiomeE, ORCHIDEE, SORTIE-NG, DeepLand, MAESTRA
Atmosphere	179	CESM/CAM, CARMA, E3SM/EAM, WRF, SURFEX/TEB, GEOS-Chem, JULES, Delft3D, Earth-2, SORTIE-NG, DeepLand, PhiSat-2, MAESTRA
Cryosphere	107	PISM, OGGM, CryoGrid, ATS, CICE, MPAS-SeaIce, SUMMA, JULES, ROMS, VIC, UVAFME, BFM, CLM snow, DeepLand, PhiSat-2
Ocean	89	XBeach, ROMS, Delft3D, MOM6, MPAS-Ocean, PISM, WW3, SURFEX/TEB, Badlands waves, PhiSat-2
Human Systems	61	APSIM, DSSAT, SURFEX/TEB, SWAT, GEOS-Chem, EwE, LANDIS-II, FORMIND, CLM, PFLOTRAN, JULES, SORTIE, FORCLIM, PhiSat-2
Geomorphology	59	Badlands, Landlab, XBeach, ROMS, Delft3D, SWAT, APSIM, MOSART
Fire	52	FATES/SPITFIRE, WRF-SFIRE, LANDIS-II SCRPPLE, CLASSIC, iLand, UVAFME, SORTIE-NG, PhiSat-2, FireBench
Radiation	46	SUMMA, CARMA, SURFEX/TEB, MARBL, BFM, GEOS-Chem, PPA, SORTIE-NG, DeepLand, CABLE, ED2, CAM, CLM/SNICAR, FATES, MAESTRA

Process Family	Count	Top Contributors
Geology	20	PFLOTRAN, PISM, CryoGrid, Badlands, Landlab, ParFlow
Evolution	1	Landlab SpeciesEvolver
Total	1195	

Table 84: Process family distribution across all 51 reference models and platforms.

The 1195 entries cataloged here represent the *initial seed* of the MAESMA Process Knowledgebase, drawn from 51 reference models and platforms and one benchmarking dataset (FireBench). The catalog spans the full Earth system — from deep-ocean biogeochemistry (MARBL, BFM) and ice sheet dynamics (PISM, OGGM) through permafrost thermodynamics (CryoGrid, ATS), coastal morphodynamics (XBeach), agricultural systems (APSIM, DSSAT), urban energy balance (SURFEX/TEB), marine food webs (EwE), aerosol microphysics (CARMA), 3D individual-tree canopy radiation and physiology (MAESTRA), GPU-native models (SORTIE-NG, DeepLand), a foundation weather model platform (NVIDIA Earth-2 / earth2studio), and autonomous observation intelligence principles (ESA PhiSat-2) that collectively exemplify the architecture’s core identity as a multi-agent system for *inferential process discovery, combination, simulation, and evolution*. Representative entries (50+ manifests from 19 source models) are implemented as typed `ProcessManifest` structures with machine-readable I/O contracts, scale envelopes, conservation properties, and cost models via the `generate_seed_manifests()` function; 40+ ontological relations (RequiresCouplingWith, IncompatibleWith, Supersedes, VariantOf, CompatibleWith) are implemented via `generate_seed_relations()`. Empirical skill records will accumulate during autonomous benchmarking. Discovered representations will *evolve* through crossover, mutation, and Pareto-optimal selection — the knowledgebase is the genome, the Pareto front is the fitness landscape. The neural inference engine begins reasoning over this graph from its first cycle, the Foundation Model Agent generates rapid ensemble atmospheric states via Earth-2, the Autonomous Observation Agent steers data acquisition toward maximal information gain via PhiSat-2-class edge-AI, and the process discovery pipeline — leveraging FNO, PINO, DeepONet, and MeshGraphNet architectures trained with Modulus physics-informed loss functions on GPU — will continuously expand and evolve the knowledgebase with learned representations from observational data.

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